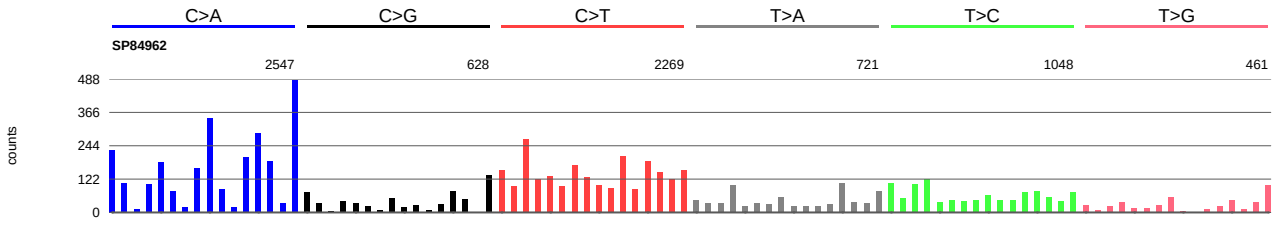
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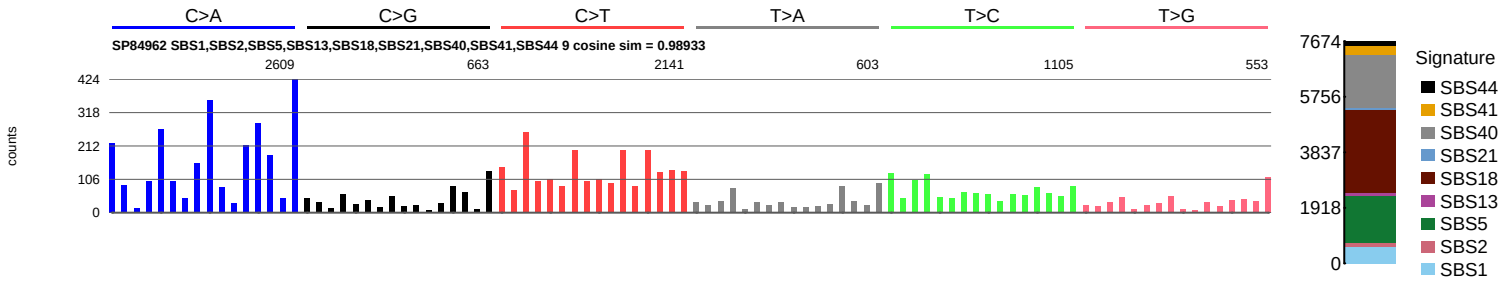
counts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SP84962 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98933
2609
663
2141
603
1105
553
424
318
212
106
0
counts
7674
5756
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Signature
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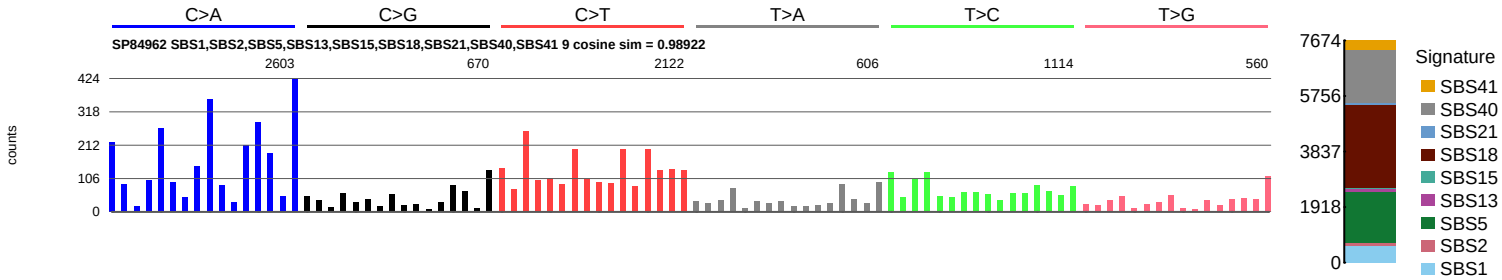
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SP84962 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98922
2603
670
2122
606
1114
560
424
318
212
106
0
counts
7674
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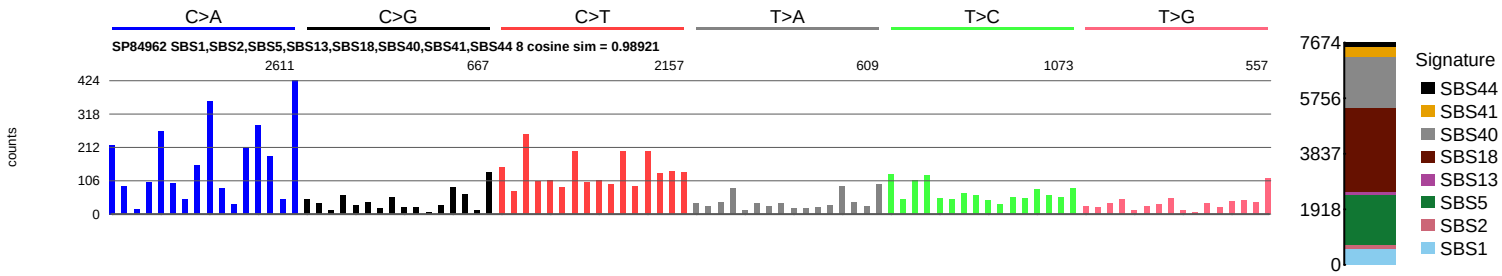
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667
2157
609
1073
557
424
318
212
106
0
counts
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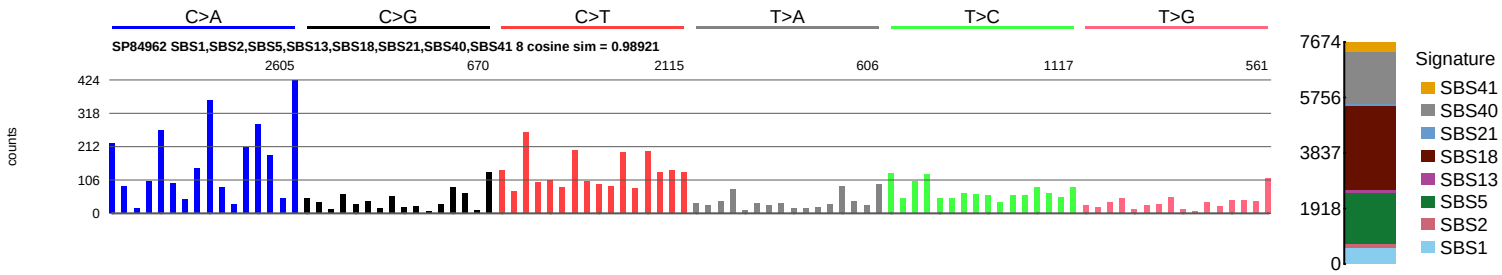
SBS1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670
2115
606
1117
561
424
318
212
106
0
counts
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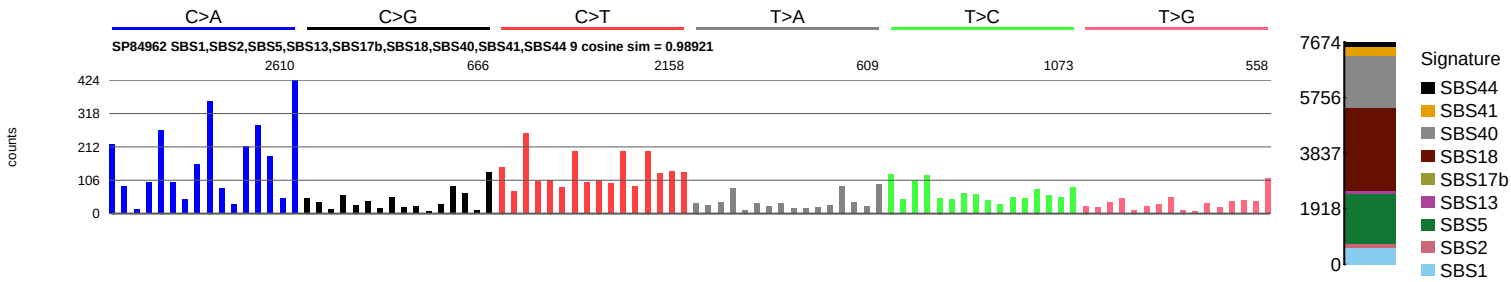
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2610
666
2158
609
1073
558
424
318
212
106
0
counts
7674
5756
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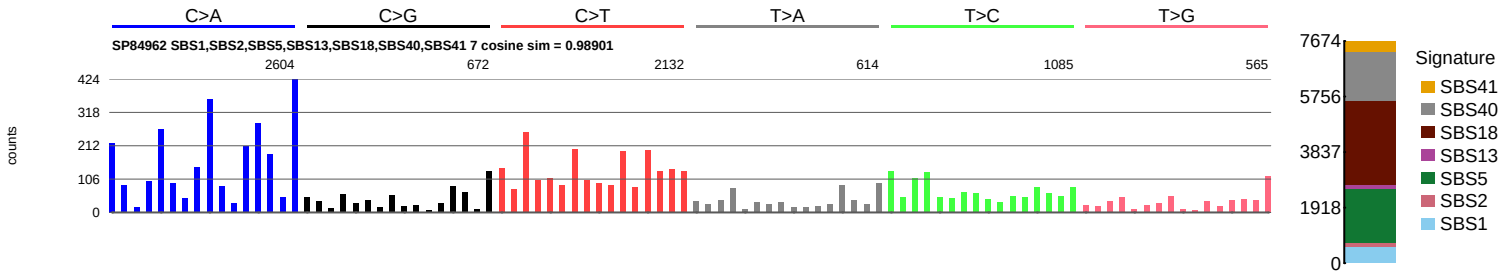
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672
2132
614
1085
565
counts
424
318
212
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3837
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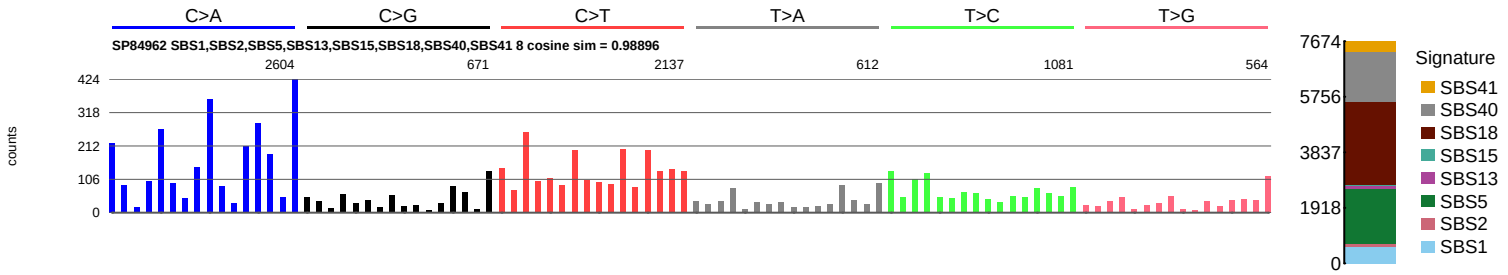
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671
2137
612
1081
564
counts
424
318
212
106
0
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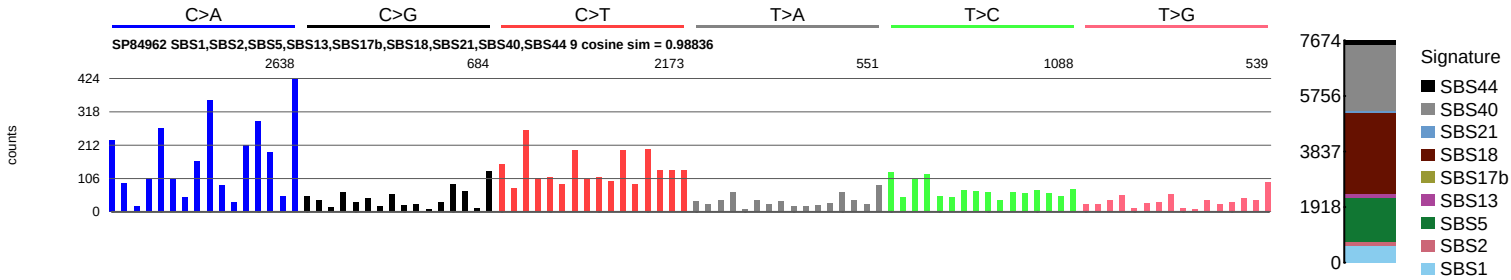
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684
2173
551
1088
539
counts
424
318
212
106
0
7674
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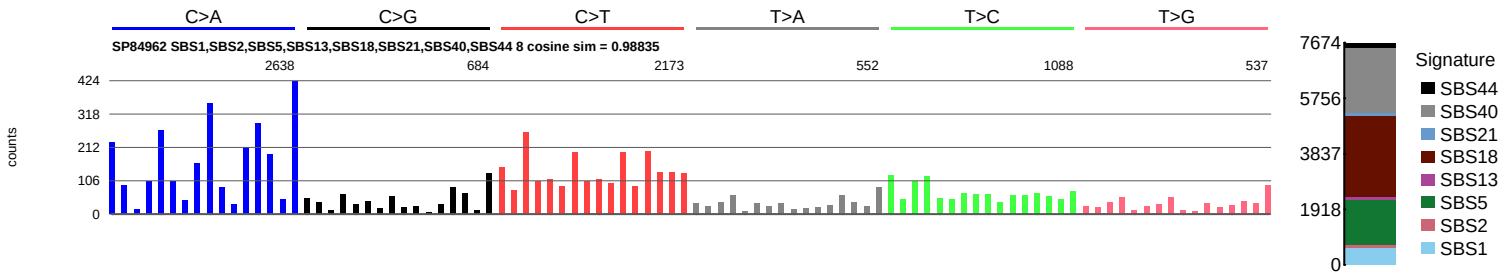
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684
2173
552
1088
537
counts
424
318
212
106
0
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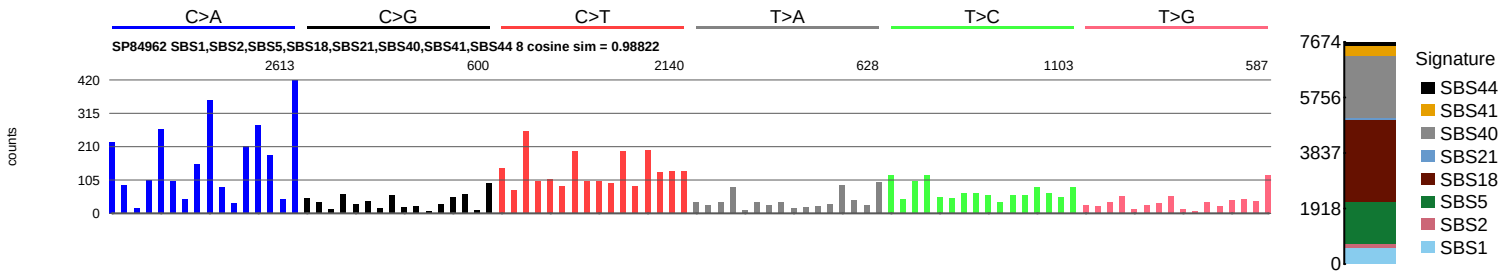
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2613
600
2140
628
1103
587
counts
420
315
210
105
0
7674
5756
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1918
0
Signature
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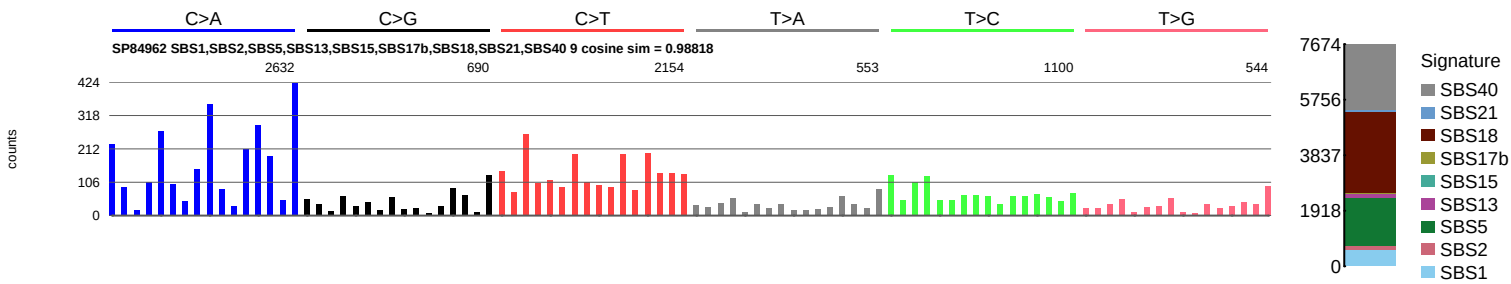
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SP84962 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98818
2632
690
2154
553
1100
544
counts
424
318
212
106
0
7674
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1918
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Signature
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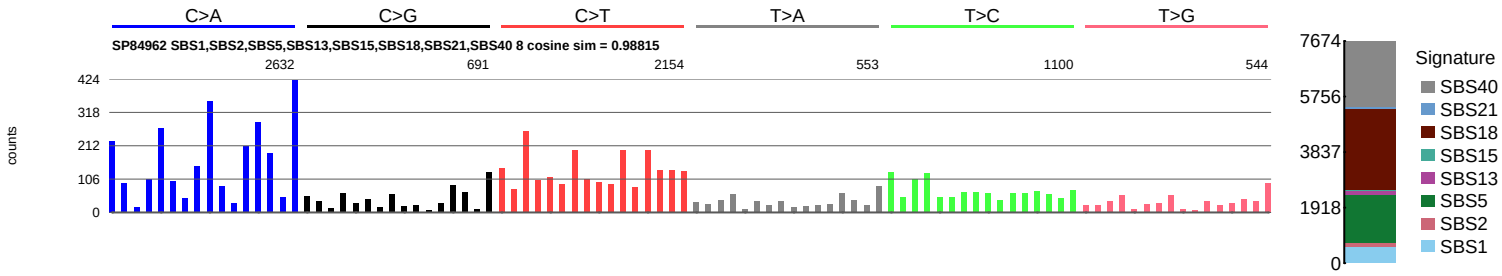
SBS1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2154
553
1100
544
counts
424
318
212
106
0
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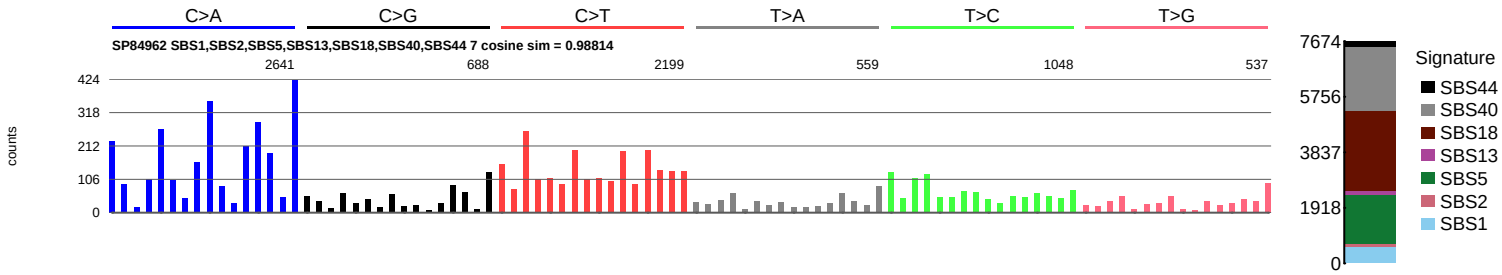
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688
2199
559
1048
537
counts
424
318
212
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3837
1918
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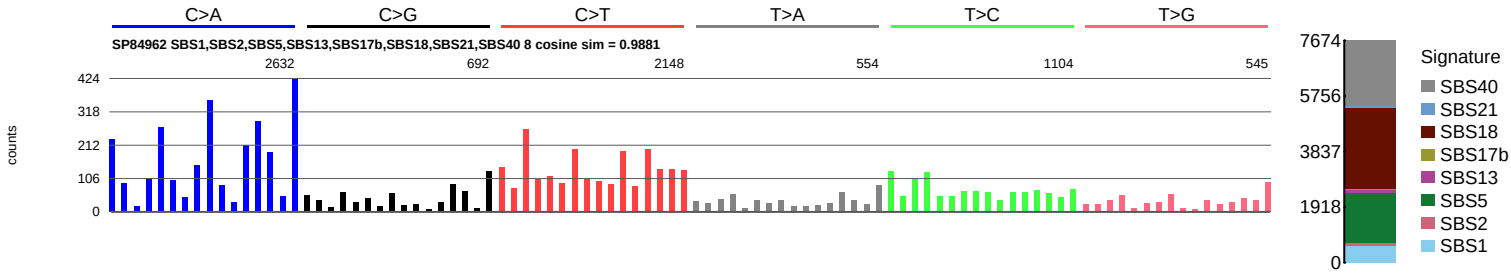
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SP84962 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.9881
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692
2148
554
1104
545
counts
424
318
212
106
0
7674
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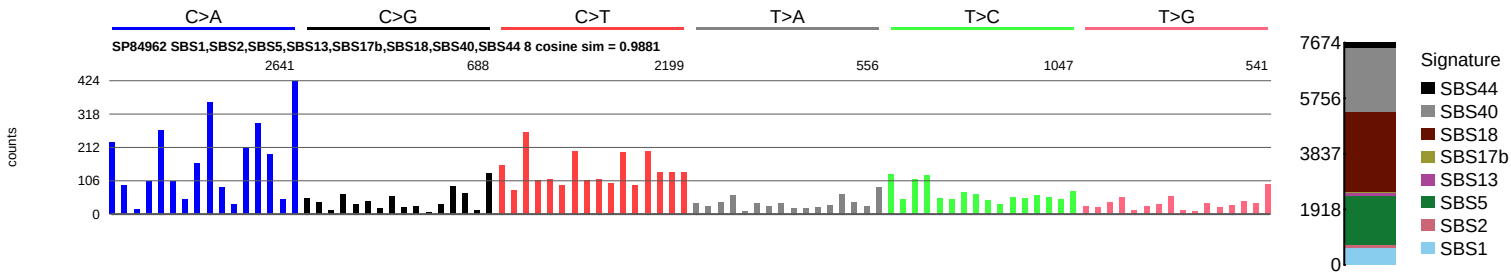
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688
2199
556
1047
541
counts
424
318
212
106
0
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3837
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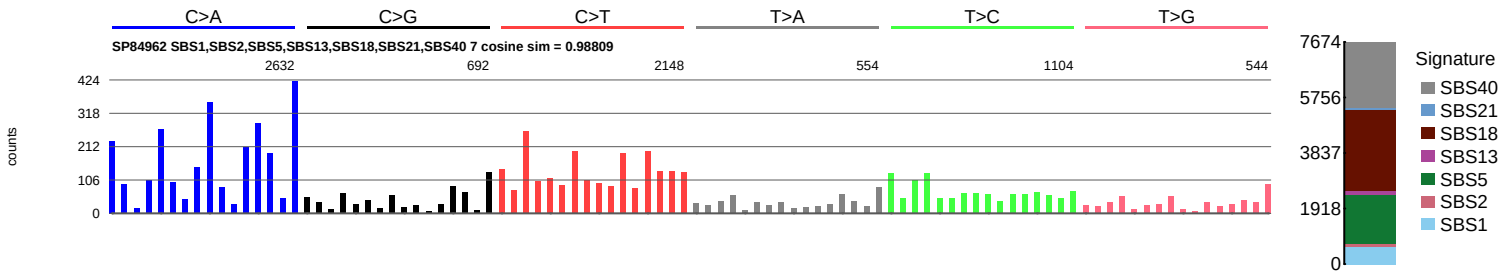
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692
2148
554
1104
544
counts
424
318
212
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1918
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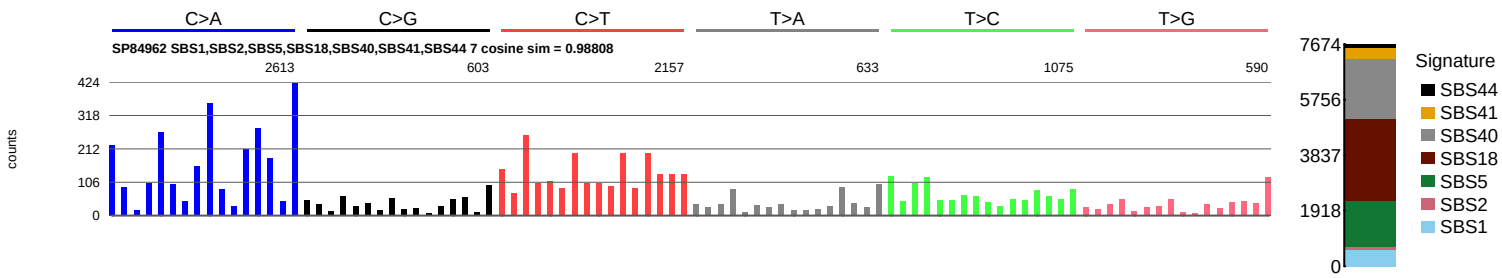
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2157
633
1075
590
counts
424
318
212
106
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5756
3837
1918
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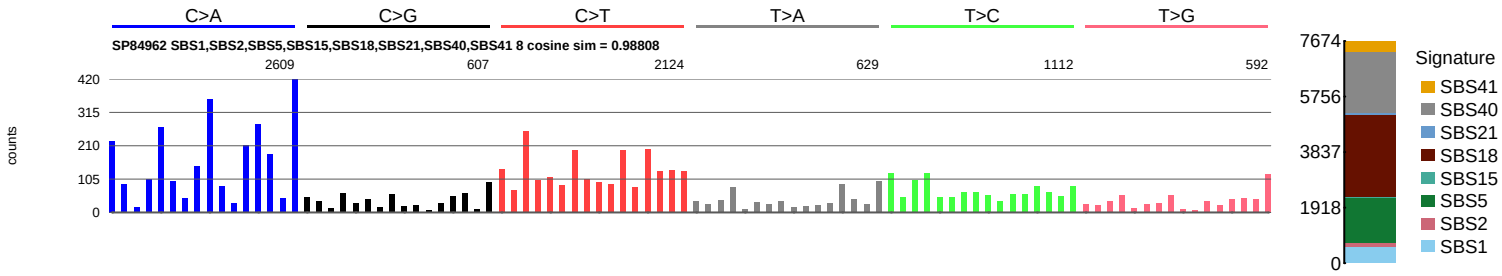
SBS1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607
2124
629
1112
592
counts
420
315
210
105
0
7674
5756
3837
1918
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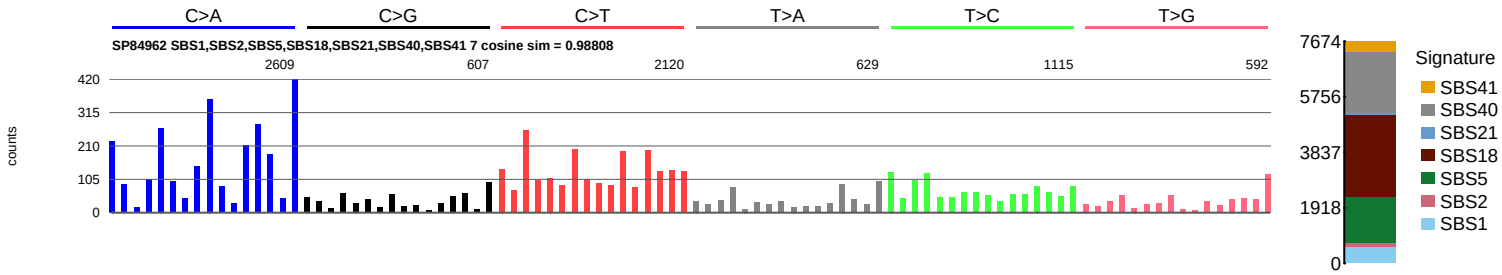
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607
2120
629
1115
592
counts
420
315
210
105
0
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5756
3837
1918
0
Signature
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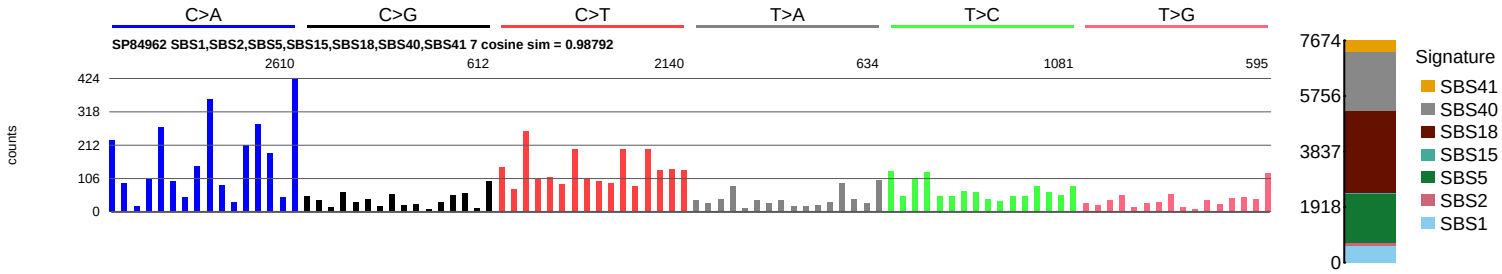
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2610
612
2140
634
1081
595
counts
424
318
212
106
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3837
1918
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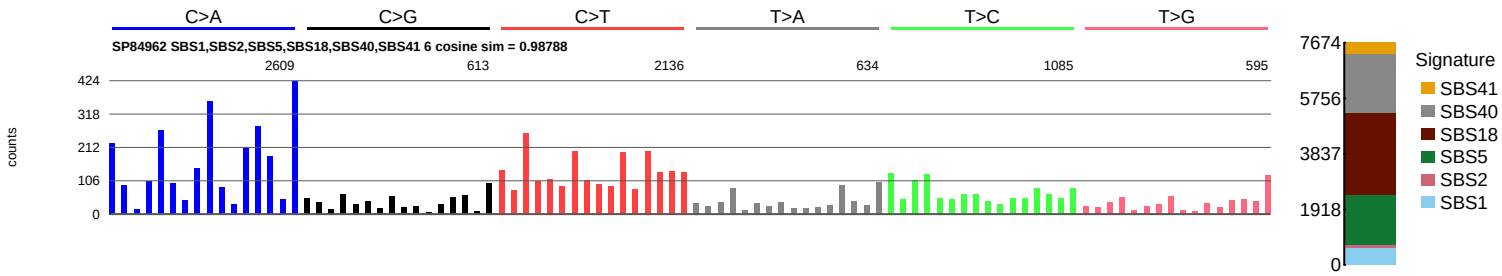
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613
2136
634
1085
595
counts
424
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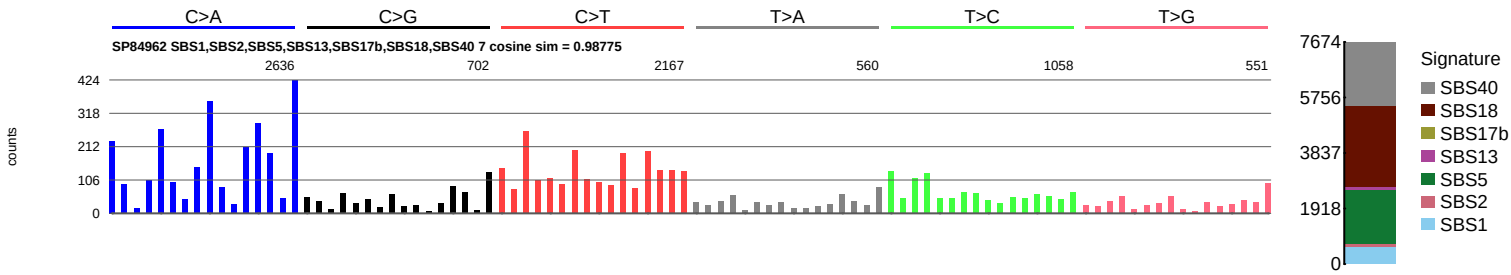
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702
2167
560
1058
551
counts
424
318
212
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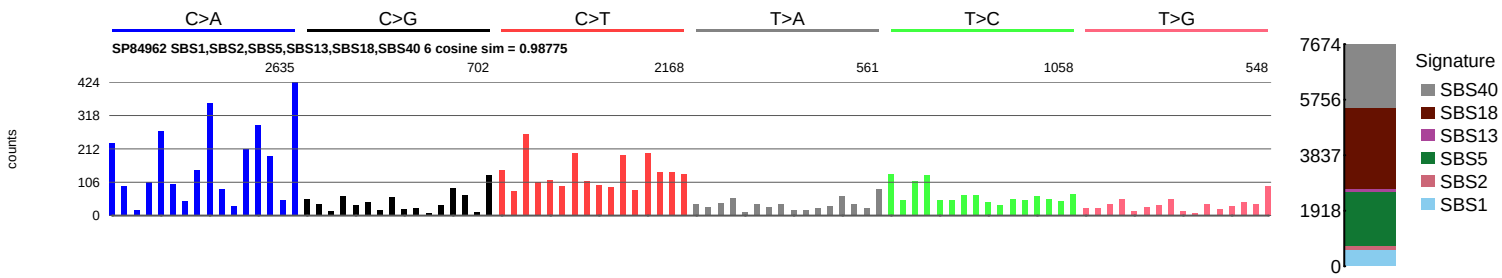
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2635
702
2168
561
1058
548
counts
424
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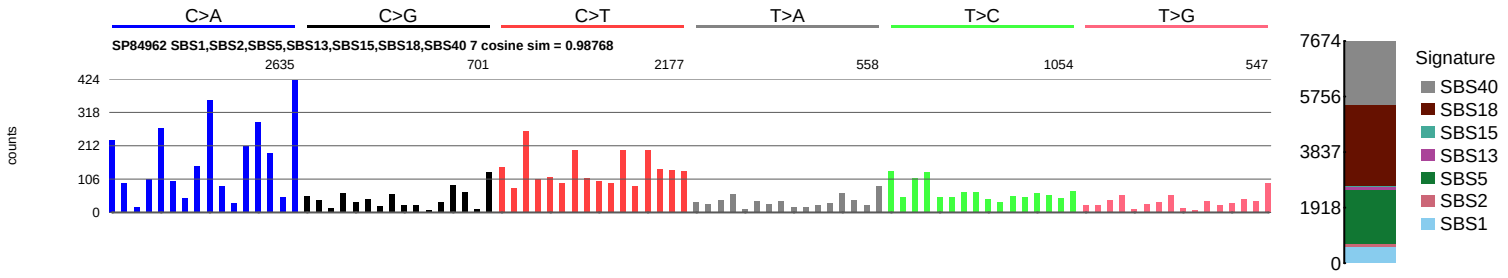
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701
2177
558
1054
547
counts
424
318
212
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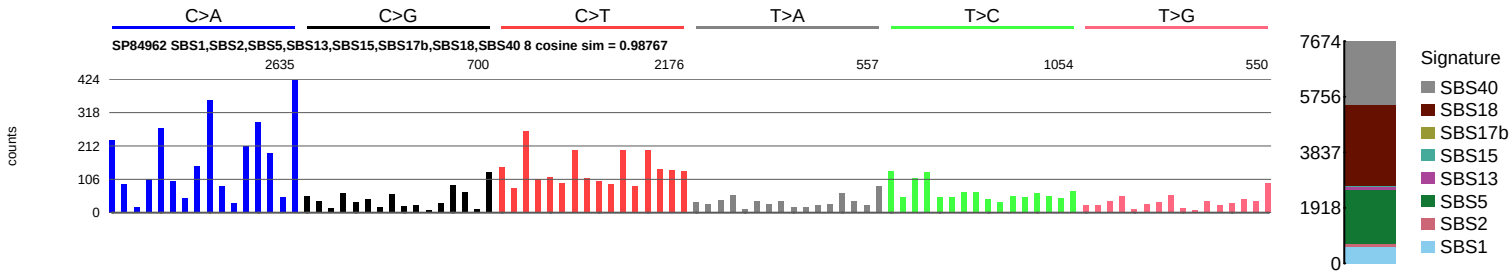
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700
2176
557
1054
550
counts
424
318
212
106
0
7674
5756
3837
1918
0
Signature
SBS40
SBS18
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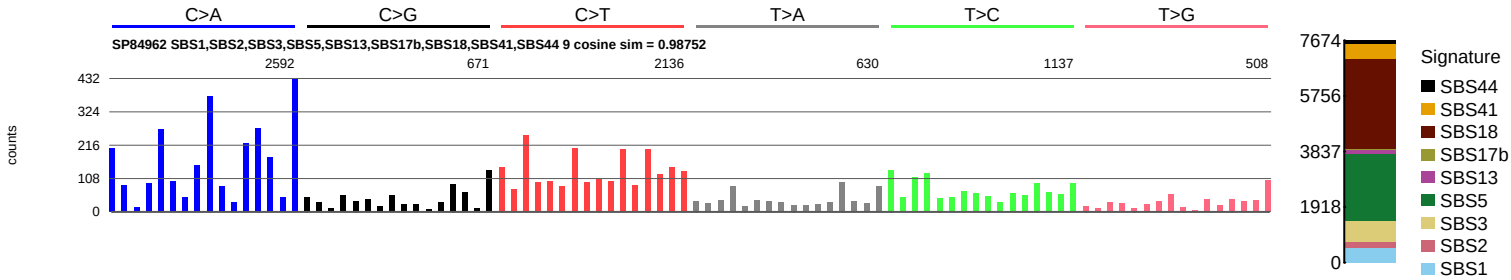
SBS13
SBS5
SBS2
SBS1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SP84962 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98752
2592
671
2136
630
1137
508
counts
432
324
216
108
0
7674
5756
3837
1918
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Signature
SBS44
SBS41
SBS18
SBS17b
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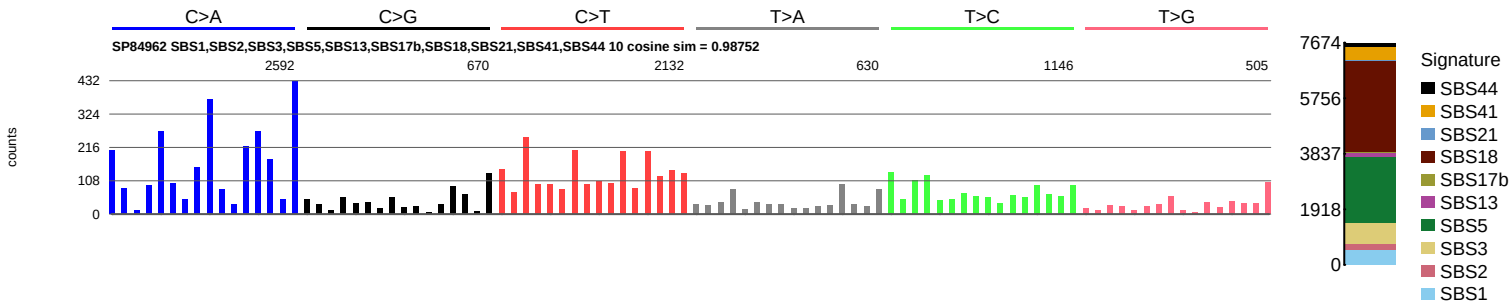
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SP84962 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98752
2592
670
2132
630
1146
505
counts
432
324
216
108
0
7674
5756
3837
1918
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Signature
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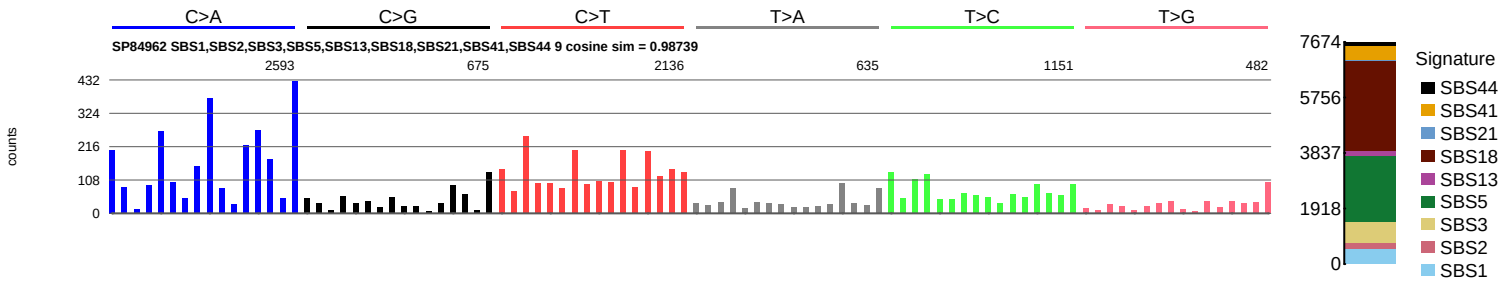
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2593
675
2136
635
1151
482
counts
432
324
216
108
0
7674
5756
3837
1918
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SBS13
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SBS2
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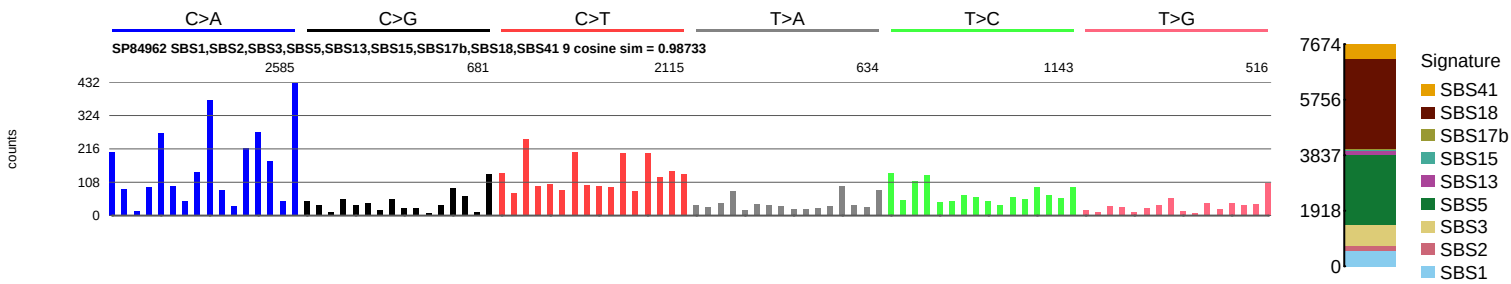
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2585
681
2115
634
1143
516
counts
432
324
216
108
0
7674
5756
3837
1918
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Signature
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SBS18
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SBS13
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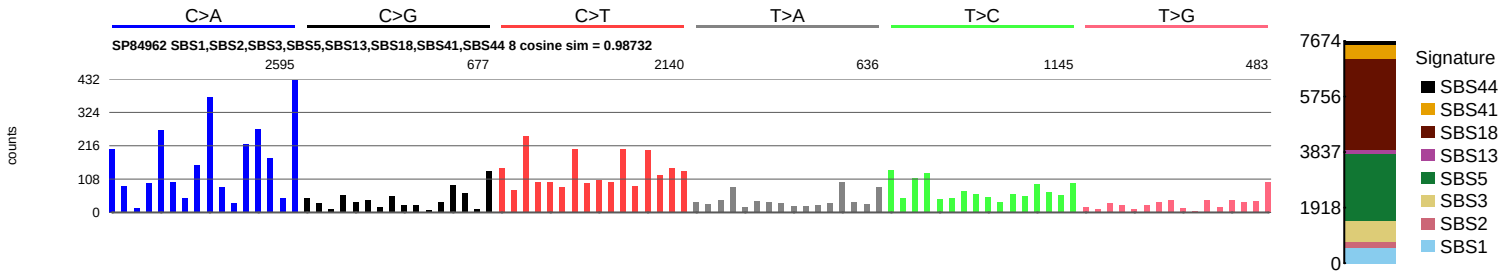
SBS3
SBS2
SBS1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SP84962 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS41,SBS44 8 cosine sim = 0.98732
2595
677
2140
636
1145
483
counts
432
324
216
108
0
7674
5756
3837
1918
0
Signature
SBS44
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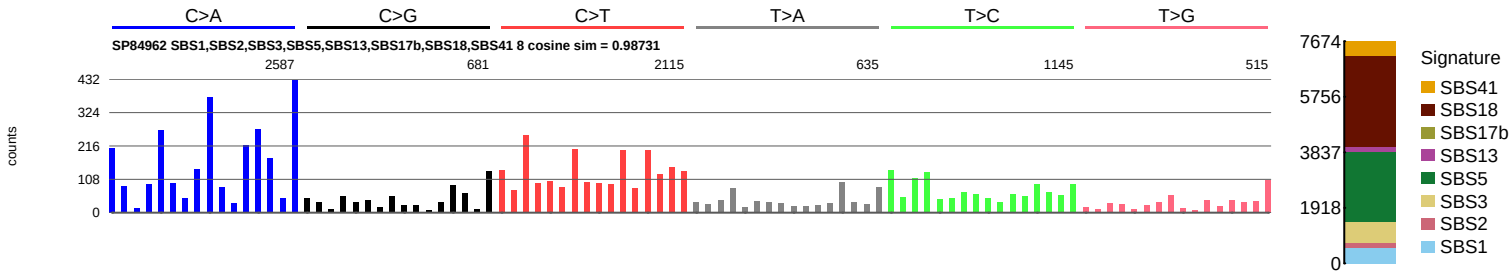
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2587
681
2115
635
1145
515
counts
432
324
216
108
0
7674
5756
3837
1918
0
Signature
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SBS17b
SBS13
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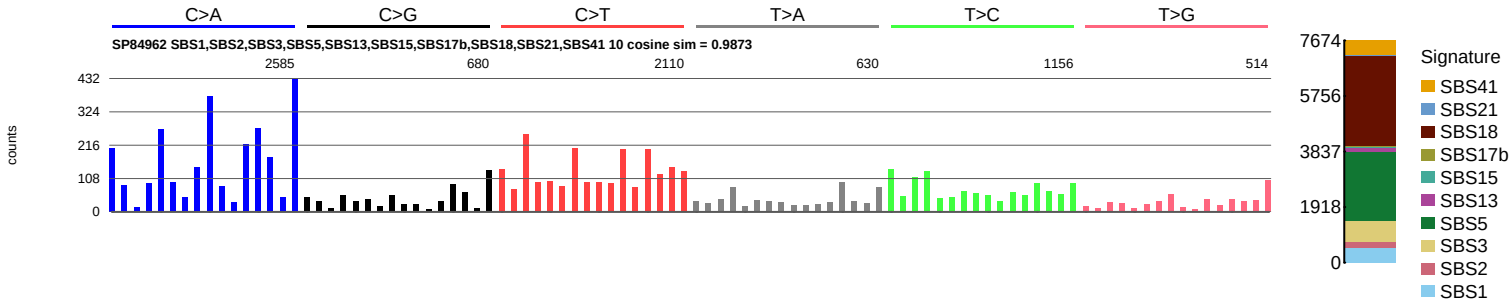
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2585
680
2110
630
1156
514
counts
432
324
216
108
0
7674
5756
3837
1918
0
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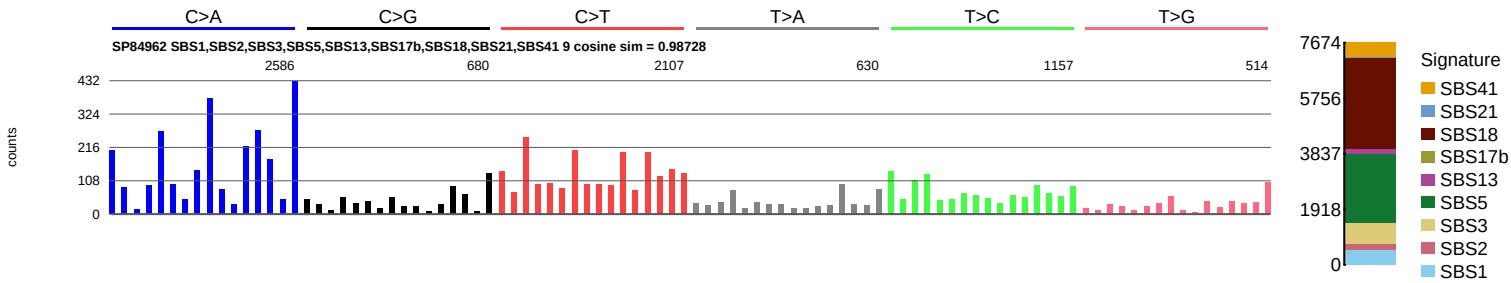
SBS2
SBS1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SP84962 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98728
2586
680
2107
630
1157
514
counts
432
324
216
108
0
7674
5756
3837
1918
0
Signature
SBS41
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SBS13
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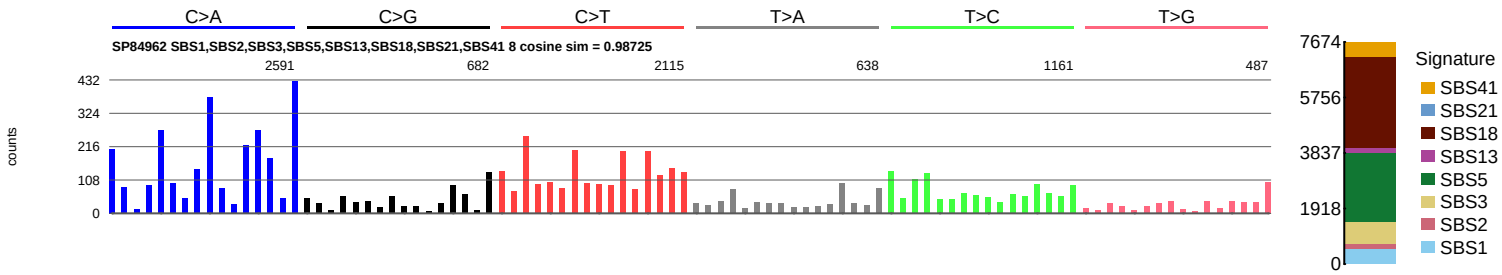
SBS2
SBS1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SP84962 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS41 8 cosine sim = 0.98725
2591
682
2115
638
1161
487
counts
432
324
216
108
0
7674
5756
3837
1918
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Signature
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SBS21
SBS18
SBS13
SBS5
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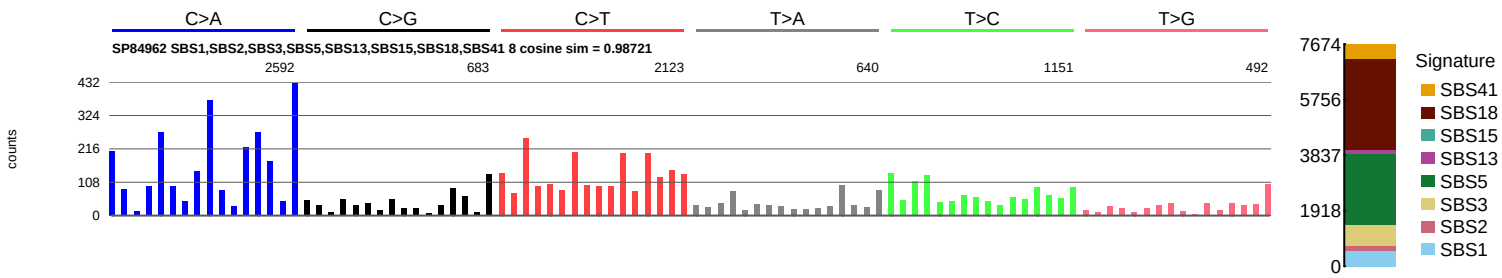
SBS2
SBS1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T>C
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SP84962 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS41 8 cosine sim = 0.98721
2592
683
2123
640
1151
492
counts
432
324
216
108
0
7674
5756
3837
1918
0
Signature
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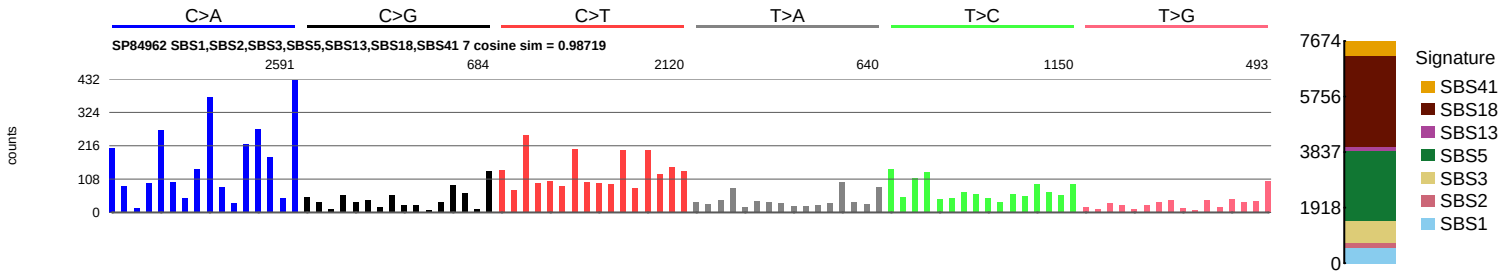
SBS2
SBS1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SP84962 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS41 7 cosine sim = 0.98719
2591
684
2120
640
1150
493
counts
432
324
216
108
0
7674
5756
3837
1918
0
Signature
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SBS13
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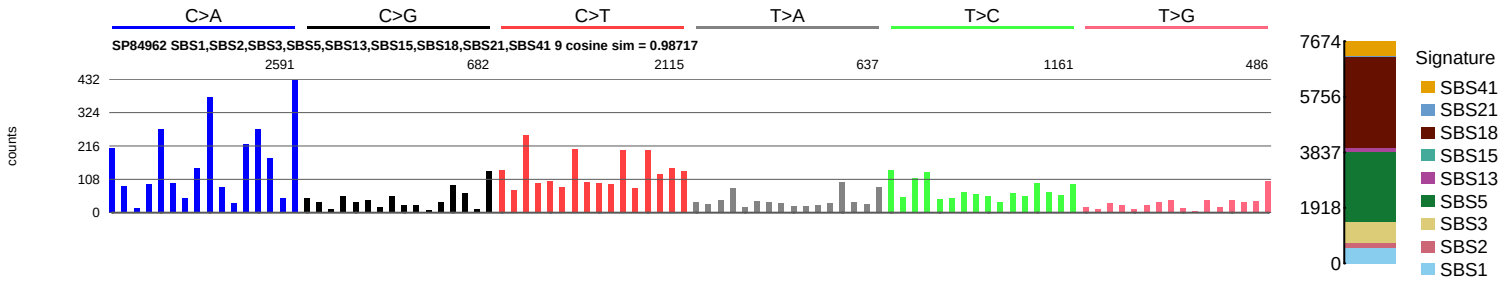
SBS3
SBS2
SBS1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SP84962 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS41 9 cosine sim = 0.98717
2591
682
2115
637
1161
486
counts
432
324
216
108
0
7674
5756
3837
1918
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Signature
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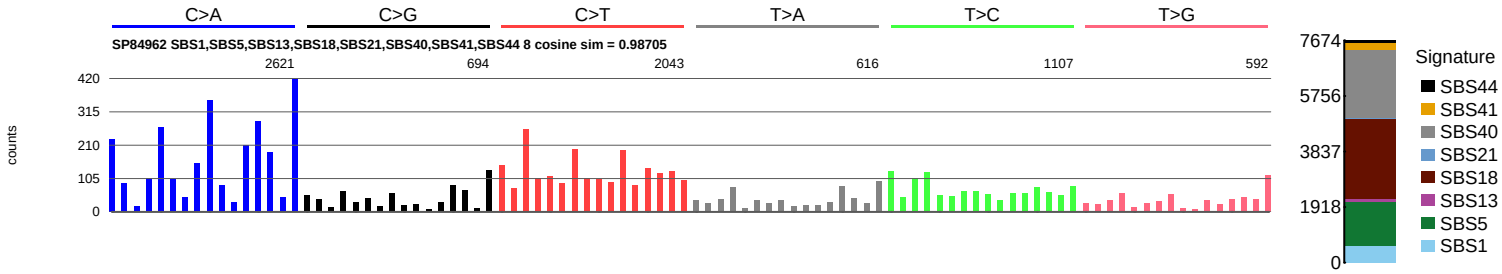
SBS2
SBS1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SP84962 SBS1,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98705
2621
694
2043
616
1107
592
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
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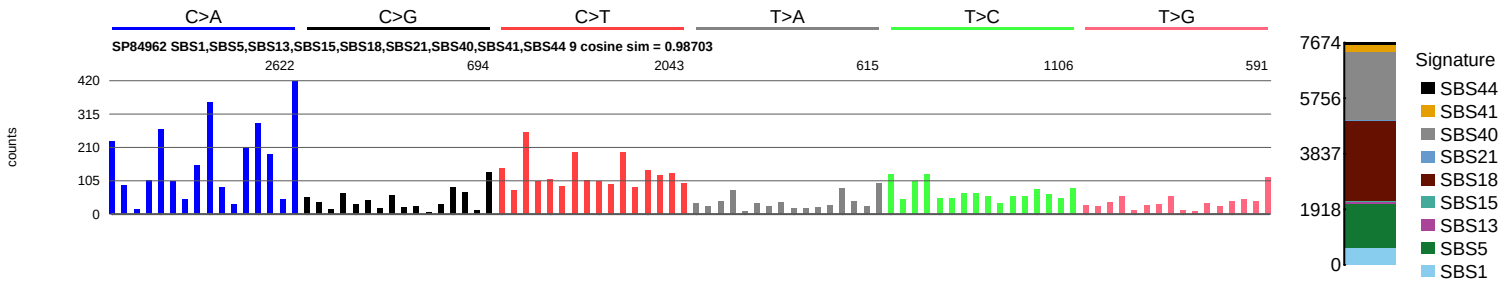
SBS21
SBS18
SBS13
SBS5
SBS1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C>G
C>T
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T>G
SP84962 SBS1,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98703
2622
694
2043
615
1106
591
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
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SBS41
SBS40
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SBS18
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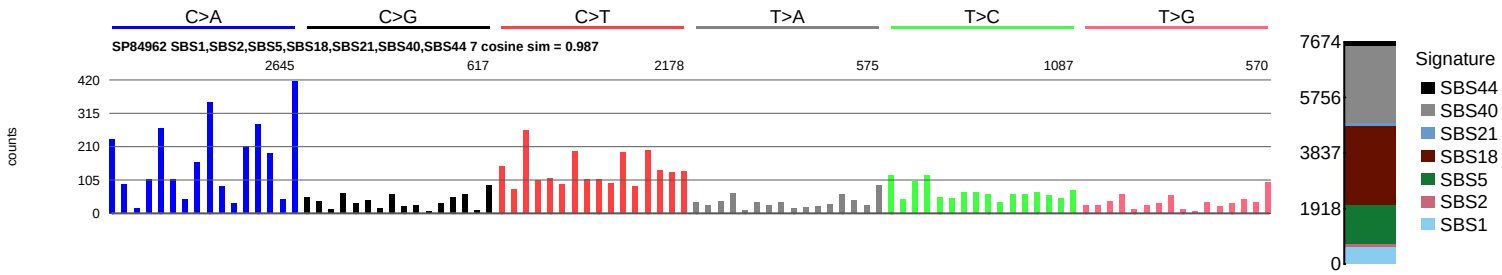
SBS13
SBS5
SBS1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C>G
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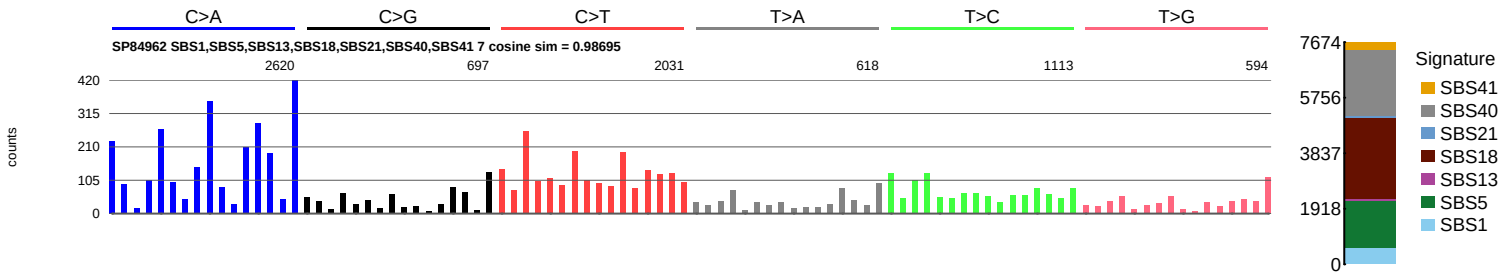
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.987
2645
617
2178
575
1087
570
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS5
SBS2
SBS1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SP84962 SBS1,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98695
2620
697
2031
618
1113
594
counts
420
315
210
105
0
7674
5756
3837
1918
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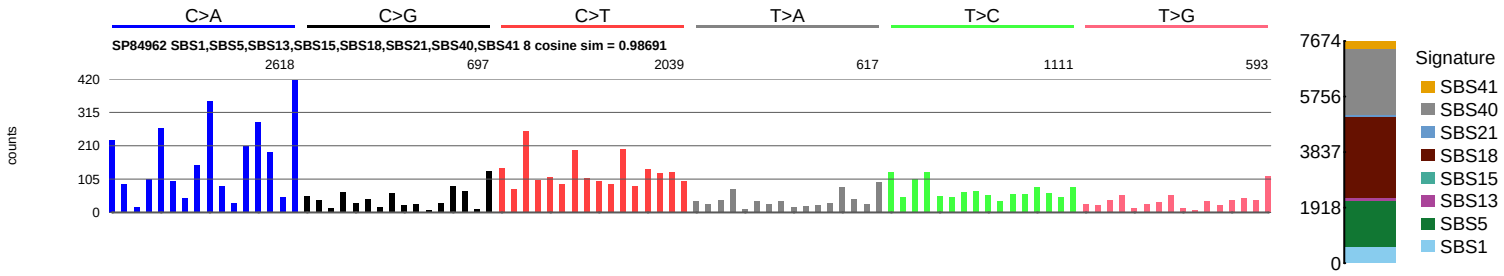
SBS5
SBS1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SP84962 SBS1,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98691
2618
697
2039
617
1111
593
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
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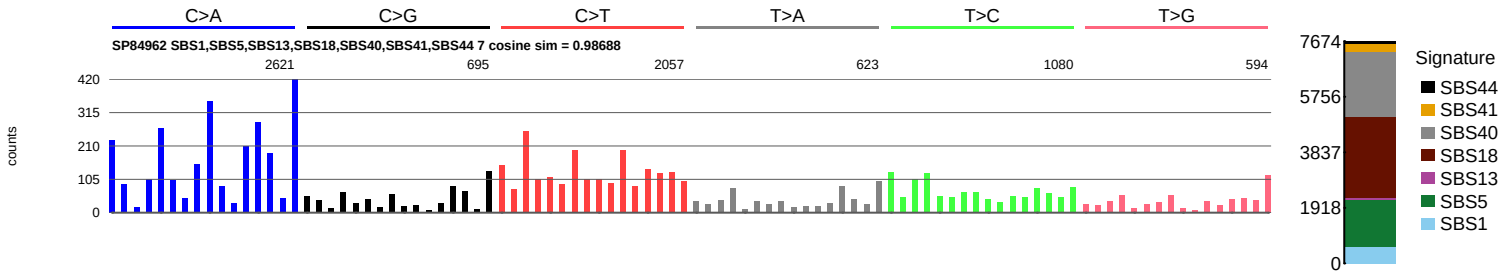
SBS5
SBS1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C>G
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SP84962 SBS1,SBS5,SBS13,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98688
2621
695
2057
623
1080
594
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
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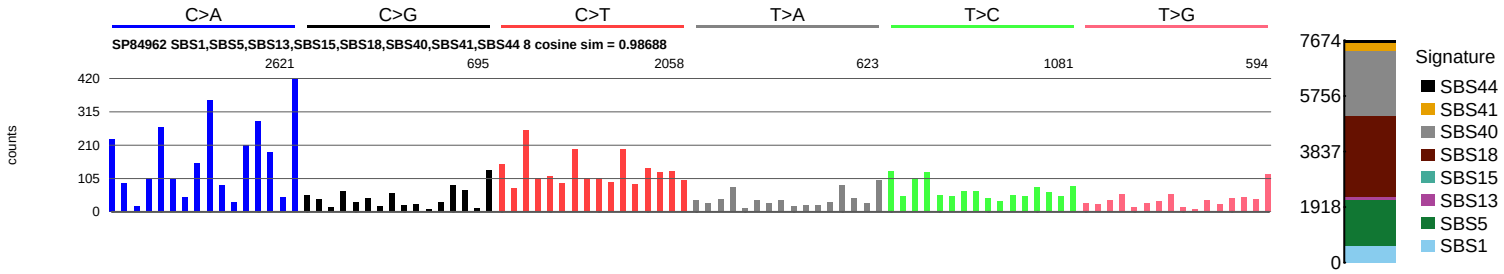
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS13,SBS15,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98688
2621
695
2058
623
1081
594
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS40
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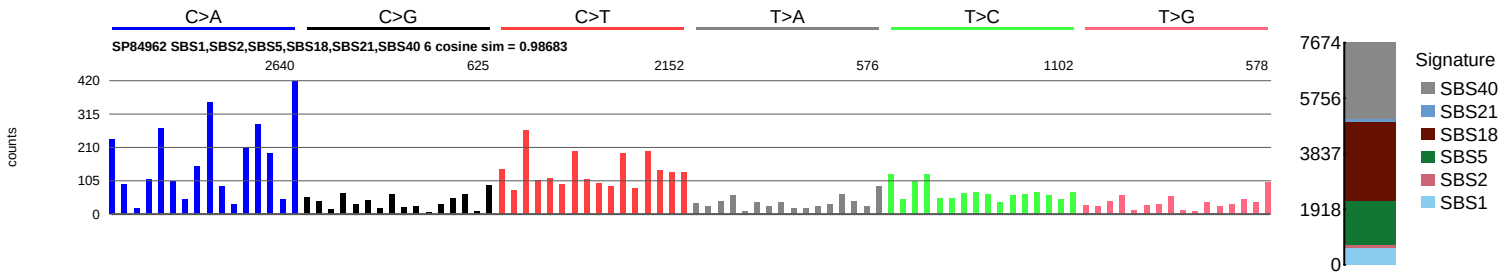
SBS5
SBS1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C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS18,SBS21,SBS40 6 cosine sim = 0.98683
2640
625
2152
576
1102
578
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
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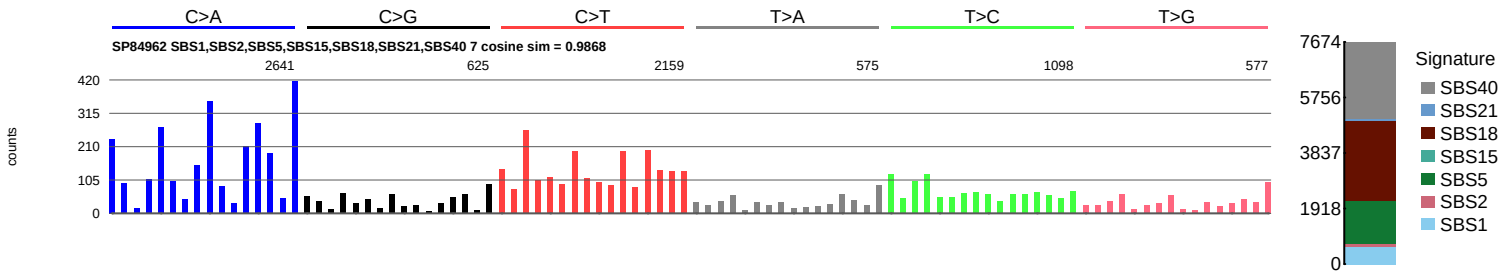
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS15,SBS18,SBS21,SBS40 7 cosine sim = 0.9868
2641
625
2159
575
1098
577
counts
420
315
210
105
0
7674
5756
3837
1918
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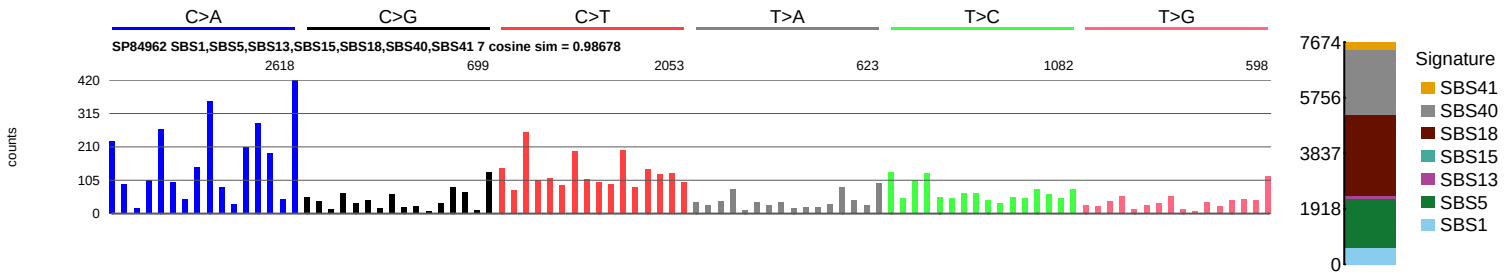
SBS1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C>G
C>T
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T>C
T>G
SP84962 SBS1,SBS5,SBS13,SBS15,SBS18,SBS40,SBS41 7 cosine sim = 0.98678
2618
699
2053
623
1082
598
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
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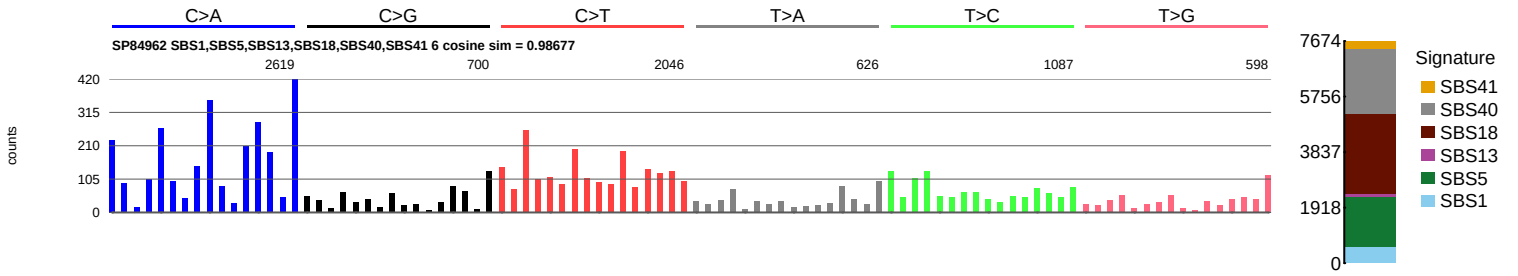
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS13,SBS18,SBS40,SBS41 6 cosine sim = 0.98677
2619
700
2046
626
1087
598
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS41
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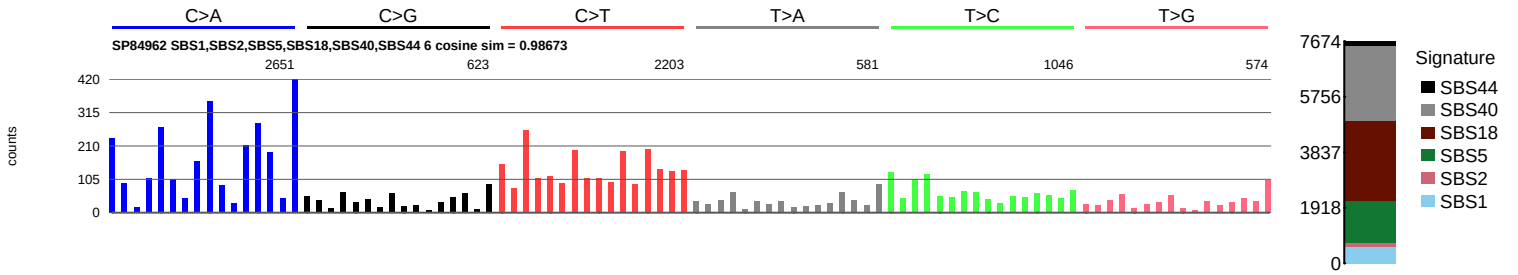
SBS5
SBS1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T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS18,SBS40,SBS44 6 cosine sim = 0.98673
2651
623
2203
581
1046
574
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS44
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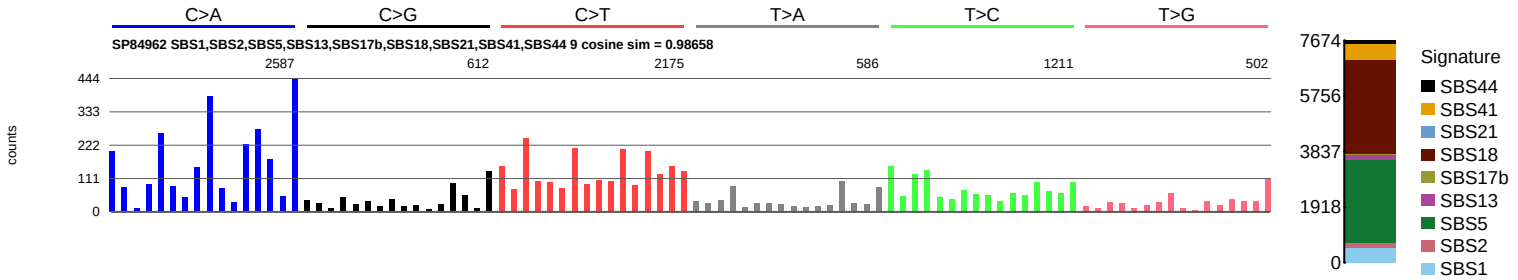
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98658
2587
612
2175
586
1211
502
counts
444
333
222
111
0
7674
5756
3837
1918
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Signature
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SBS21
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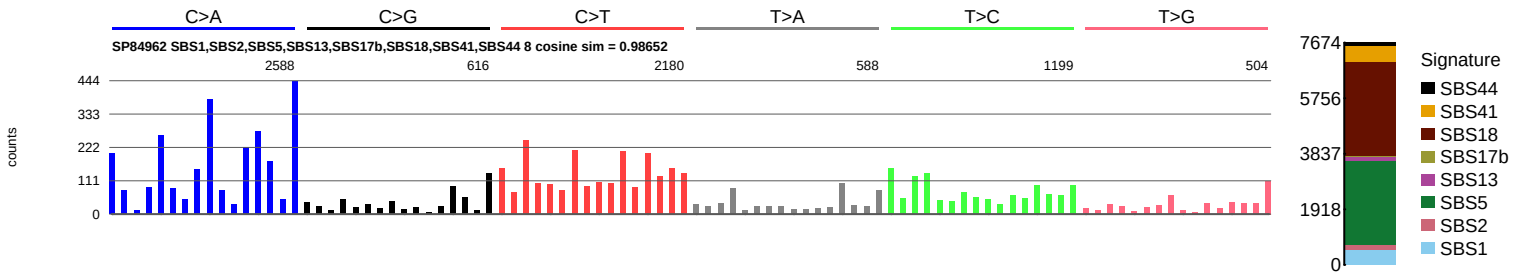
SBS2
SBS1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SP84962 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98652
2588
616
2180
588
1199
504
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS13
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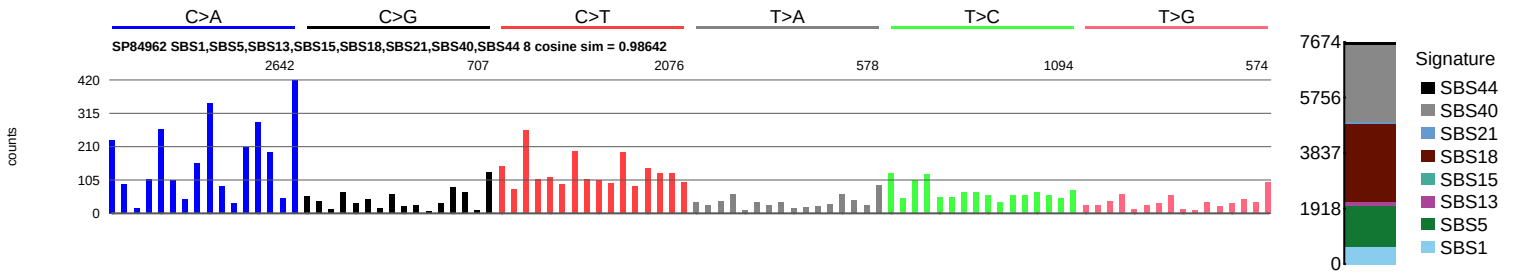
SBS2
SBS1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C>T
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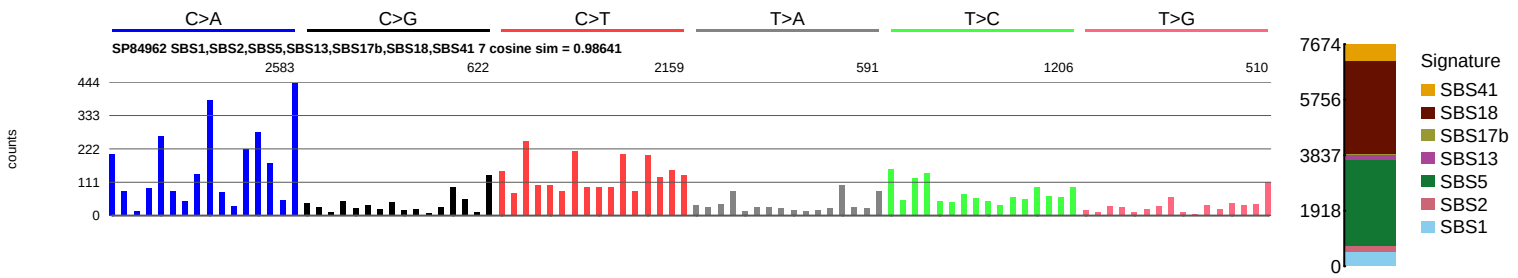
T>G
SP84962 SBS1,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98642
2642
707
2076
578
1094
574
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS15
SBS13
SBS5
SBS1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C>G
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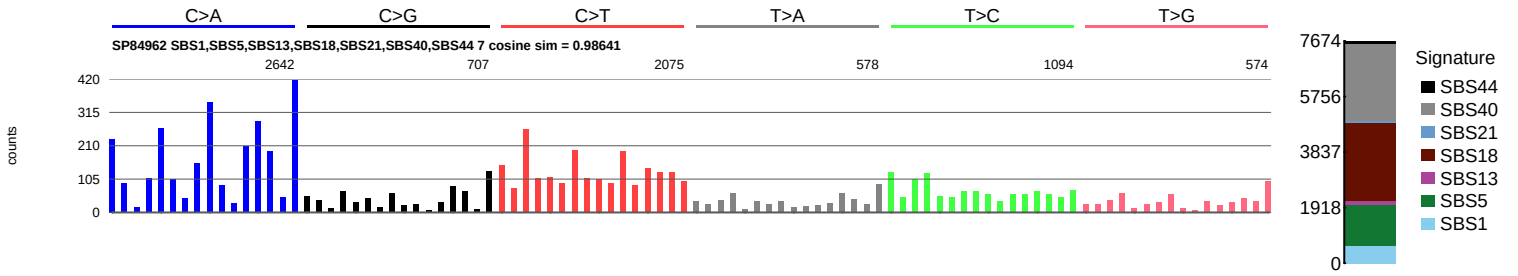
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS41 7 cosine sim = 0.98641
2583
622
2159
591
1206
510
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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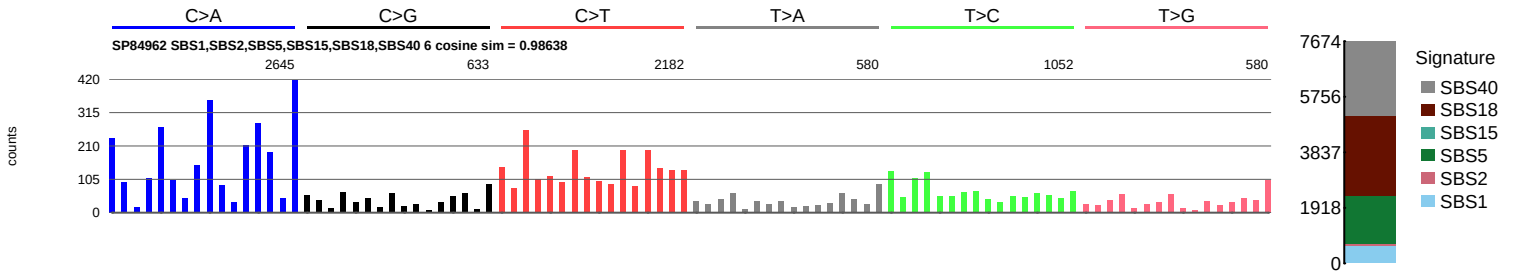
SP84962 SBS1,SBS5,SBS13,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98641
2642
707
2075
578
1094
574
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS13
SBS5
SBS1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SP84962 SBS1,SBS2,SBS5,SBS15,SBS18,SBS40 6 cosine sim = 0.98638
2645
633
2182
580
1052
580
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
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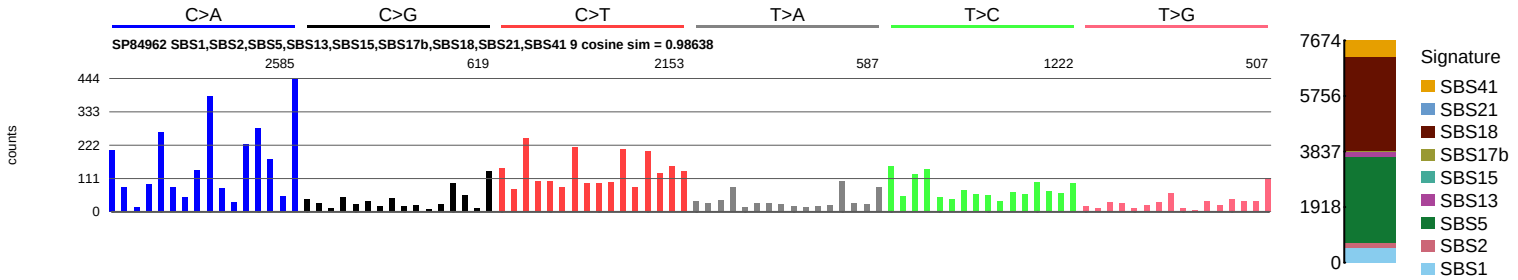
SBS2
SBS1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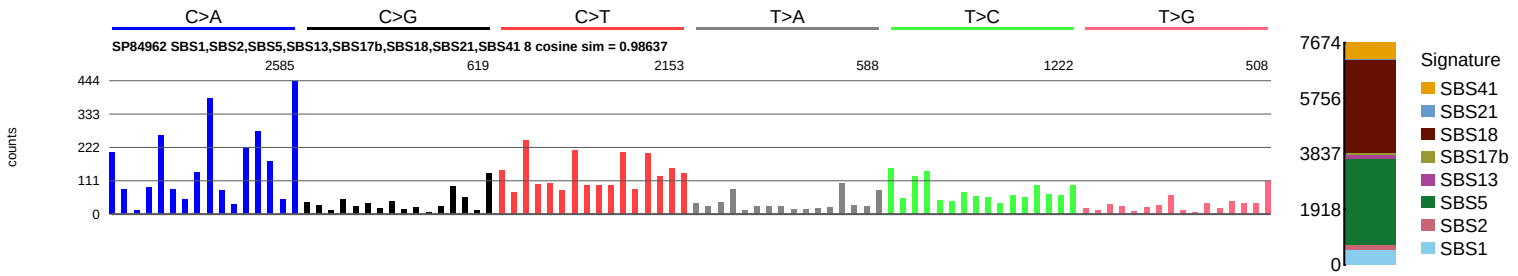
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98638
2585
619
2153
587
1222
507
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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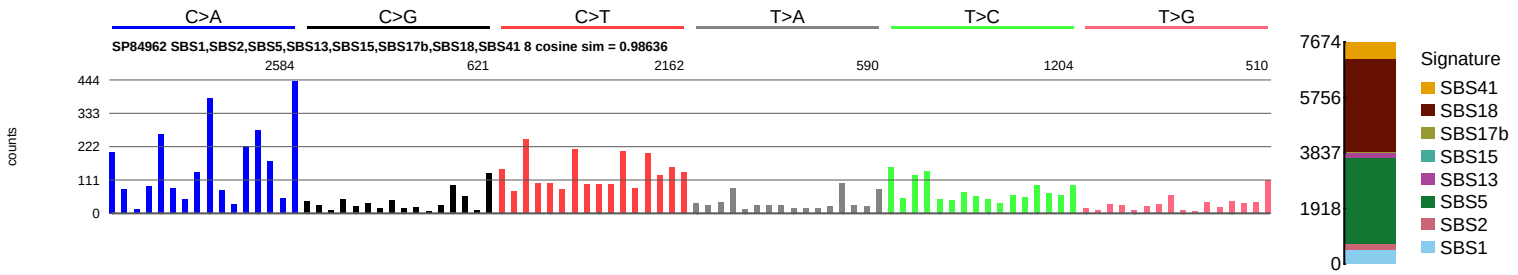
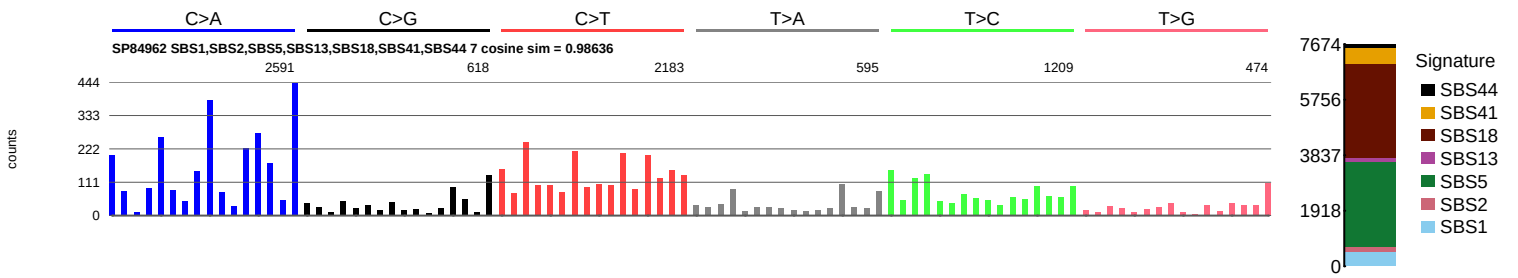
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98637
2585
619
2153
588
1222
508
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.98636
2584
621
2162
590
1204
510
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS18,SBS41,SBS44 7 cosine sim = 0.98636
2591
618
2183
595
1209
474
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
SBS44
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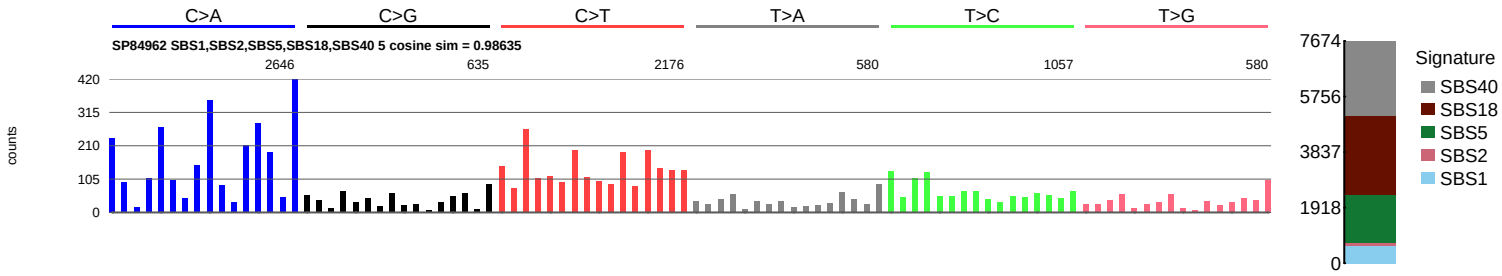
SBS2
SBS1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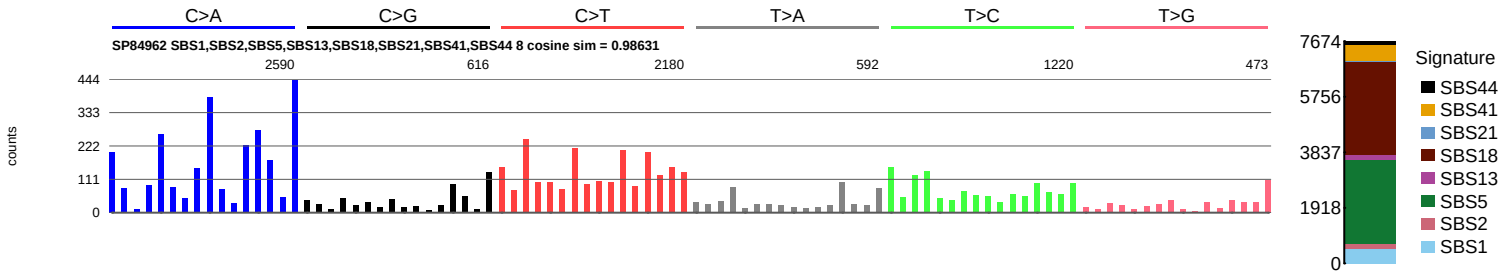
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS18,SBS40 5 cosine sim = 0.98635
2646
635
2176
580
1057
580
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS40
SBS18
SBS5
SBS2
SBS1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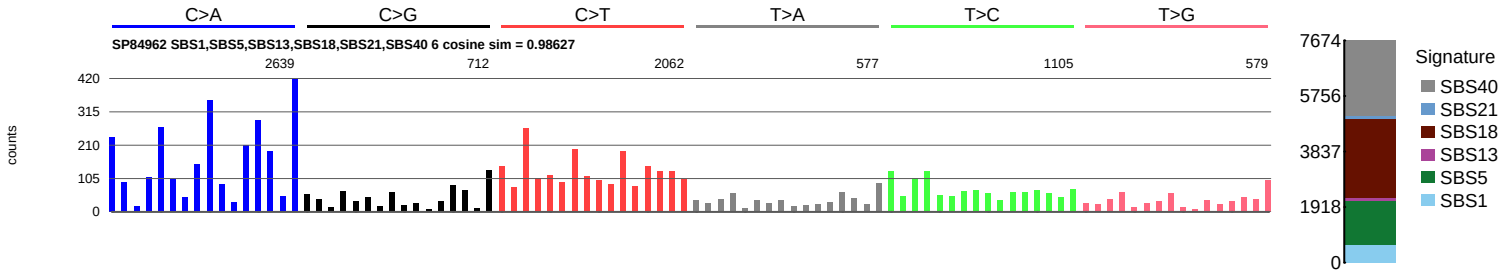
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98631
2590
616
2180
592
1220
473
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS13
SBS5
SBS2
SBS1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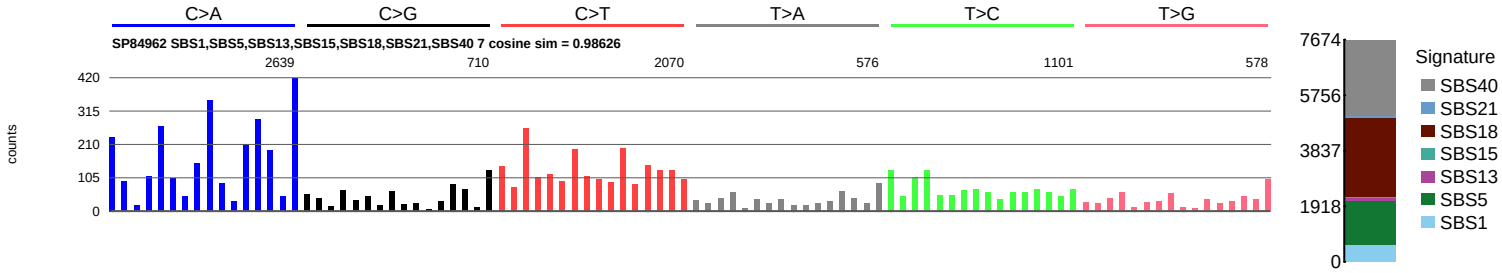
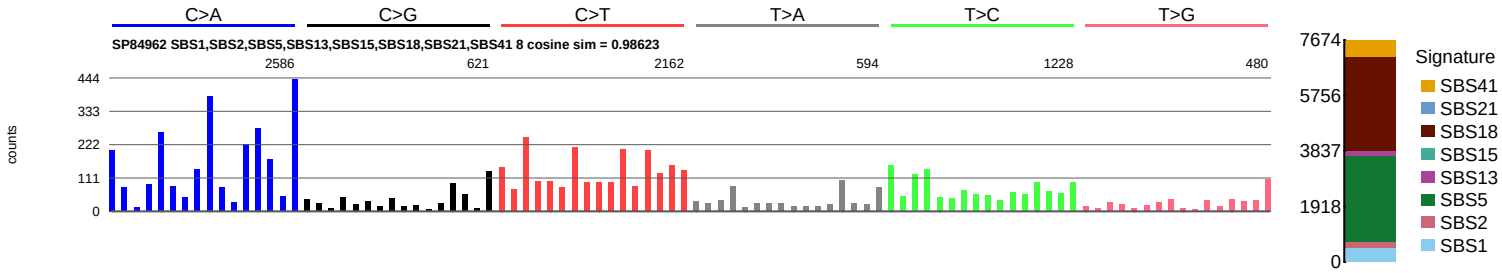
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS13,SBS18,SBS21,SBS40 6 cosine sim = 0.98627
2639
712
2062
577
1105
579
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS40
SBS21
SBS18
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40 7 cosine sim = 0.98626
2639
710
2070
576
1101
578
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS40
SBS21
SBS18
SBS15
SBS13
SBS5
SBS1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C>G
C>T
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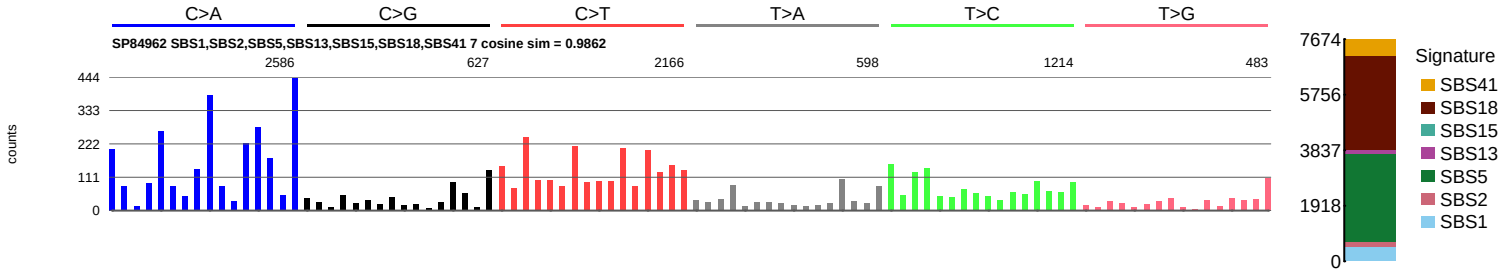
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS41 8 cosine sim = 0.98623
2586
621
2162
594
1228
480
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
SBS41
SBS21
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
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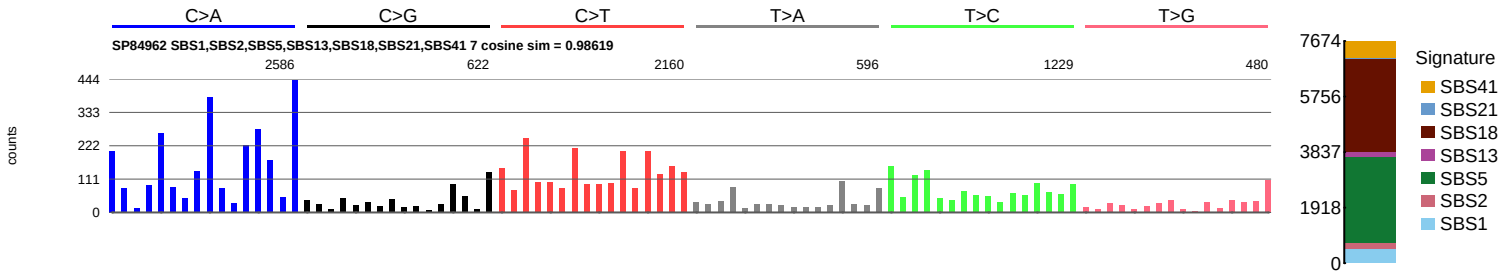
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS41 7 cosine sim = 0.9862
2586
627
2166
598
1214
483
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
SBS41
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
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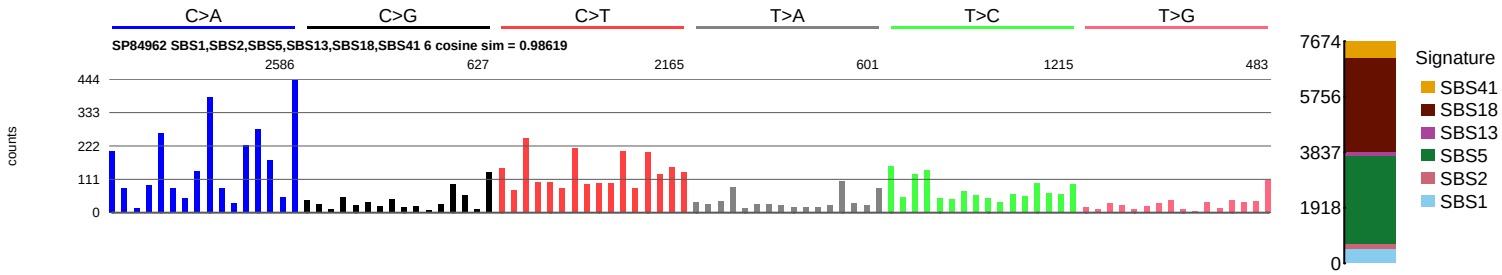
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS41 7 cosine sim = 0.98619
2586
622
2160
596
1229
480
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
SBS41
SBS21
SBS18
SBS13
SBS5
SBS2
SBS1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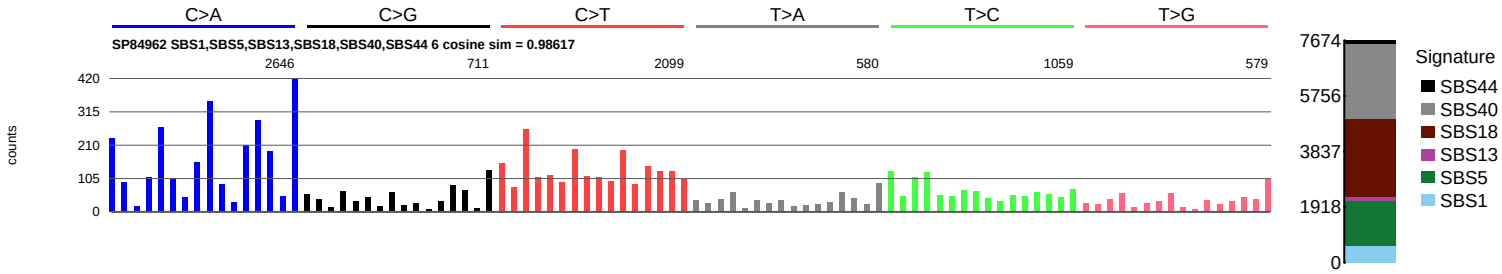
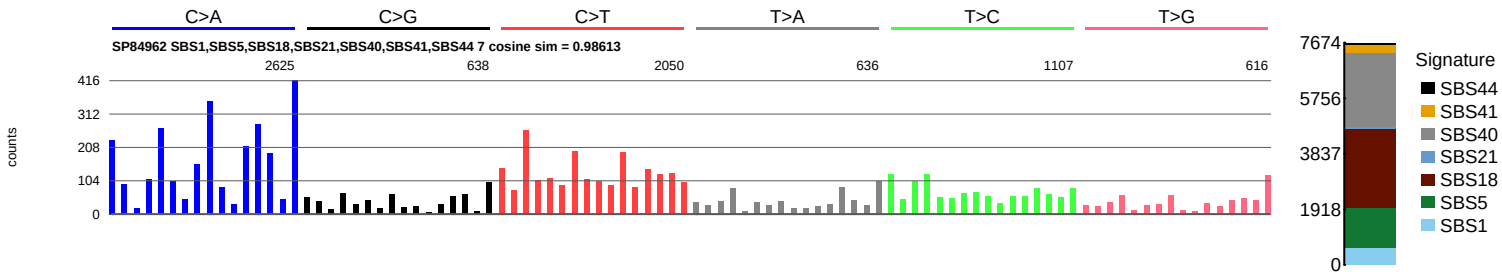
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS18,SBS41 6 cosine sim = 0.98619
2586
627
2165
601
1215
483
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
SBS41
SBS18
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS13,SBS18,SBS40,SBS44 6 cosine sim = 0.98617
2646
711
2099
580
1059
579
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS18
SBS13
SBS5
SBS1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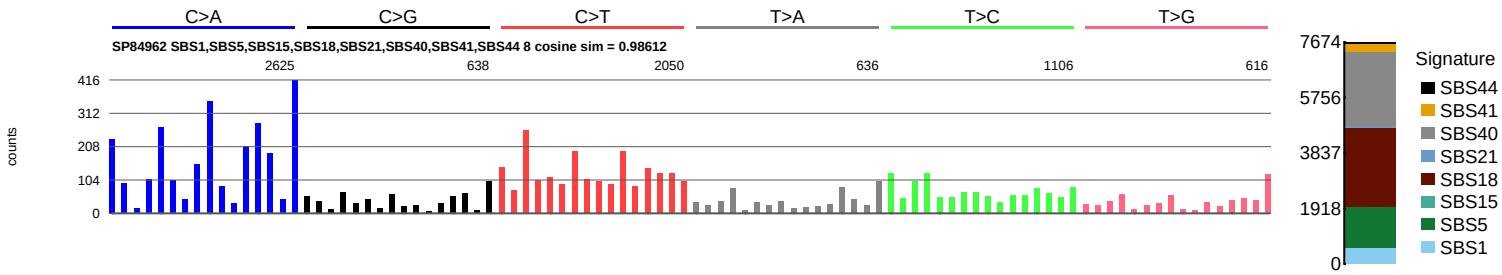
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS18,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98613
2625
638
2050
636
1107
616
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS5
SBS1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C>G
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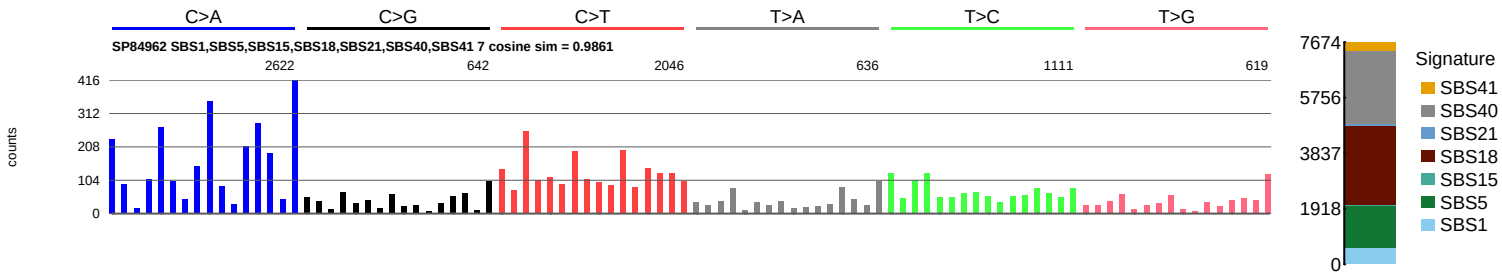
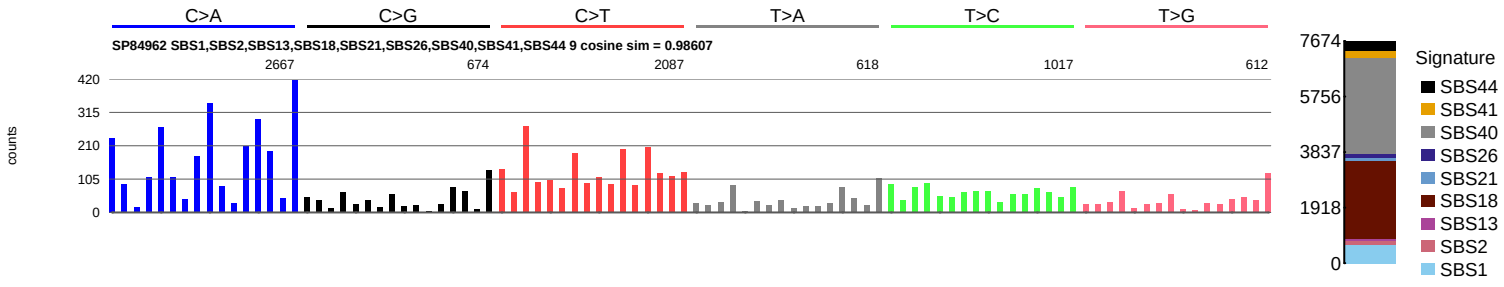
T>C
T>G
SP84962 SBS1,SBS5,SBS15,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98612
2625
638
2050
636
1106
616
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS15,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.9861
2622
642
2046
636
1111
619
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS13,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98607
2667
674
2087
618
1017
612
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
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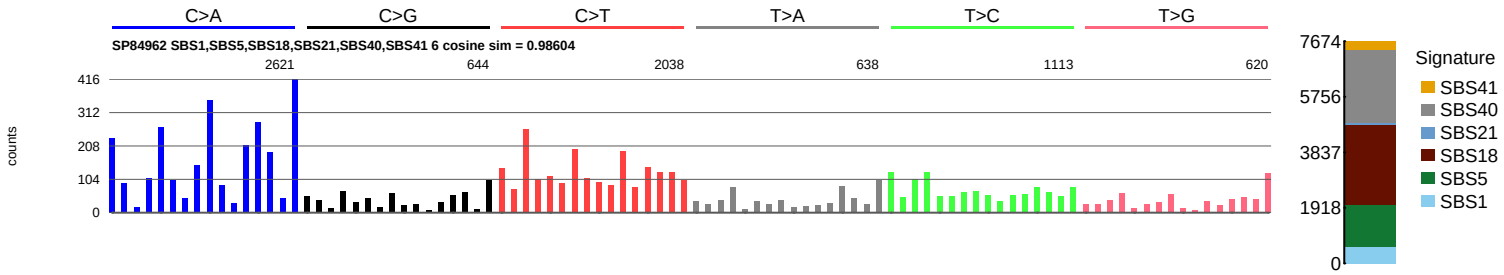
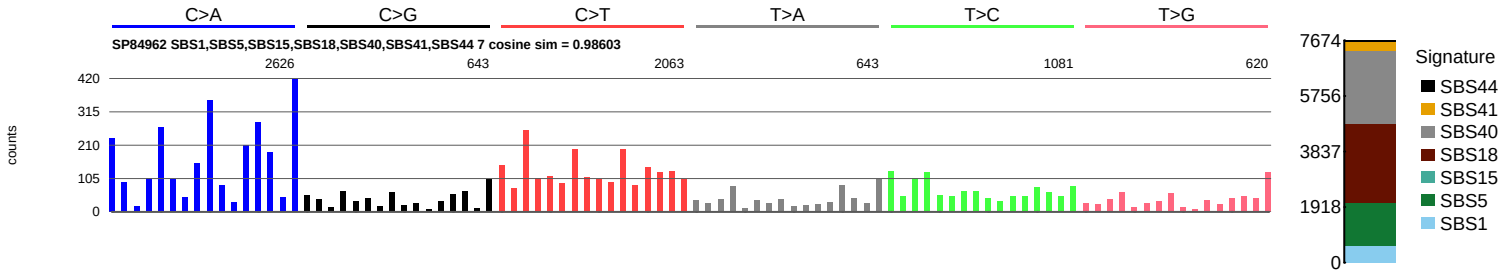
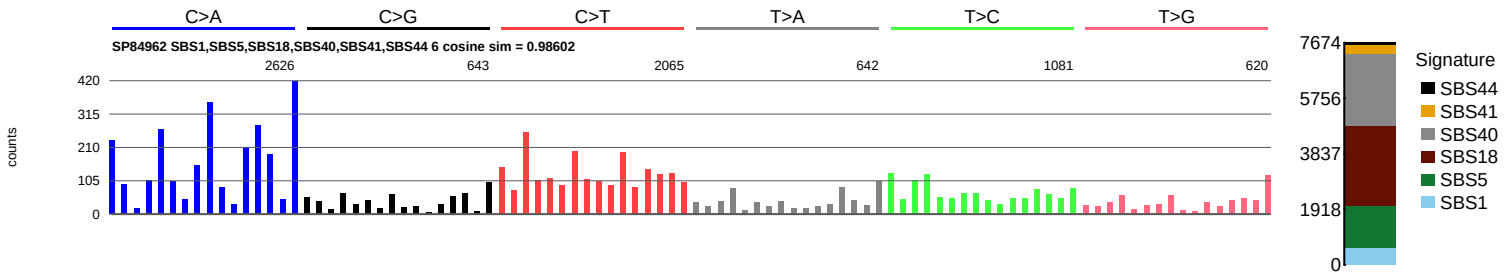
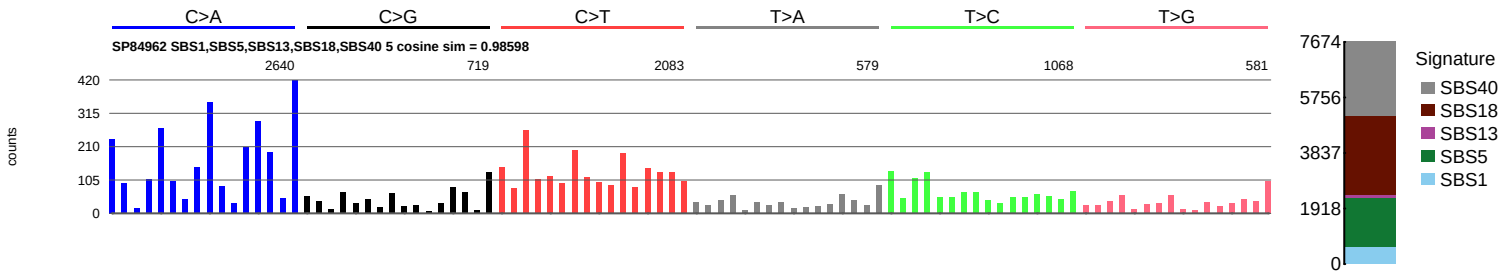
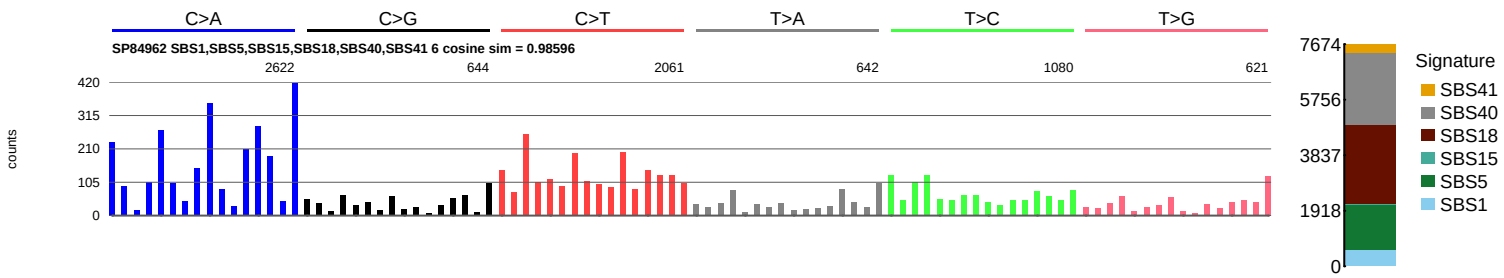
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS18,SBS21,SBS40,SBS41 6 cosine sim = 0.98604
2621
644
2038
638
1113
620
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS15,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98603
2626
643
2063
643
1081
620
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS18,SBS40,SBS41,SBS44 6 cosine sim = 0.98602
2626
643
2065
642
1081
620
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS13,SBS18,SBS40 5 cosine sim = 0.98598
2640
719
2083
579
1068
581
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS40
SBS18
SBS13
SBS5
SBS1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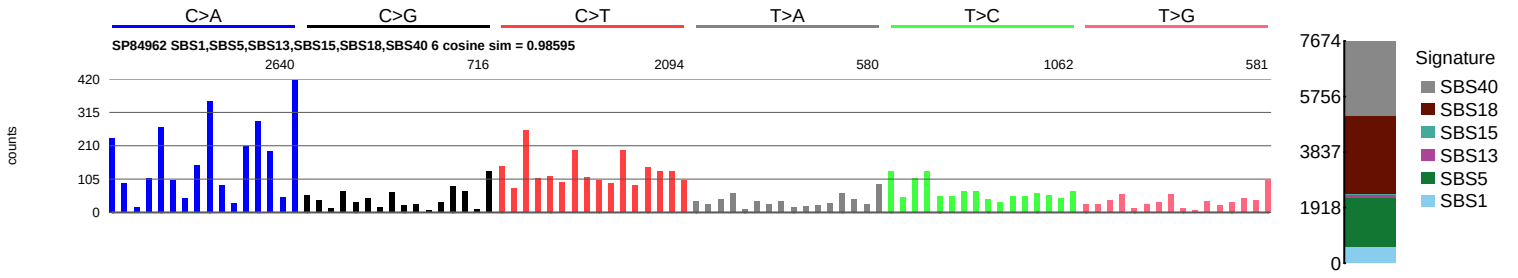
SP84962 SBS1,SBS5,SBS15,SBS18,SBS40,SBS41 6 cosine sim = 0.98596
2622
644
2061
642
1080
621
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS5
SBS1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SP84962 SBS1,SBS5,SBS13,SBS15,SBS18,SBS40 6 cosine sim = 0.98595
2640
716
2094
580
1062
581
counts
420
315
210
105
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5756
3837
1918
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Signature
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SBS18
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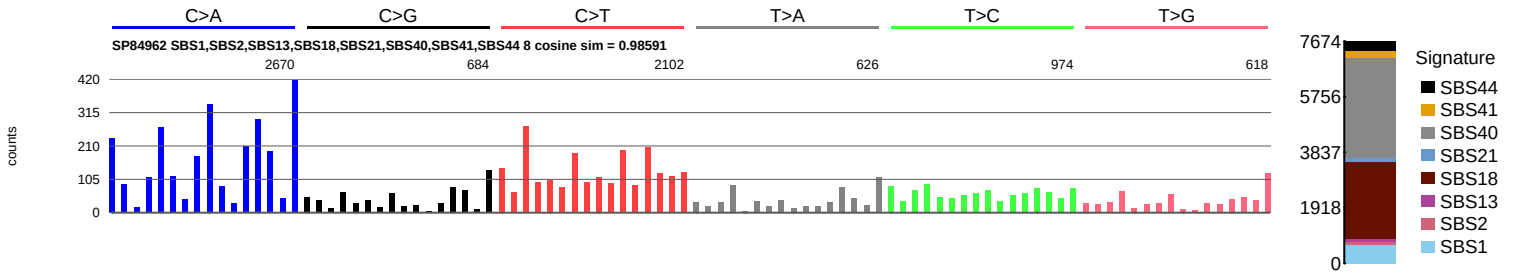
SBS13
SBS5
SBS1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C>T
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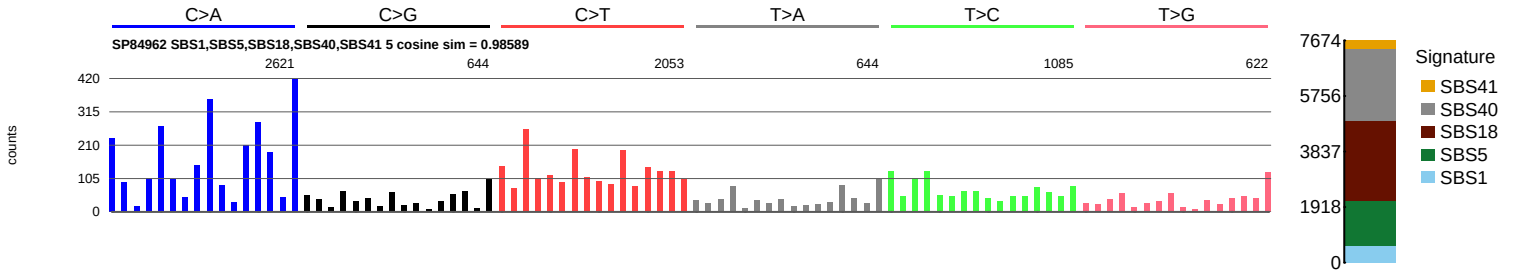
SP84962 SBS1,SBS2,SBS13,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98591
2670
684
2102
626
974
618
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS13
SBS2
SBS1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SP84962 SBS1,SBS5,SBS18,SBS40,SBS41 5 cosine sim = 0.98589
2621
644
2053
644
1085
622
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
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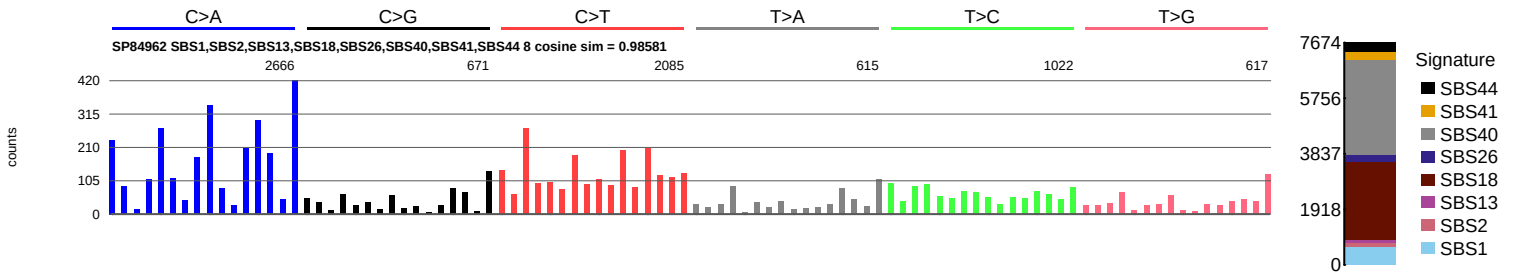
SBS5
SBS1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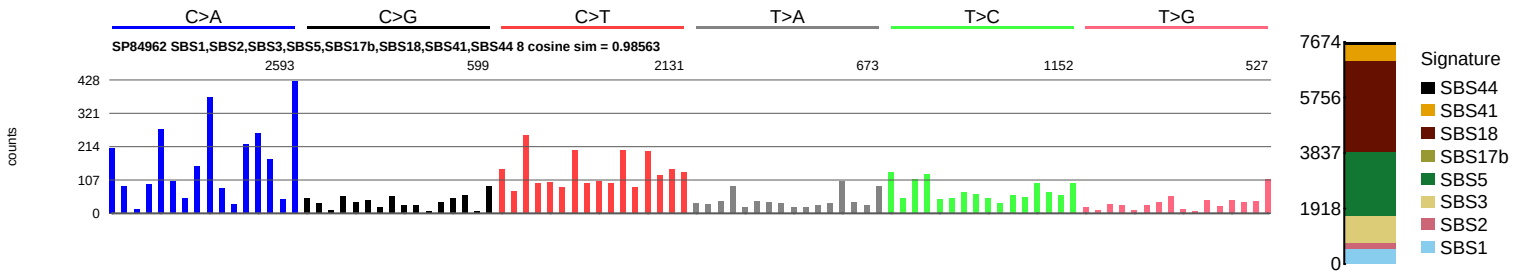
SP84962 SBS1,SBS2,SBS13,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98581
2666
671
2085
615
1022
617
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS13
SBS2
SBS1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C>G
C>T
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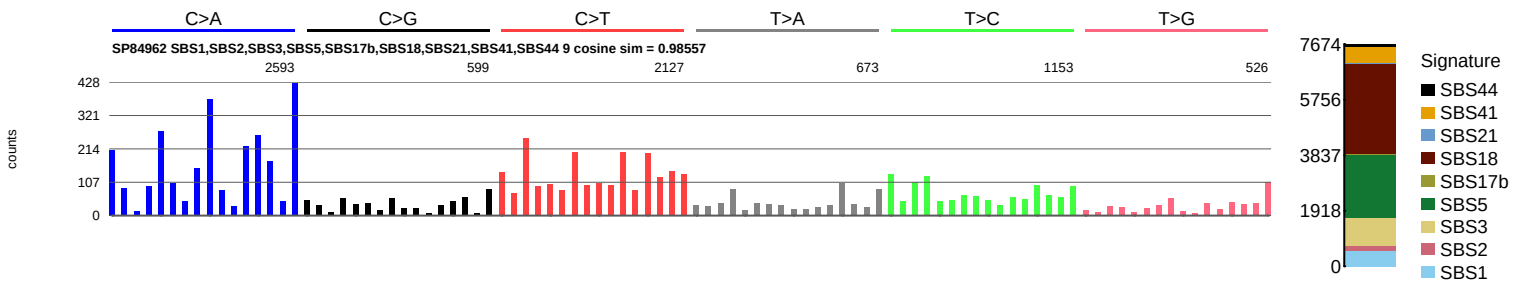
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98563
2593
599
2131
673
1152
527
counts
428
321
214
107
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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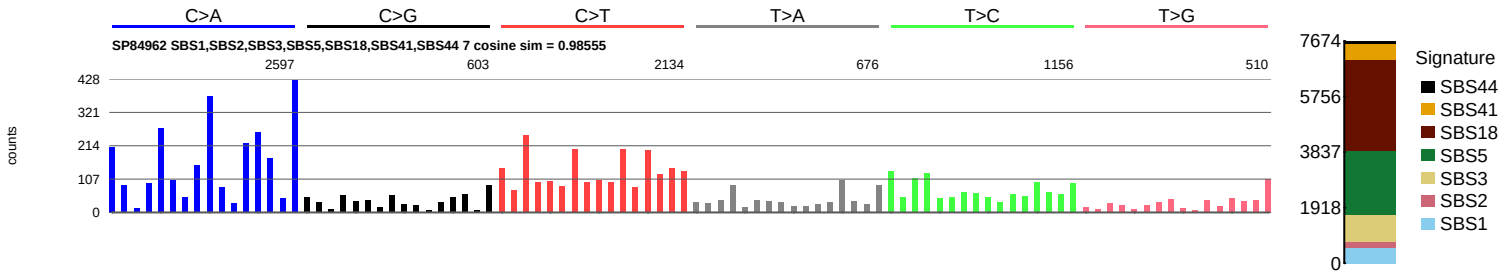
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98557
2593
599
2127
673
1153
526
counts
428
321
214
107
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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C>G
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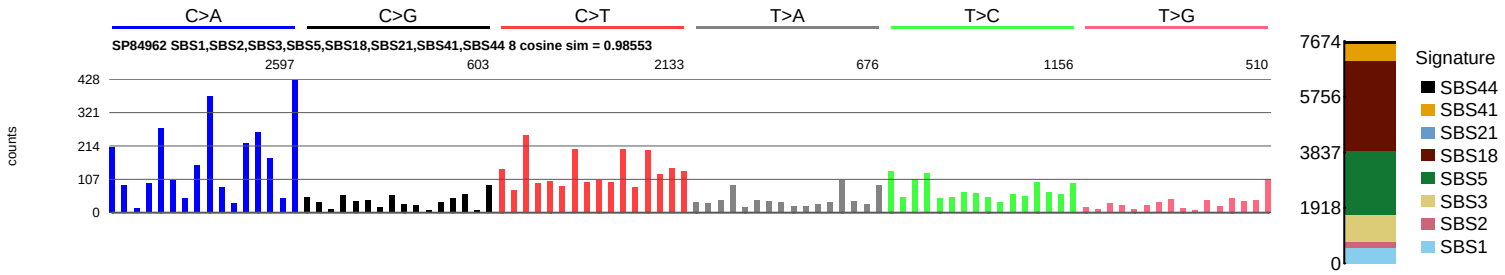
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS18,SBS41,SBS44 7 cosine sim = 0.98555
2597
603
2134
676
1156
510
counts
428
321
214
107
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS18
SBS5
SBS3
SBS2
SBS1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C>G
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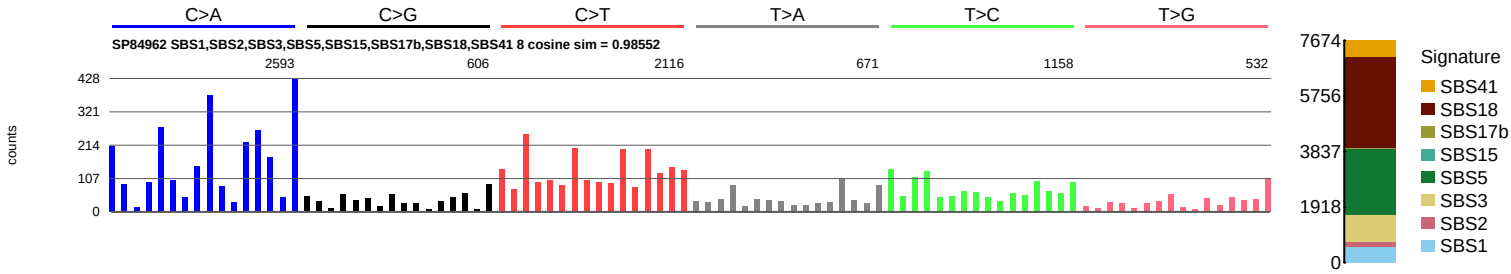
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98553
2597
603
2133
676
1156
510
counts
428
321
214
107
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS5
SBS3
SBS2
SBS1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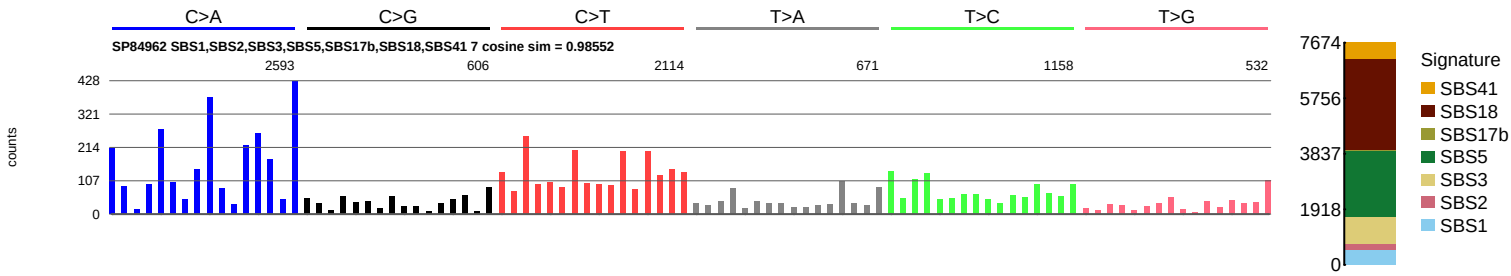
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.98552
2593
606
2116
671
1158
532
counts
428
321
214
107
0
7674
5756
3837
1918
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
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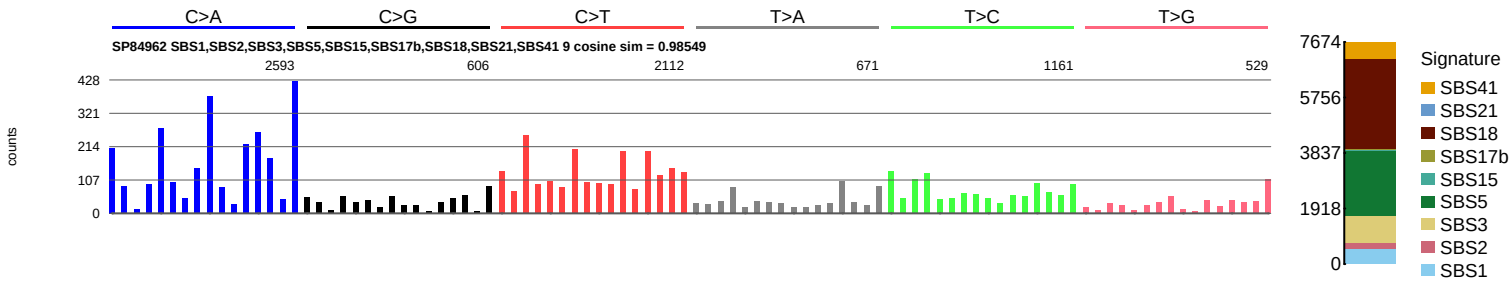
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS41 7 cosine sim = 0.98552
2593
606
2114
671
1158
532
counts
428
321
214
107
0
7674
5756
3837
1918
0
Signature
SBS41
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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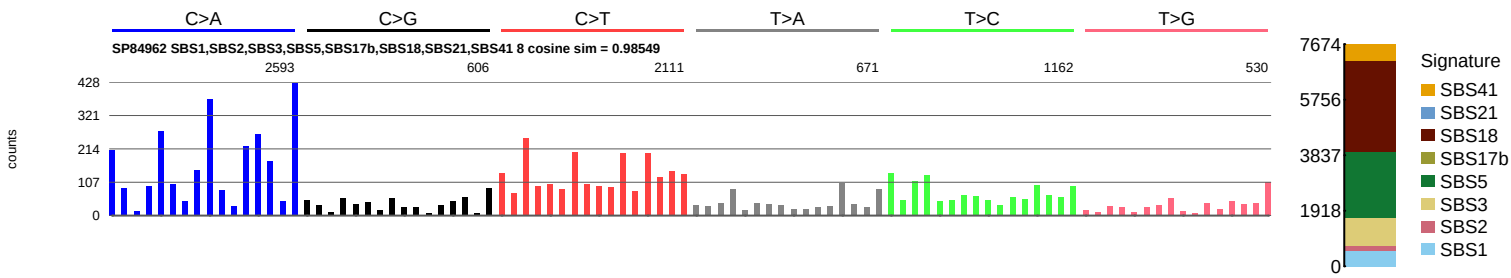
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98549
2593
606
2112
671
1161
529
counts
428
321
214
107
0
7674
5756
3837
1918
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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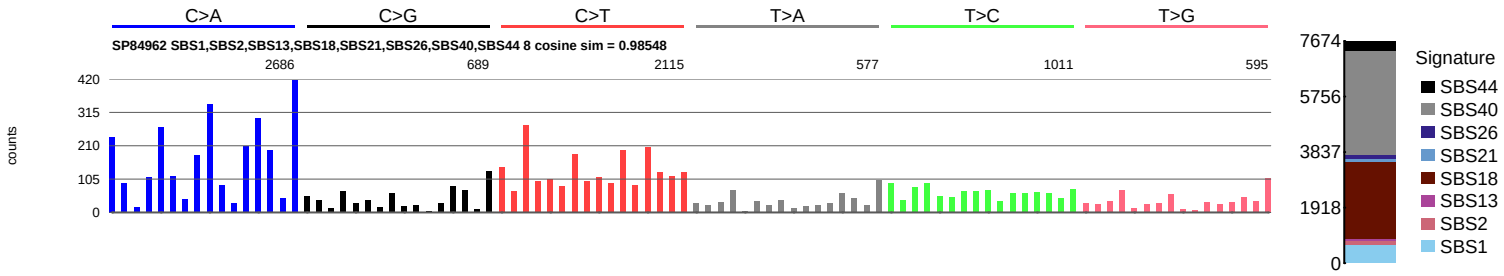
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98549
2593
606
2111
671
1162
530
counts
428
321
214
107
0
7674
5756
3837
1918
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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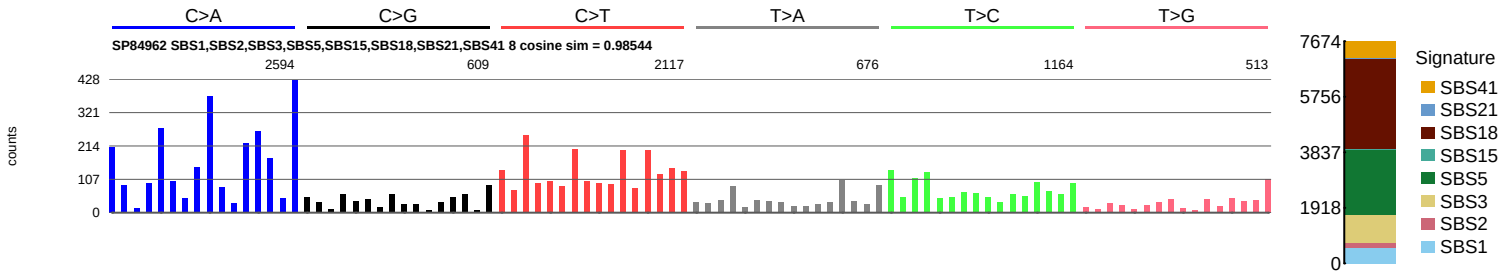
SP84962 SBS1,SBS2,SBS13,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98548
2686
689
2115
577
1011
595
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS13
SBS2
SBS1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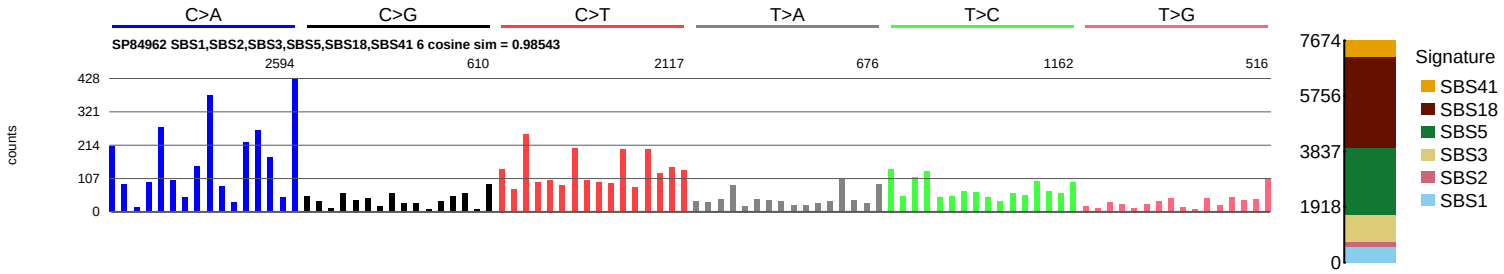
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS21,SBS41 8 cosine sim = 0.98544
2594
609
2117
676
1164
513
counts
428
321
214
107
0
7674
5756
3837
1918
0
Signature
SBS41
SBS21
SBS18
SBS15
SBS5
SBS3
SBS2
SBS1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T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS18,SBS41 6 cosine sim = 0.98543
2594
610
2117
676
1162
516
counts
428
321
214
107
0
7674
5756
3837
1918
0
Signature
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SBS18
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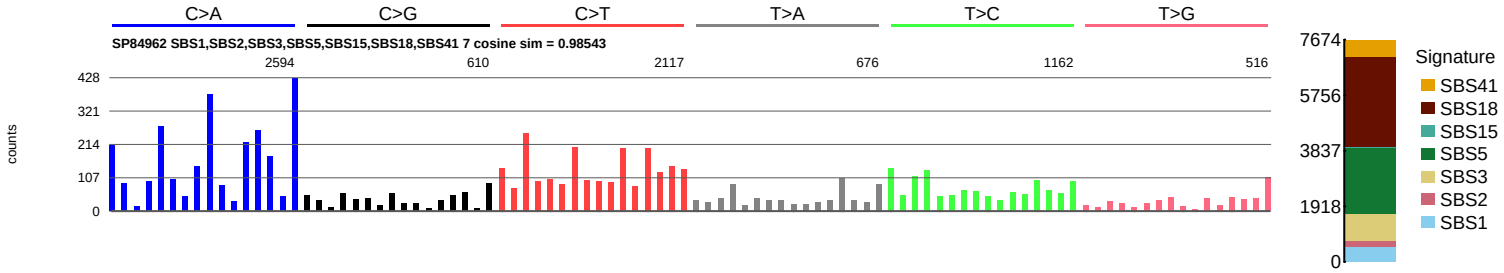
SBS3
SBS2
SBS1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T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS41 7 cosine sim = 0.98543
2594
610
2117
676
1162
516
counts
428
321
214
107
0
7674
5756
3837
1918
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Signature
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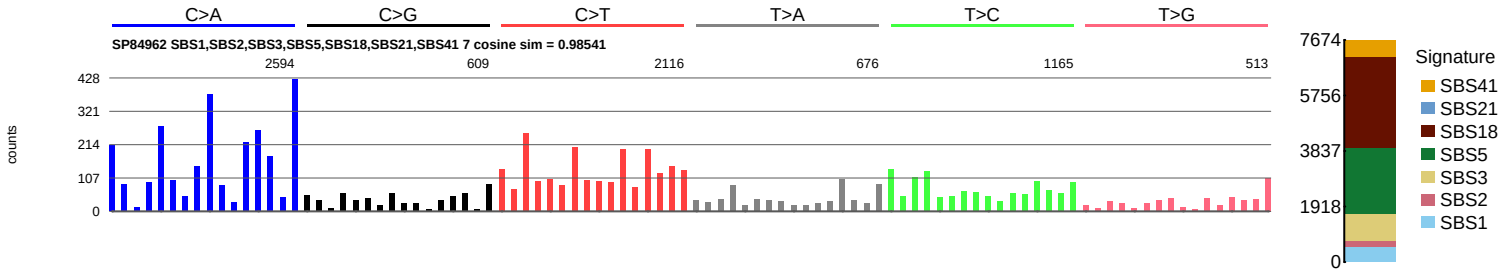
SBS2
SBS1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C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS18,SBS21,SBS41 7 cosine sim = 0.98541
2594
609
2116
676
1165
513
counts
428
321
214
107
0
7674
5756
3837
1918
0
Signature
SBS41
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SBS18
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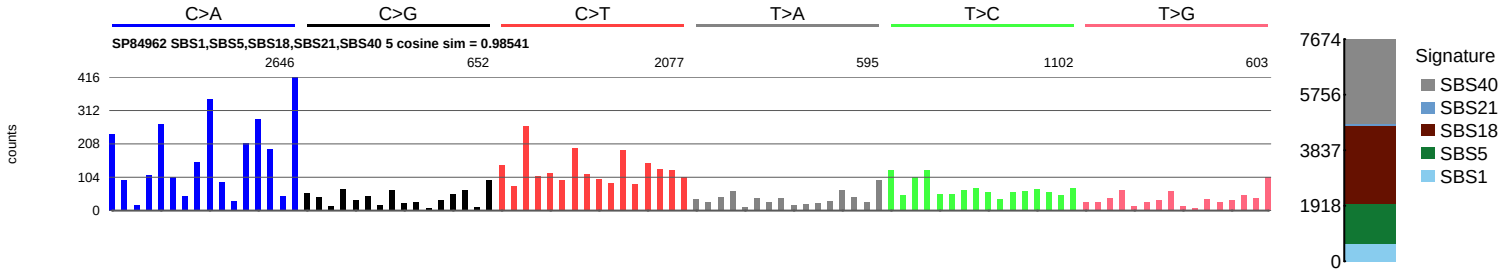
SBS2
SBS1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C>G
C>T
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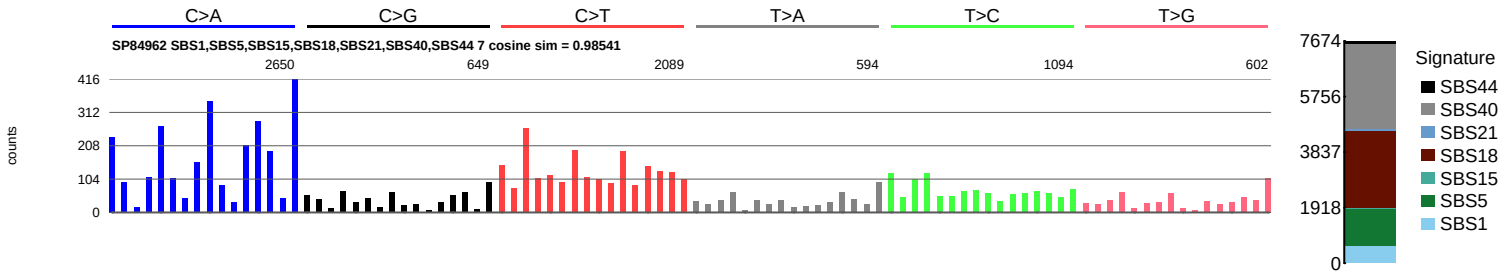
T>C
T>G
SP84962 SBS1,SBS5,SBS18,SBS21,SBS40 5 cosine sim = 0.98541
2646
652
2077
595
1102
603
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS40
SBS21
SBS18
SBS5
SBS1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C>G
C>T
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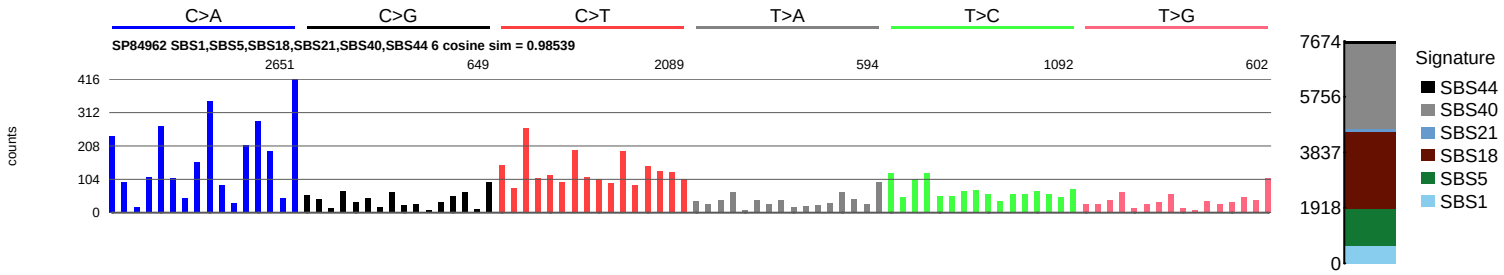
T>C
T>G
SP84962 SBS1,SBS5,SBS15,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98541
2650
649
2089
594
1094
602
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS15
SBS5
SBS1


C>A
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C>T
T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS18,SBS21,SBS40,SBS44 6 cosine sim = 0.98539
2651
649
2089
594
1092
602
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
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SBS40
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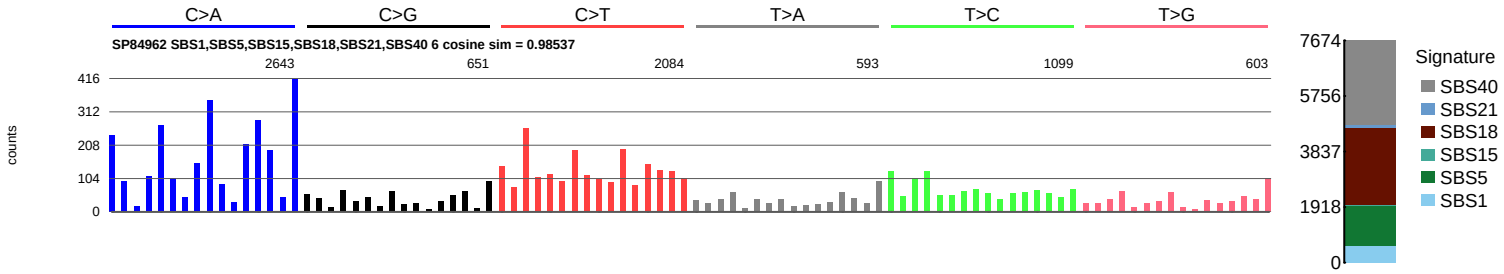
SBS5
SBS1


C>A
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C>T
T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS15,SBS18,SBS21,SBS40 6 cosine sim = 0.98537
2643
651
2084
593
1099
603
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS40
SBS21
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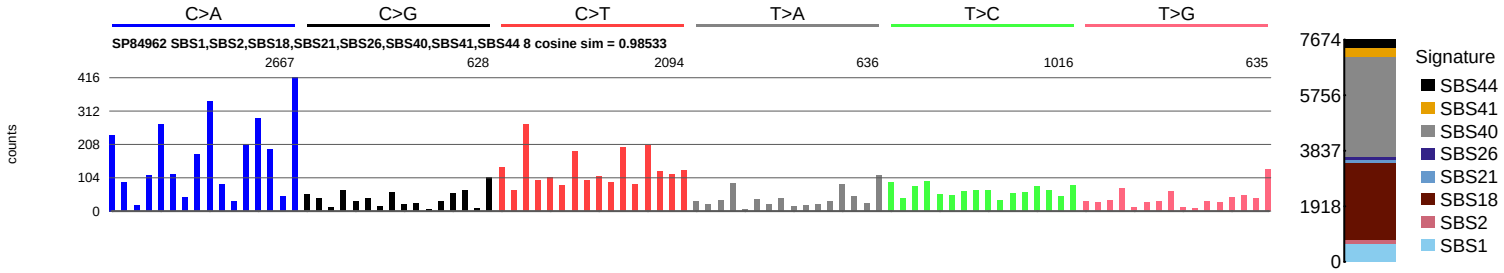
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98533
2667
628
2094
636
1016
635
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
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SBS40
SBS26
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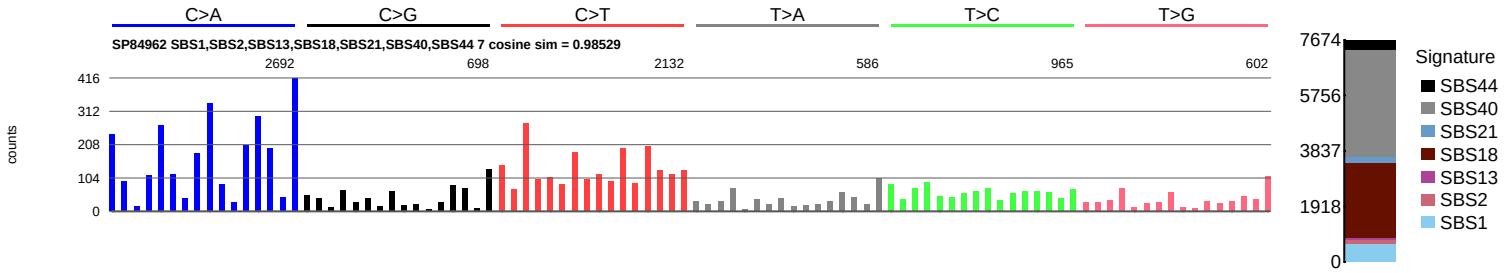
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS13,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98529
2692
698
2132
586
965
602
counts
416
312
208
104
0
7674
5756
3837
1918
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Signature
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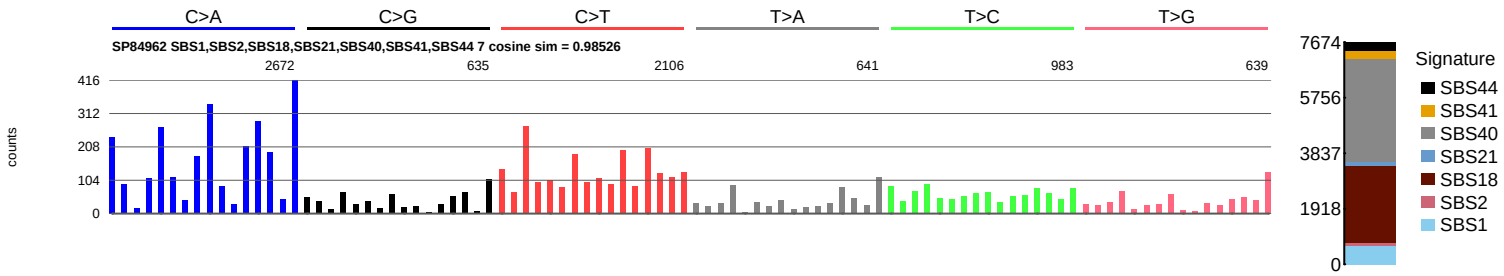
C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS18,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98526
2672
635
2106
641
983
639
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
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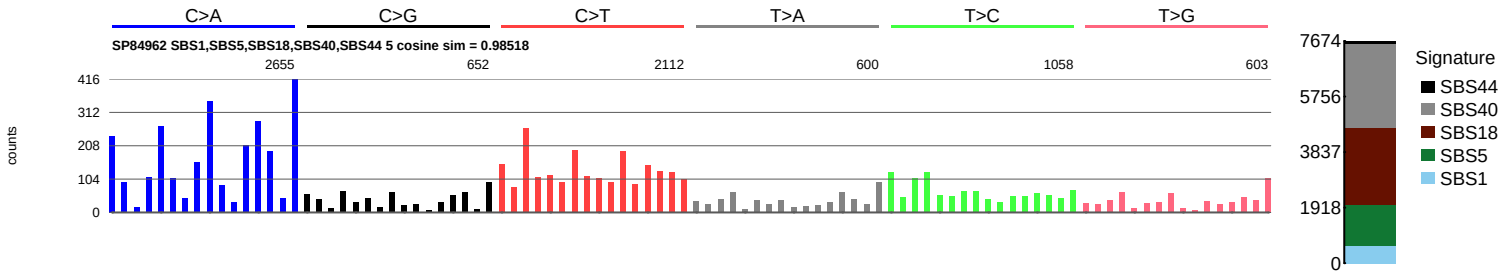
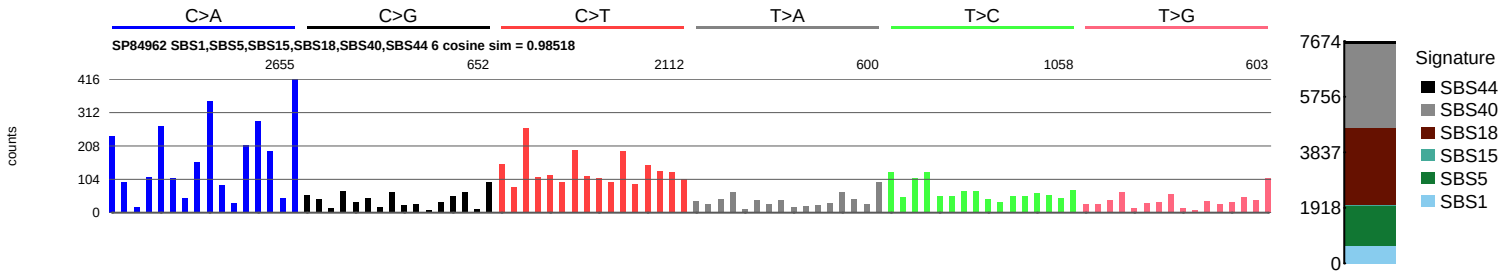
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS18,SBS40,SBS44 5 cosine sim = 0.98518
2655
652
2112
600
1058
603
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS18
SBS5
SBS1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C>G
C>T
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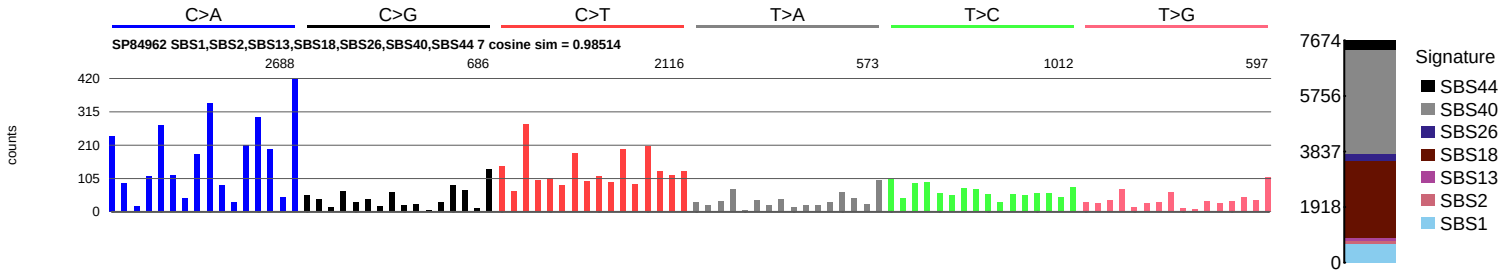
T>C
T>G
SP84962 SBS1,SBS5,SBS15,SBS18,SBS40,SBS44 6 cosine sim = 0.98518
2655
652
2112
600
1058
603
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS18
SBS15
SBS5
SBS1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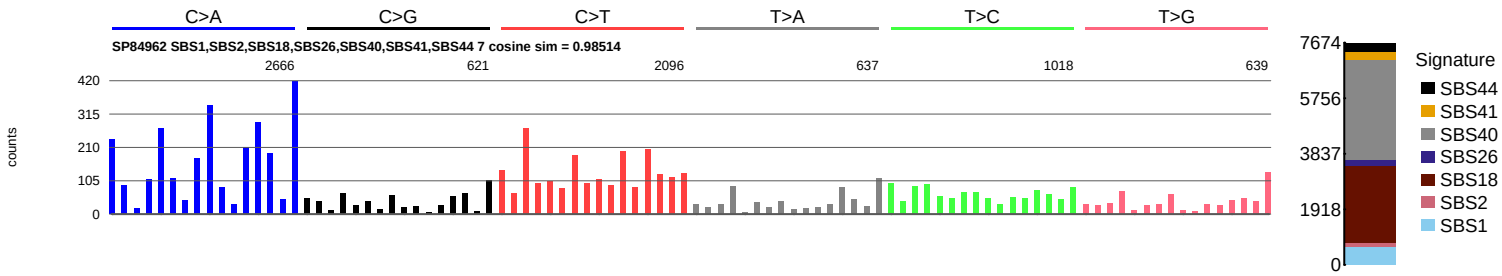
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS13,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98514
2688
686
2116
573
1012
597
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS13
SBS2
SBS1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C>G
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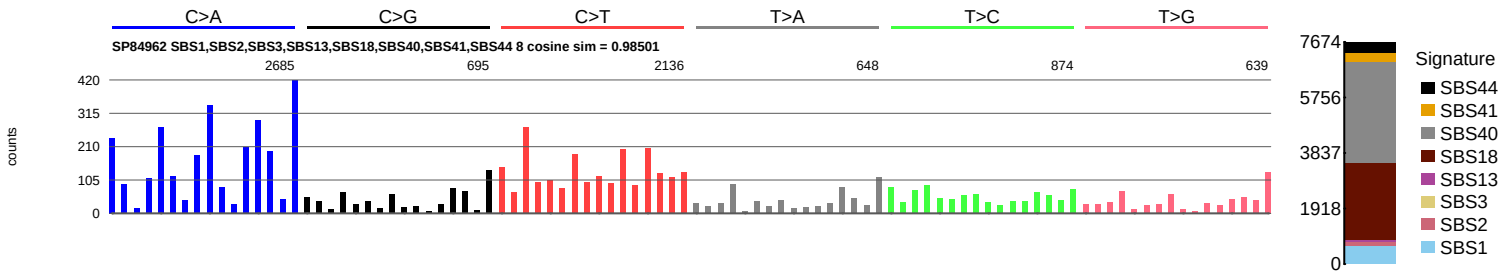
T>C
T>G
SP84962 SBS1,SBS2,SBS18,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98514
2666
621
2096
637
1018
639
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS2
SBS1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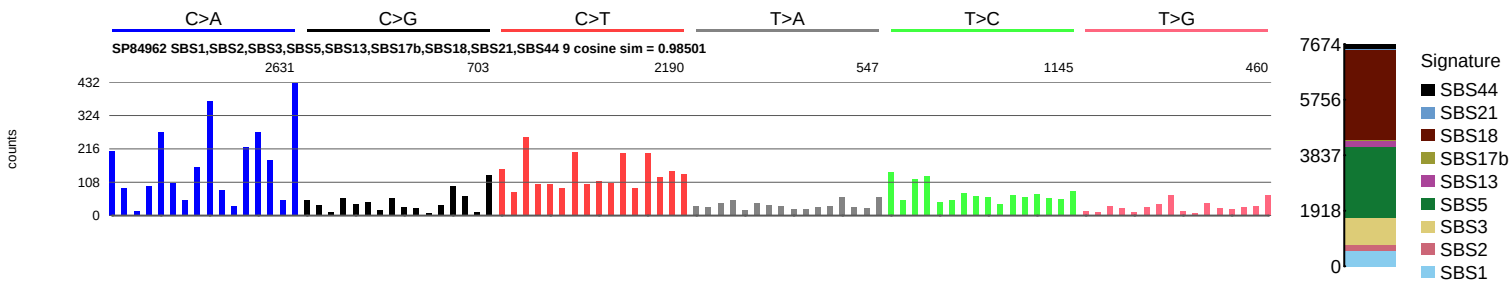
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS13,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98501
2685
695
2136
648
874
639
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS13
SBS3
SBS2
SBS1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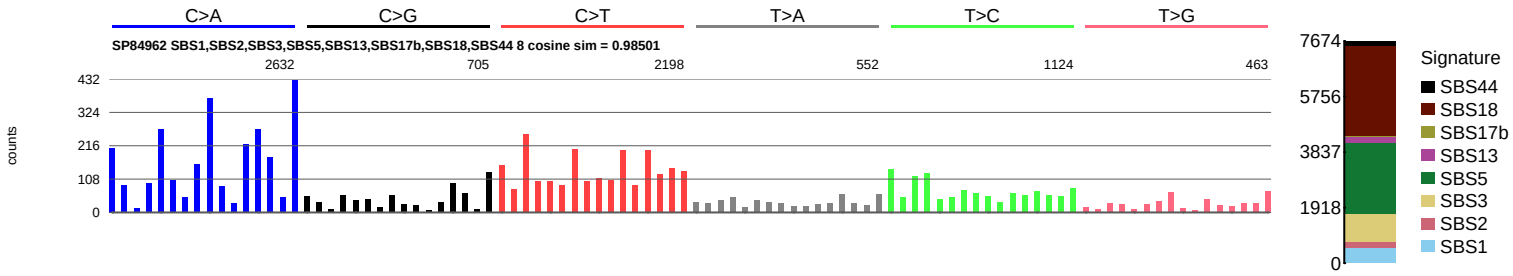
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98501
2631
703
2190
547
1145
460
counts
432
324
216
108
0
7674
5756
3837
1918
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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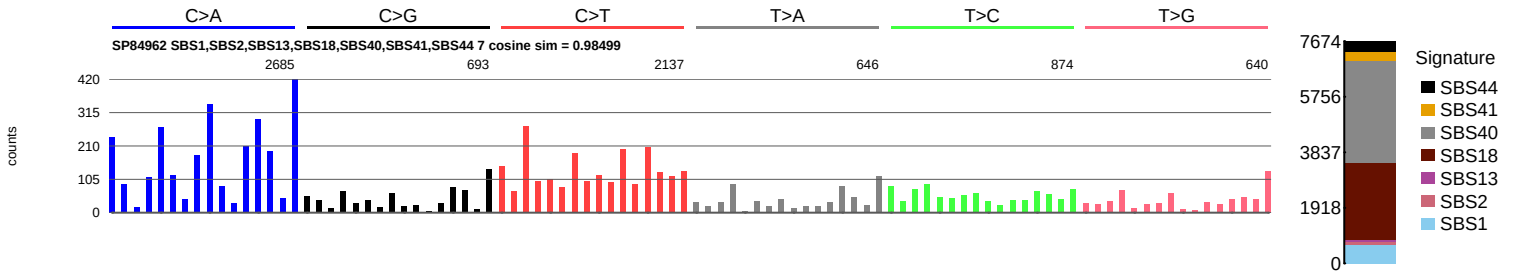
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS44 8 cosine sim = 0.98501
2632
705
2198
552
1124
463
counts
432
324
216
108
0
7674
5756
3837
1918
0
Signature
SBS44
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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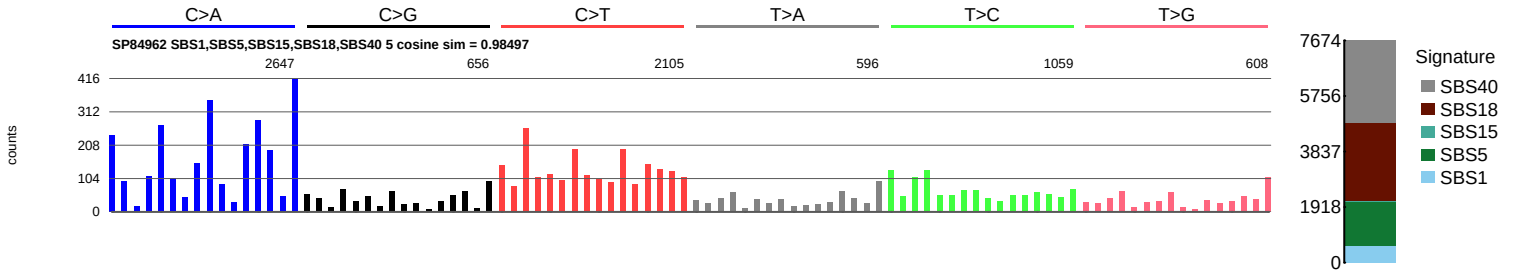
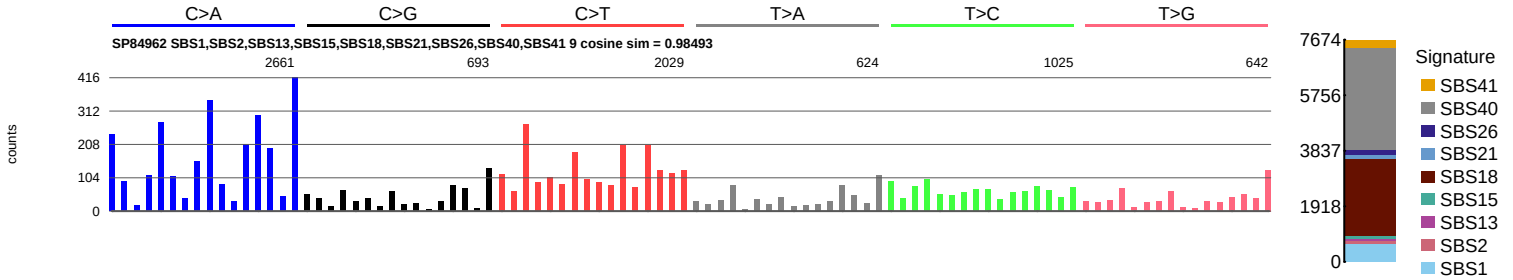
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS13,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98499
2685
693
2137
646
874
640
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS15,SBS18,SBS40 5 cosine sim = 0.98497
2647
656
2105
596
1059
608
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS40
SBS18
SBS15
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98493
2661
693
2029
624
1025
642
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
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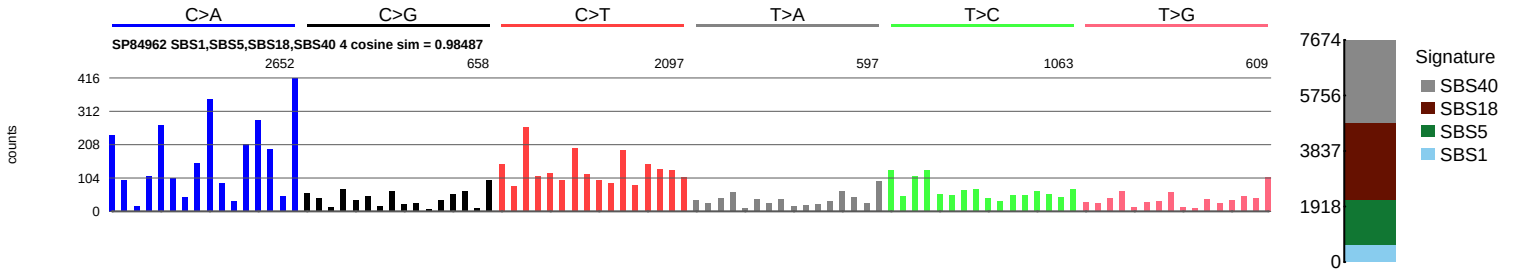
SBS2
SBS1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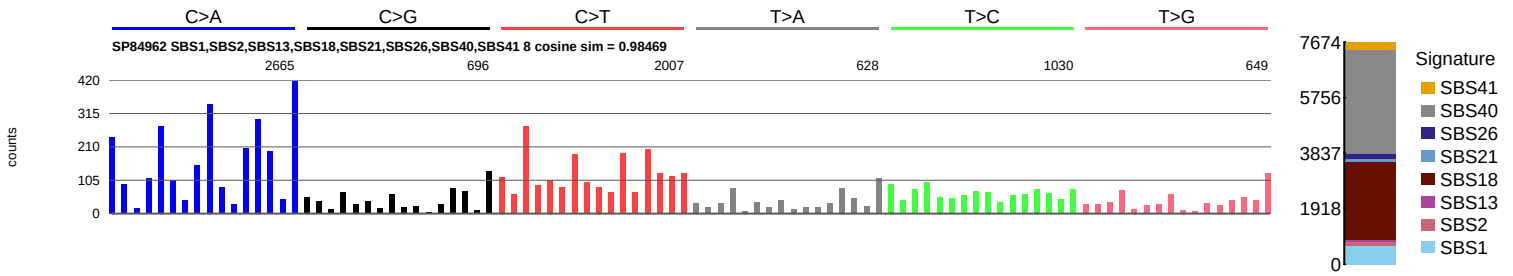
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS18,SBS40 4 cosine sim = 0.98487
2652
658
2097
597
1063
609
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS40
SBS18
SBS5
SBS1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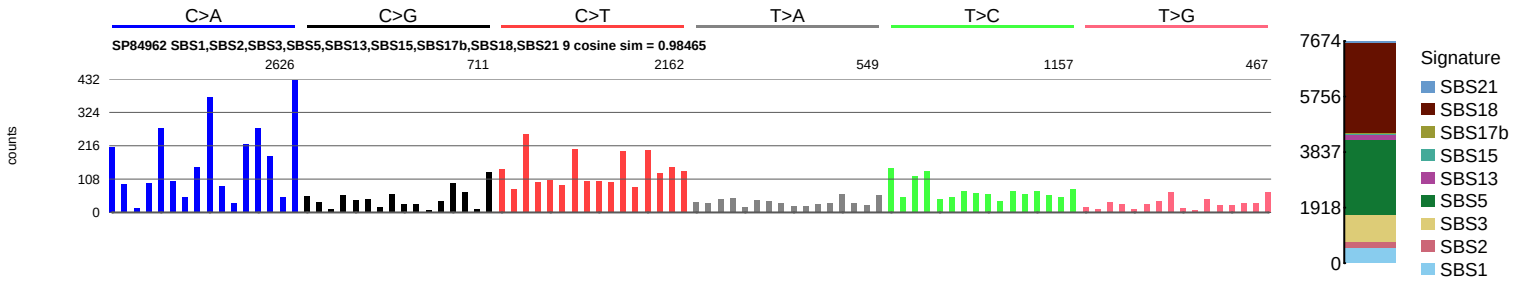
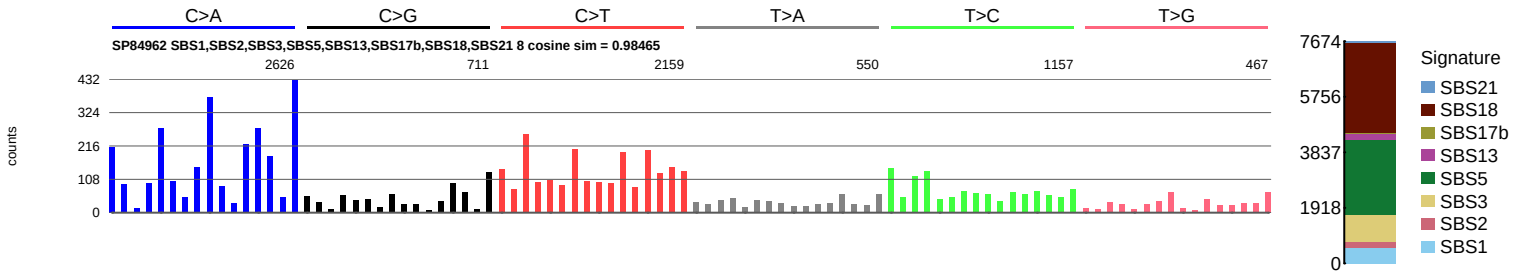
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS13,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98469
2665
696
2007
628
1030
649
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21 9 cosine sim = 0.98465
2626
711
2162
549
1157
467
counts
432
324
216
108
0
7674
5756
3837
1918
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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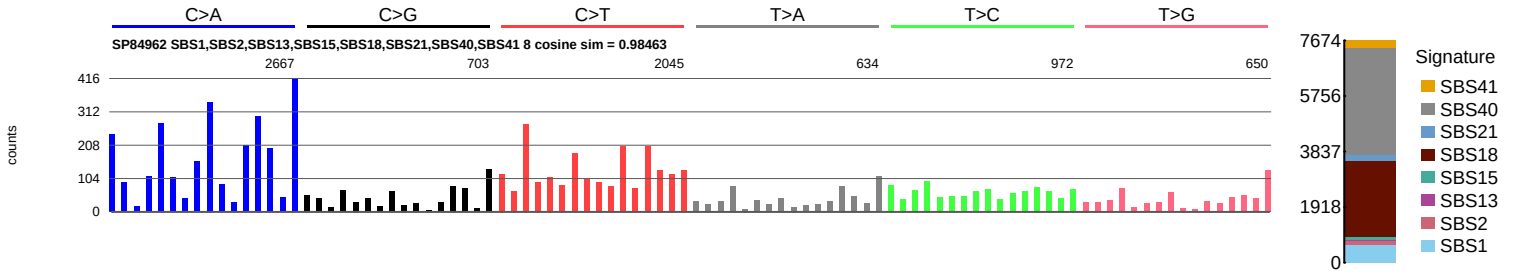
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21 8 cosine sim = 0.98465
2626
711
2159
550
1157
467
counts
432
324
216
108
0
7674
5756
3837
1918
0
Signature
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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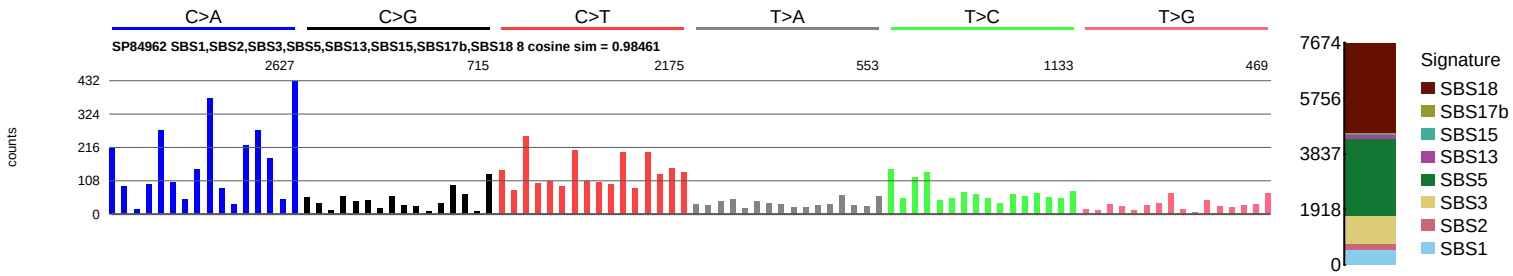
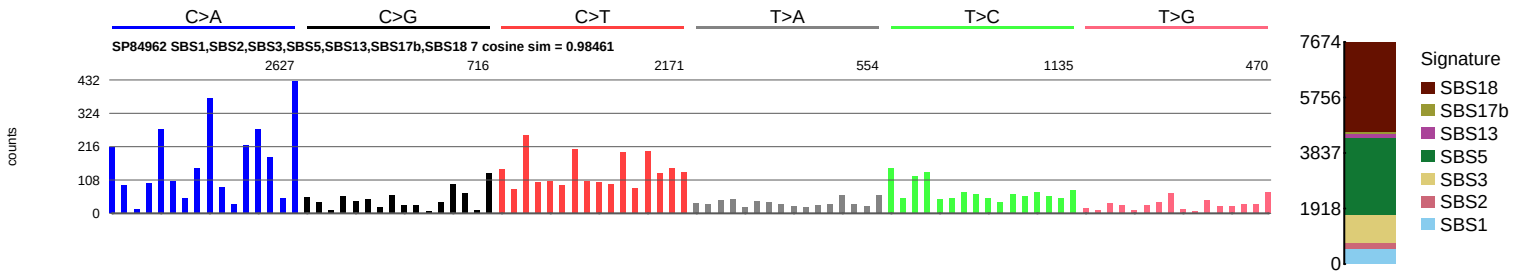
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98463
2667
703
2045
634
972
650
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18 8 cosine sim = 0.98461
2627
715
2175
553
1133
469
counts
432
324
216
108
0
7674
5756
3837
1918
0
Signature
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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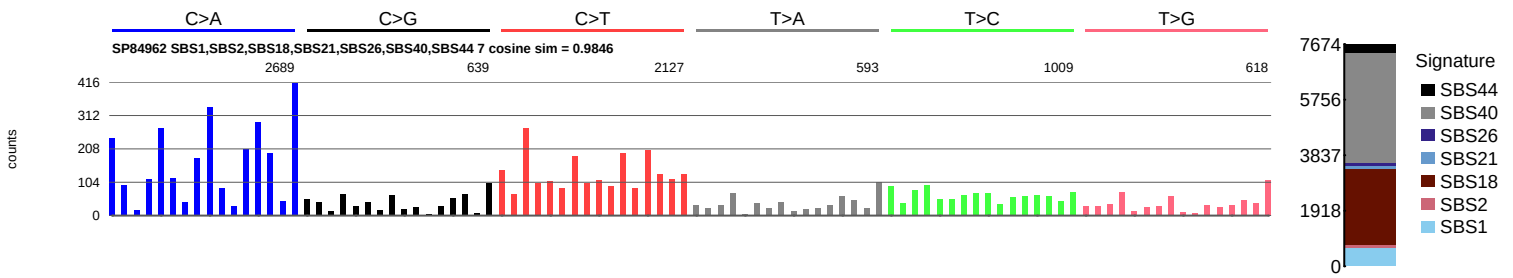
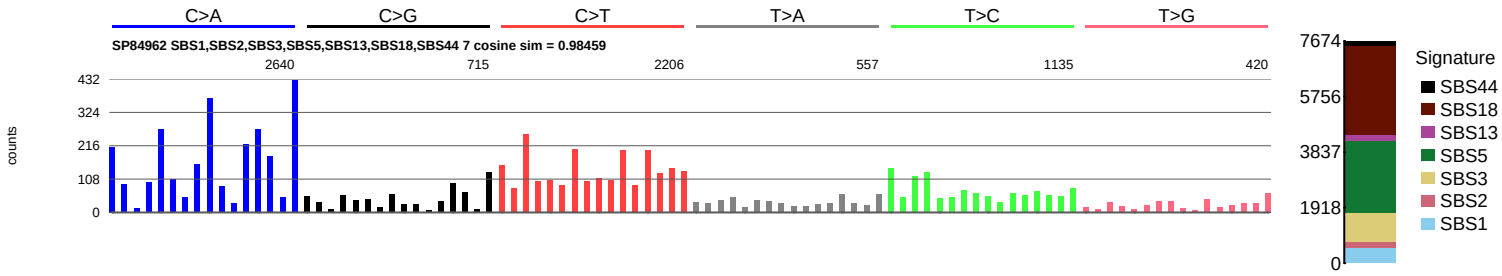
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18 7 cosine sim = 0.98461
2627
716
2171
554
1135
470
counts
432
324
216
108
0
7674
5756
3837
1918
0
Signature
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS18,SBS21,SBS26,SBS40,SBS44 7 cosine sim = 0.9846
2689
639
2127
593
1009
618
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS44 7 cosine sim = 0.98459
2640
715
2206
557
1135
420
counts
432
324
216
108
0
7674
5756
3837
1918
0
Signature
SBS44
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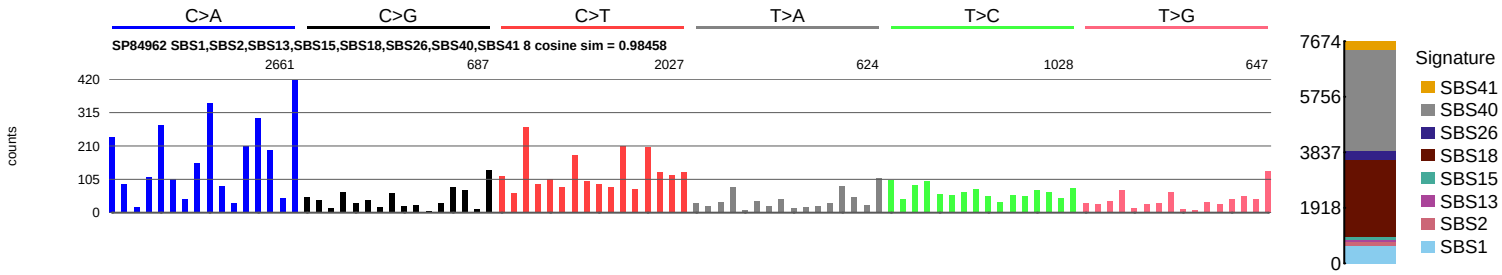
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98458
2661
687
2027
624
1028
647
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
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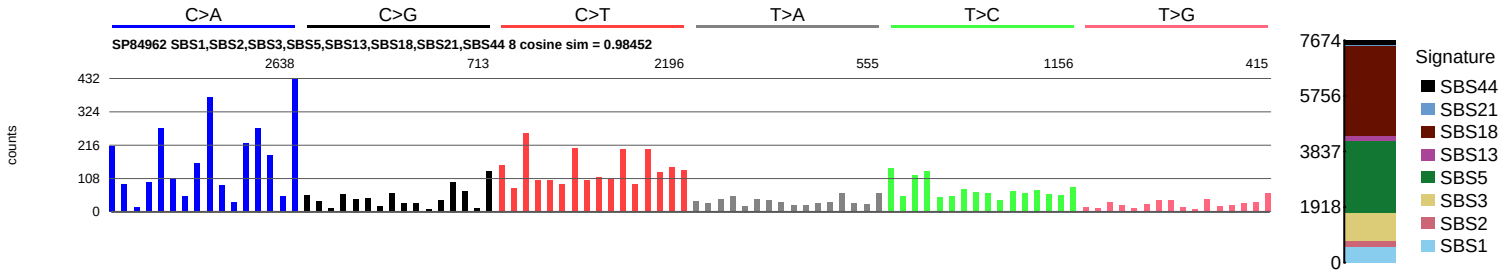
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS44 8 cosine sim = 0.98452
2638
713
2196
555
1156
415
counts
432
324
216
108
0
7674
5756
3837
1918
0
Signature
SBS44
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SBS18
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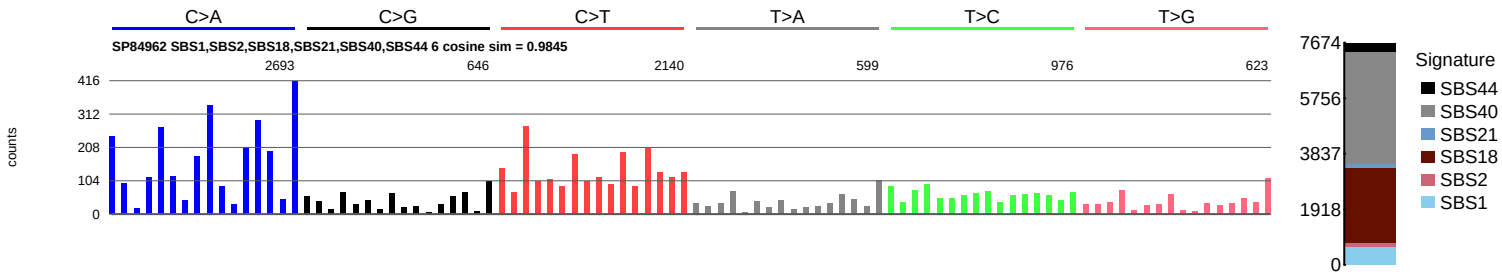
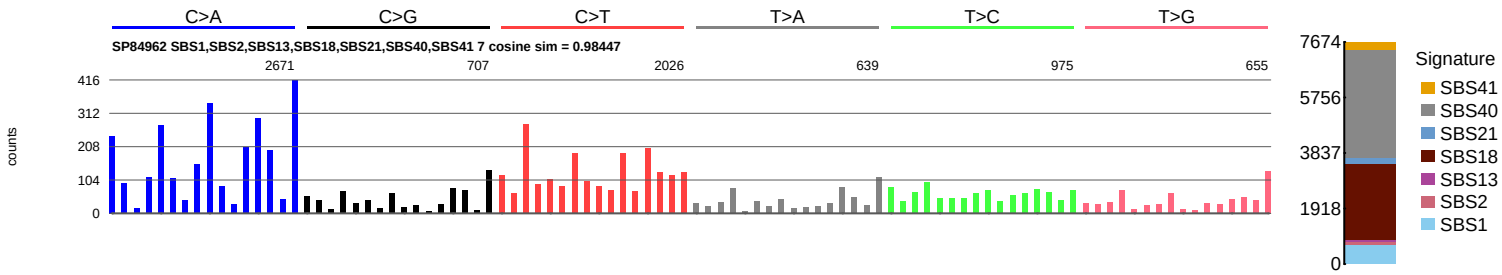
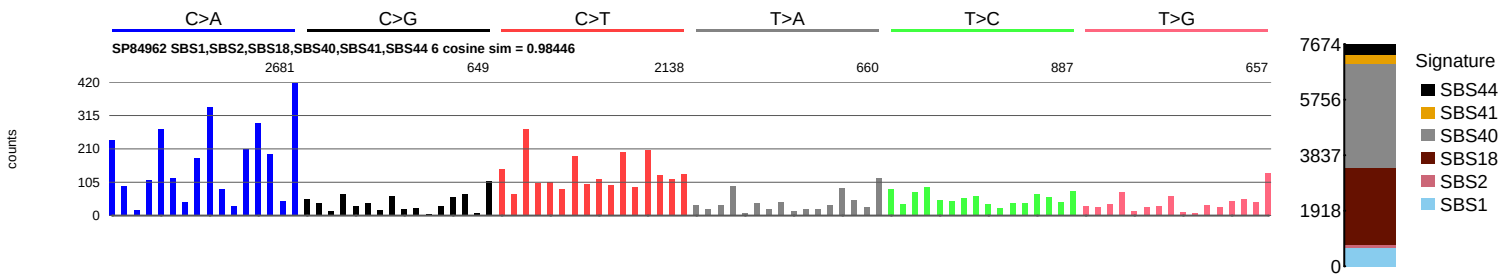
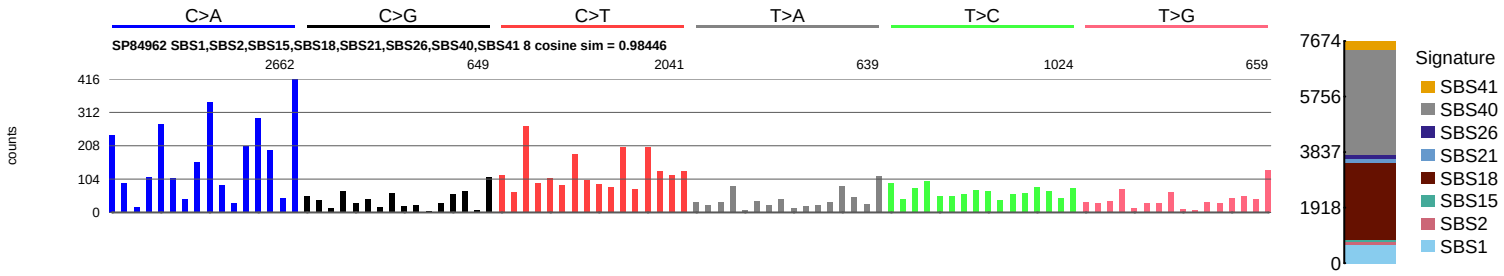
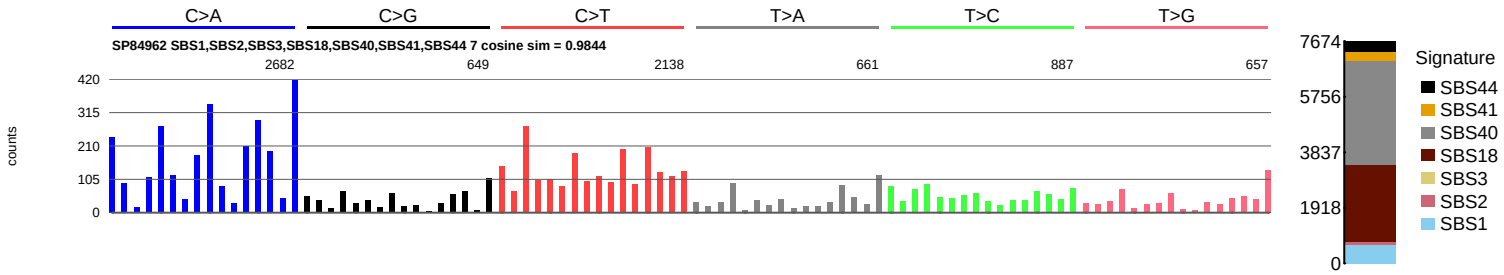
SBS2
SBS1

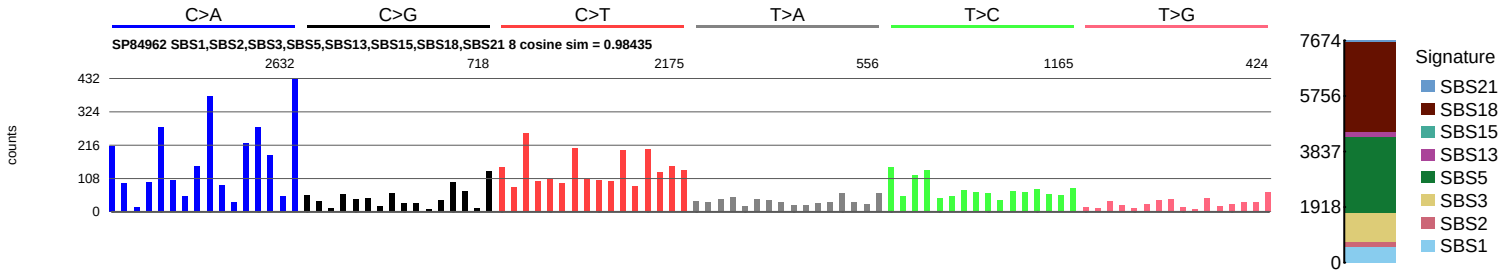

C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS18,SBS21,SBS40,SBS44 6 cosine sim = 0.9845
2693
646
2140
599
976
623
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS2
SBS1

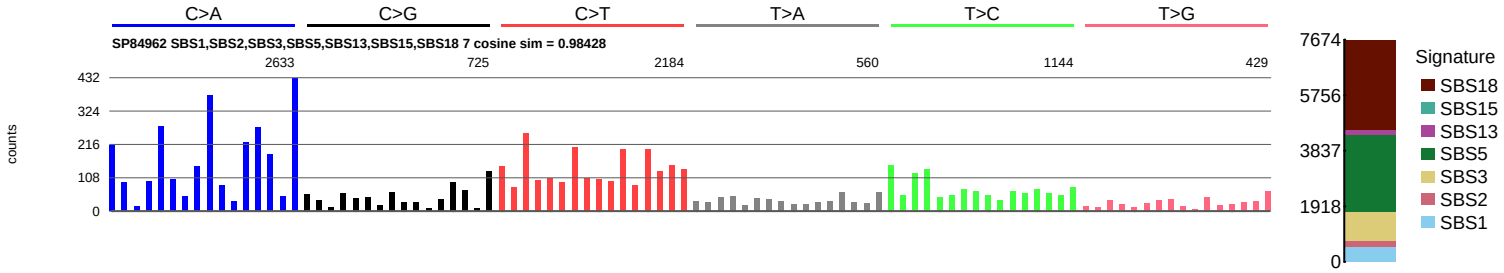

C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS13,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98447
2671
707
2026
639
975
655
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS18,SBS40,SBS41,SBS44 6 cosine sim = 0.98446
2681
649
2138
660
887
657
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98446
2662
649
2041
639
1024
659
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.9844
2682
649
2138
661
887
657
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21 8 cosine sim = 0.98435
2632
718
2175
556
1165
424
counts
432
324
216
108
0
7674
5756
3837
1918
0
Signature
SBS21
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18 7 cosine sim = 0.98428
2633
725
2184
560
1144
429
counts
432
324
216
108
0
7674
5756
3837
1918
0
Signature
SBS18
SBS15
SBS13
SBS5
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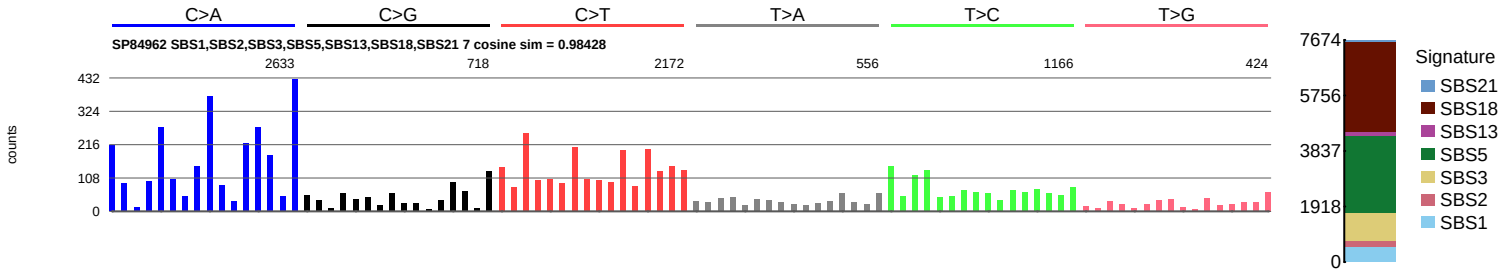
SBS2
SBS1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C>G
C>T
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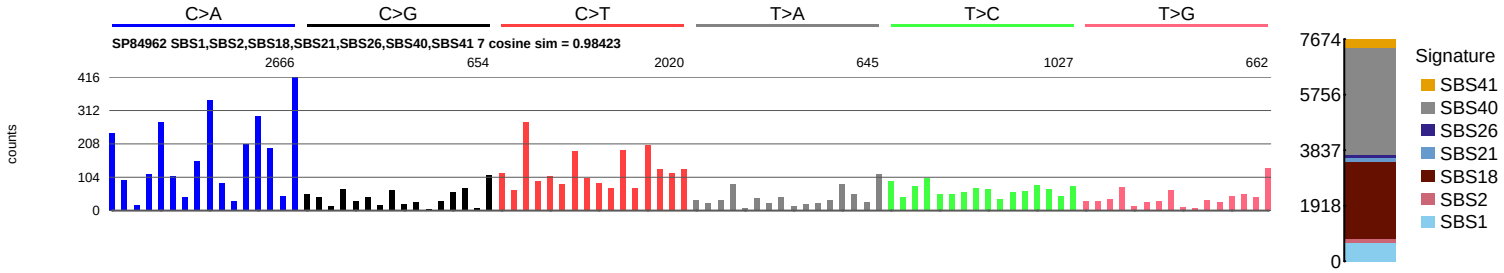
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21 7 cosine sim = 0.98428
2633
718
2172
556
1166
424
counts
432
324
216
108
0
7674
5756
3837
1918
0
Signature
SBS21
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
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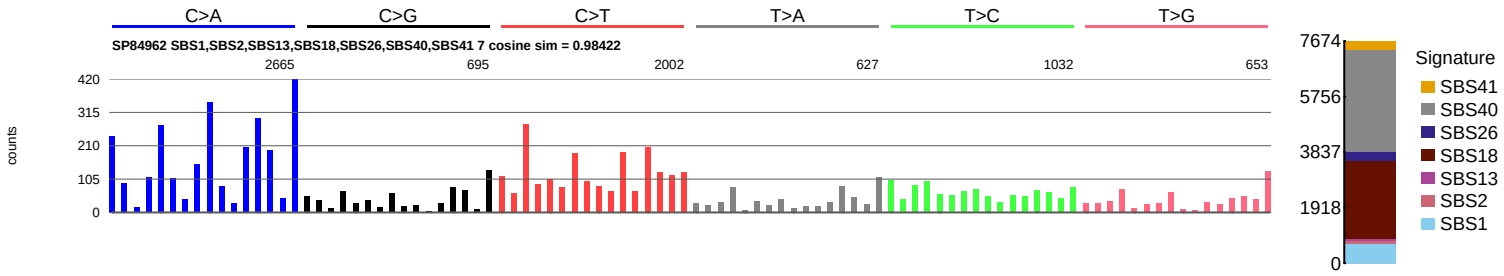
SP84962 SBS1,SBS2,SBS18,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.98423
2666
654
2020
645
1027
662
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS2
SBS1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C>G
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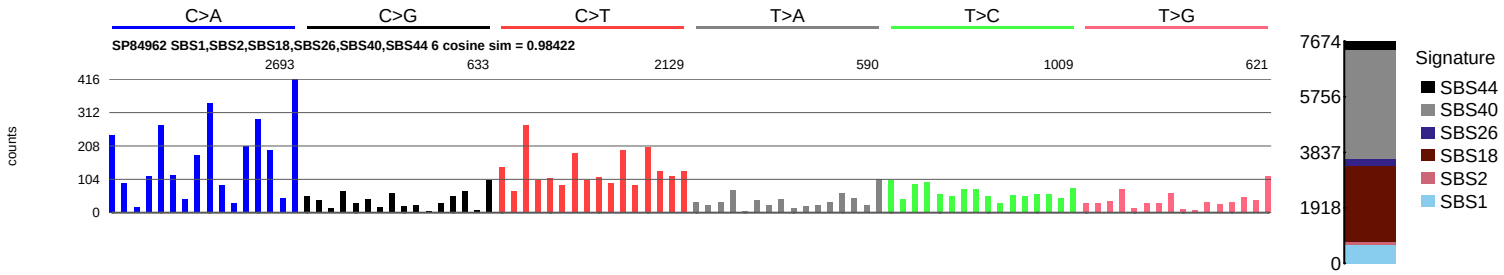
SP84962 SBS1,SBS2,SBS13,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98422
2665
695
2002
627
1032
653
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS13
SBS2
SBS1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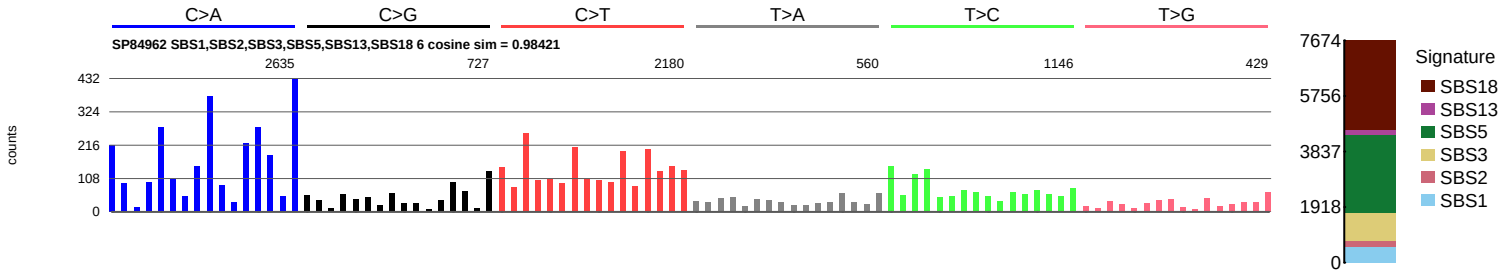
T>C
T>G
SP84962 SBS1,SBS2,SBS18,SBS26,SBS40,SBS44 6 cosine sim = 0.98422
2693
633
2129
590
1009
621
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18 6 cosine sim = 0.98421
2635
727
2180
560
1146
429
counts
432
324
216
108
0
7674
5756
3837
1918
0
Signature
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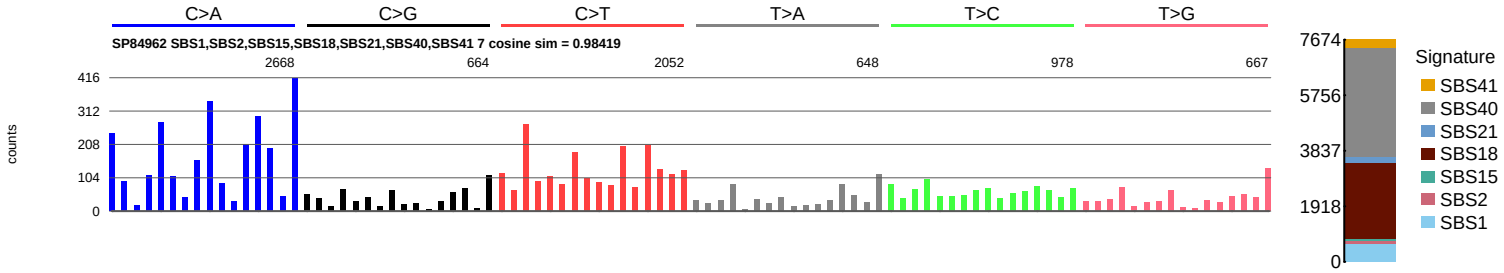
SBS2
SBS1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C>G
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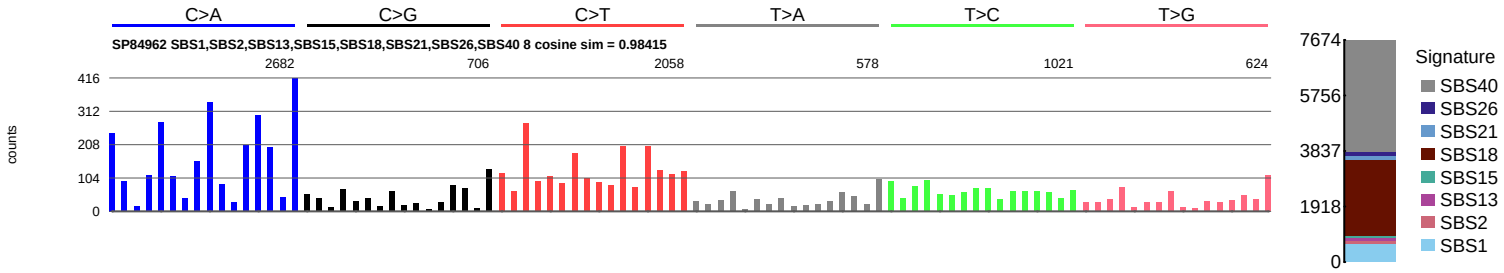
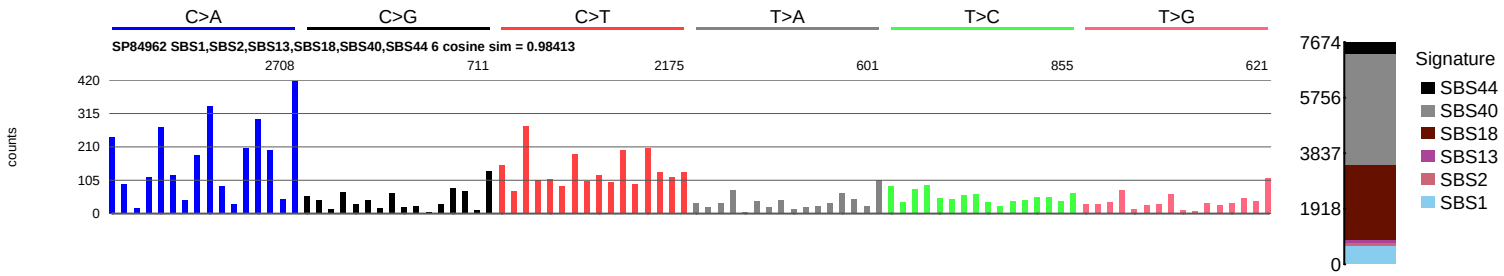
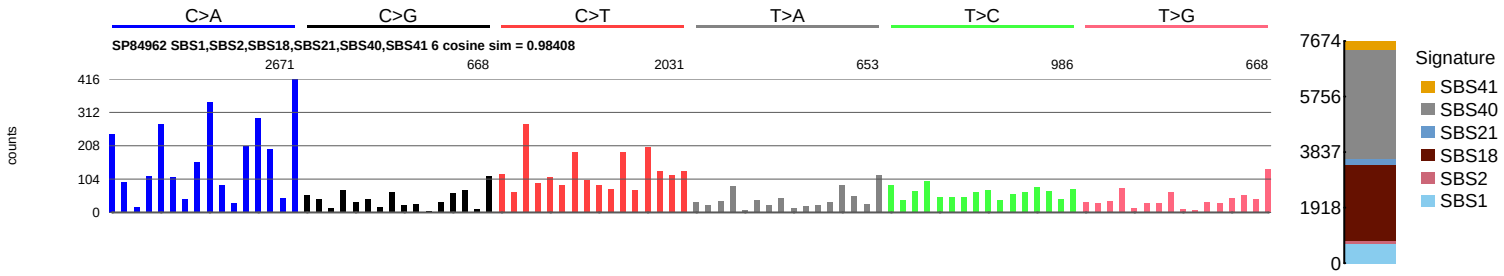
T>C
T>G
SP84962 SBS1,SBS2,SBS15,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98419
2668
664
2052
648
978
667
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98415
2682
706
2058
578
1021
624
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS13,SBS18,SBS40,SBS44 6 cosine sim = 0.98413
2708
711
2175
601
855
621
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS18
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS18,SBS21,SBS40,SBS41 6 cosine sim = 0.98408
2671
668
2031
653
986
668
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
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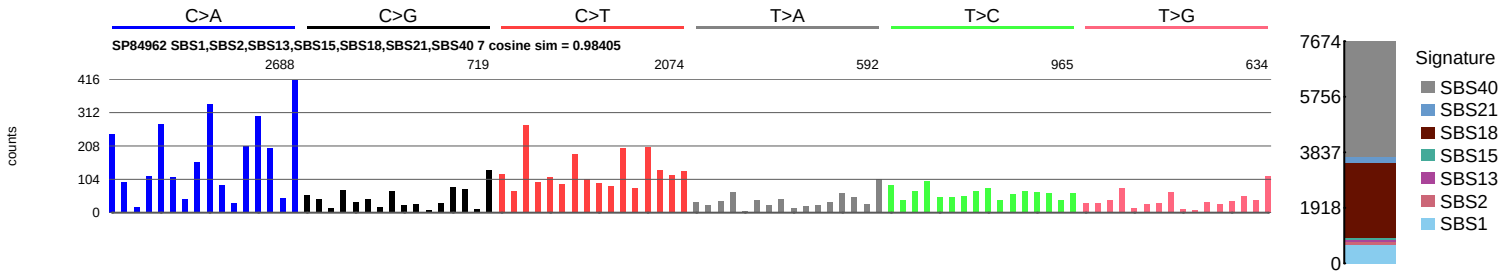
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS40 7 cosine sim = 0.98405
2688
719
2074
592
965
634
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
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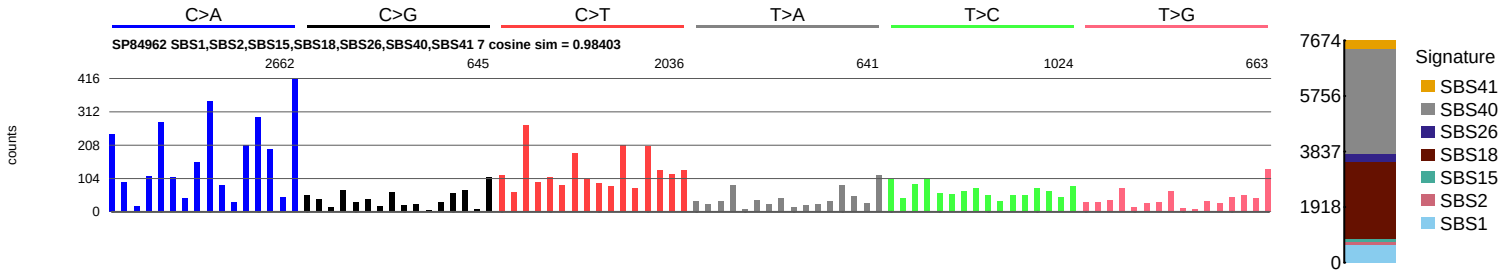
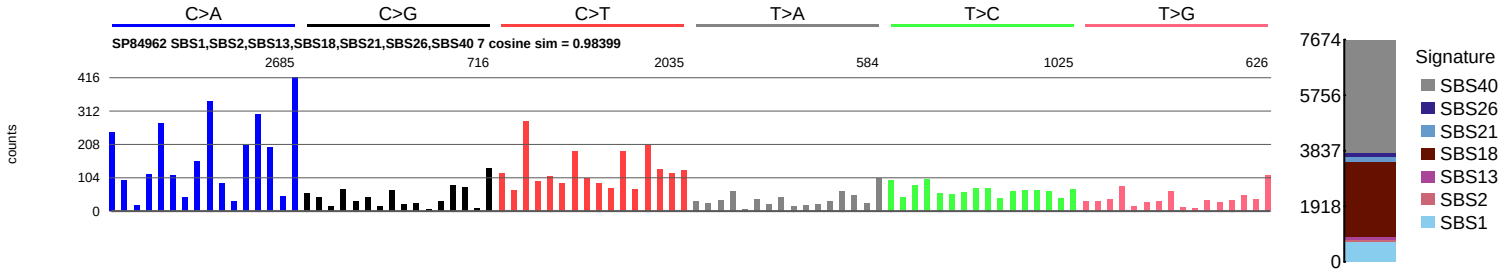
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS15,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98403
2662
645
2036
641
1024
663
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS13,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98399
2685
716
2035
584
1025
626
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
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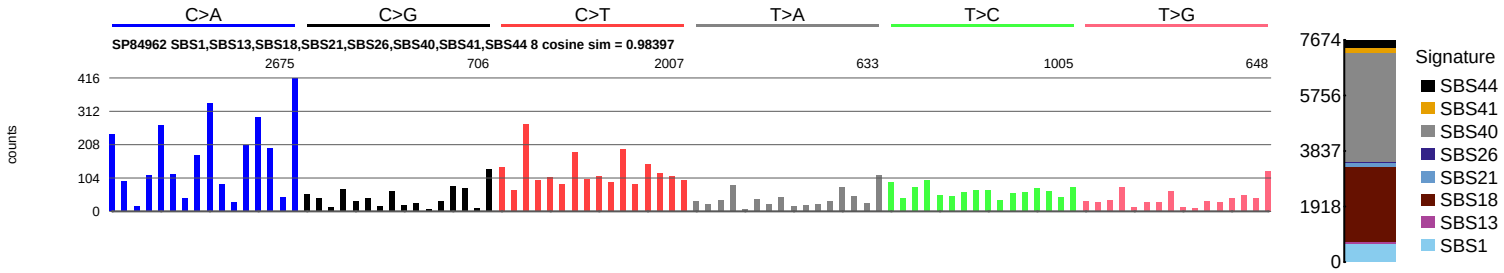
SBS2
SBS1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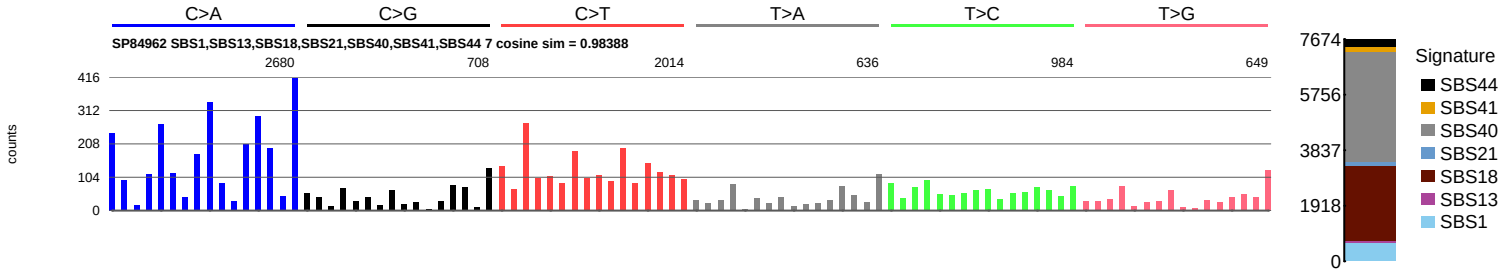
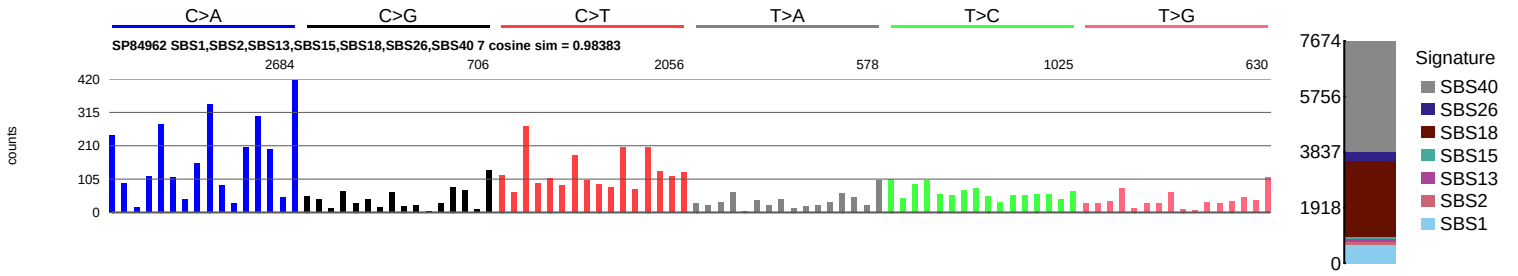
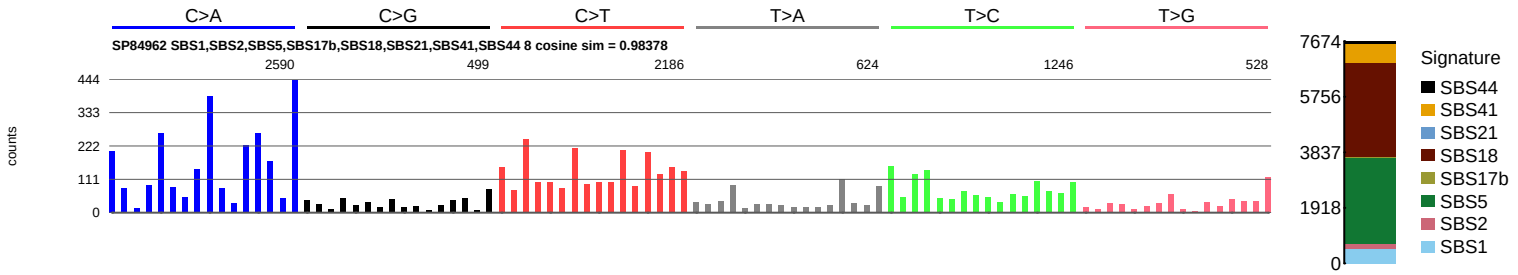
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98397
2675
706
2007
633
1005
648
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS18,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98388
2680
708
2014
636
984
649
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS13,SBS15,SBS18,SBS26,SBS40 7 cosine sim = 0.98383
2684
706
2056
578
1025
630
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS13
SBS2
SBS1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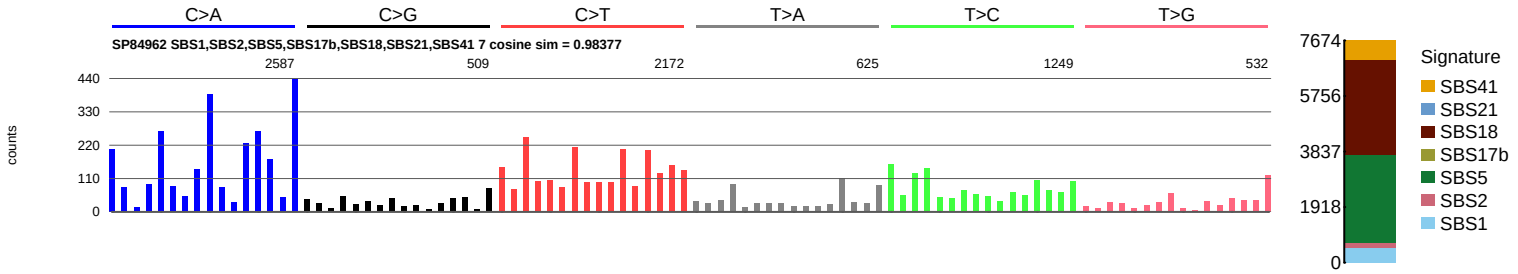
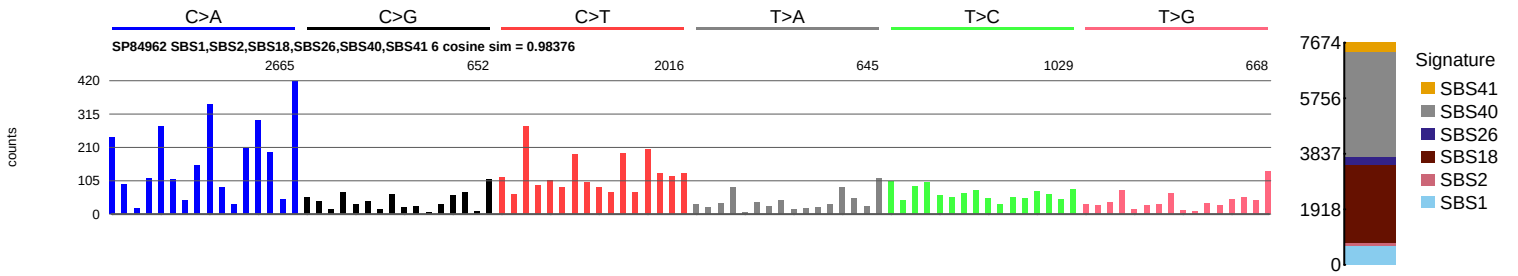
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98378
2590
499
2186
624
1246
528
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98377
2587
509
2172
625
1249
532
counts
440
330
220
110
0
7674
5756
3837
1918
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1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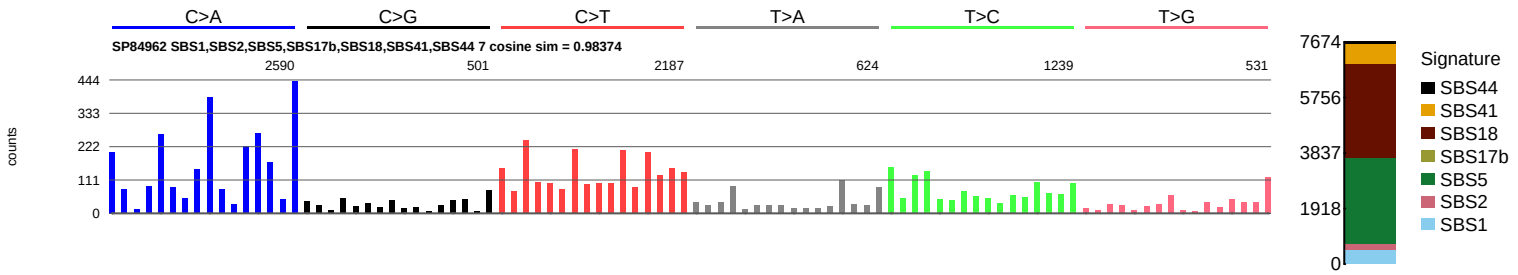
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS18,SBS26,SBS40,SBS41 6 cosine sim = 0.98376
2665
652
2016
645
1029
668
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98374
2590
501
2187
624
1239
531
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
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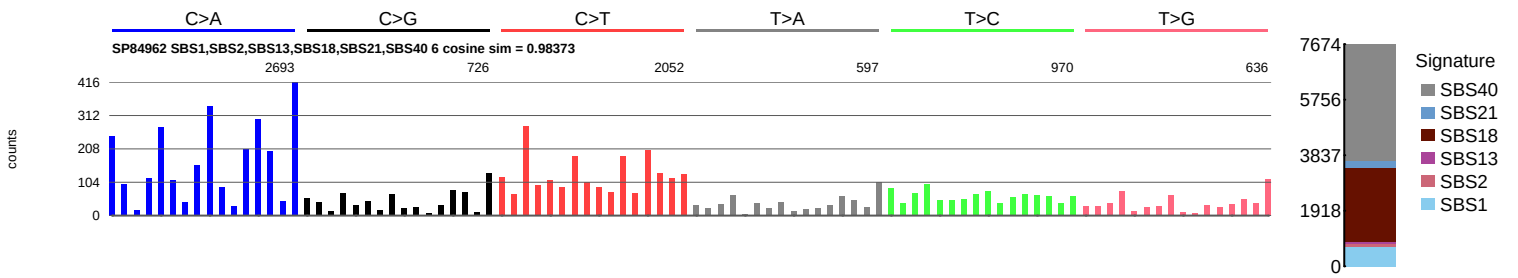
SBS5
SBS2
SBS1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T>C
T>G
SP84962 SBS1,SBS2,SBS13,SBS18,SBS21,SBS40 6 cosine sim = 0.98373
2693
726
2052
597
970
636
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS40
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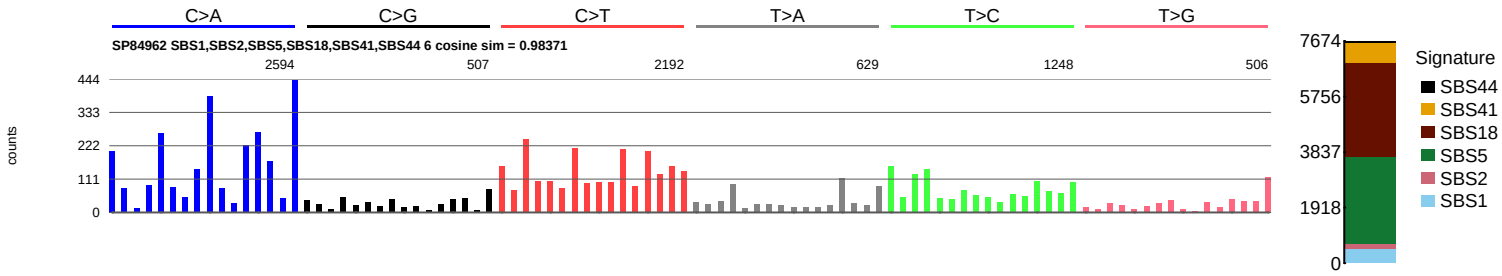
SBS2
SBS1


C>A
C>G
C>T
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T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS18,SBS41,SBS44 6 cosine sim = 0.98371
2594
507
2192
629
1248
506
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
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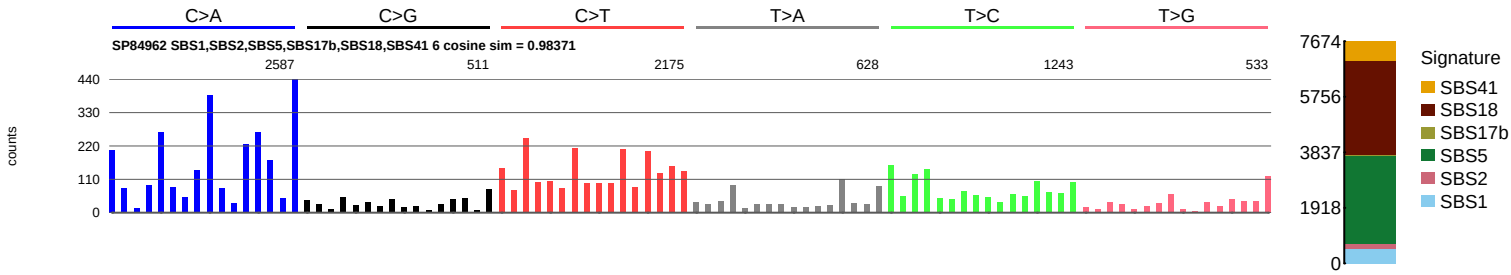
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS41 6 cosine sim = 0.98371
2587
511
2175
628
1243
533
counts
440
330
220
110
0
7674
5756
3837
1918
0
Signature
SBS41
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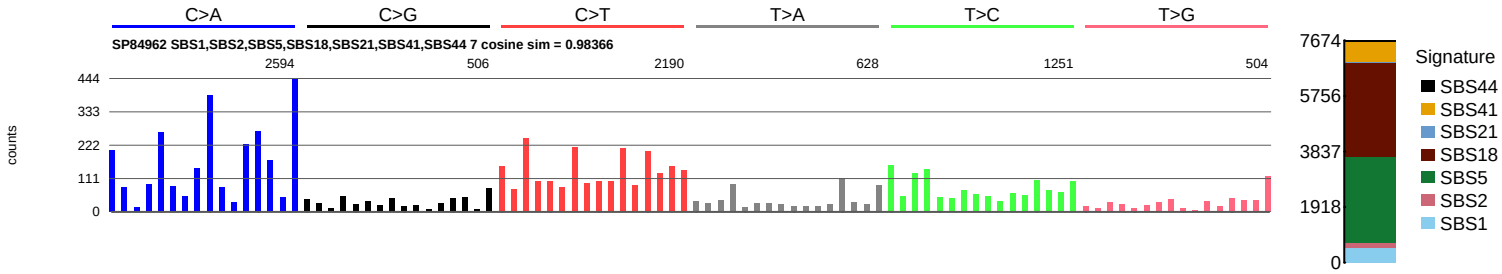
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98366
2594
506
2190
628
1251
504
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
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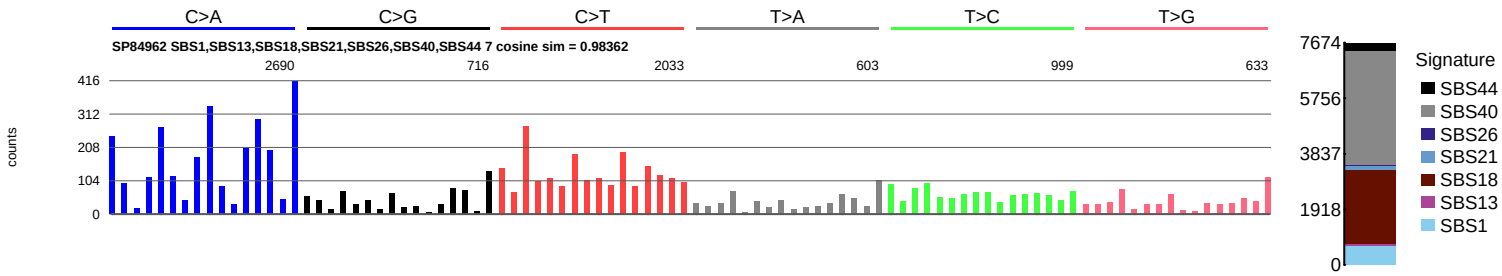
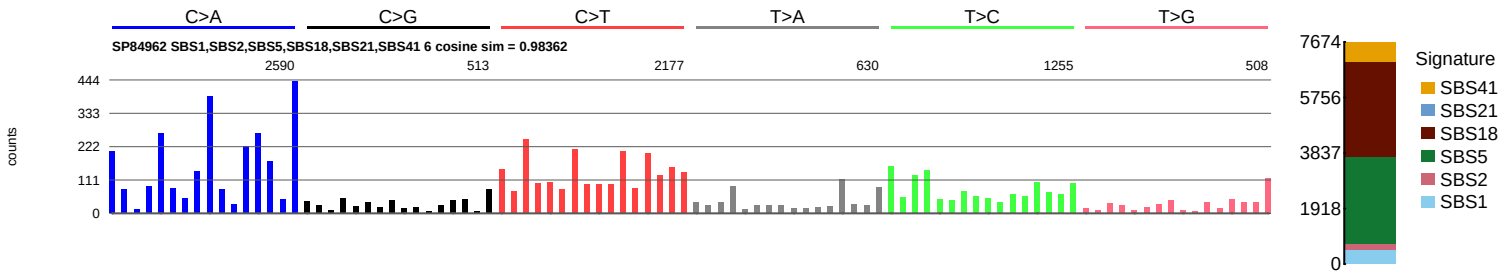
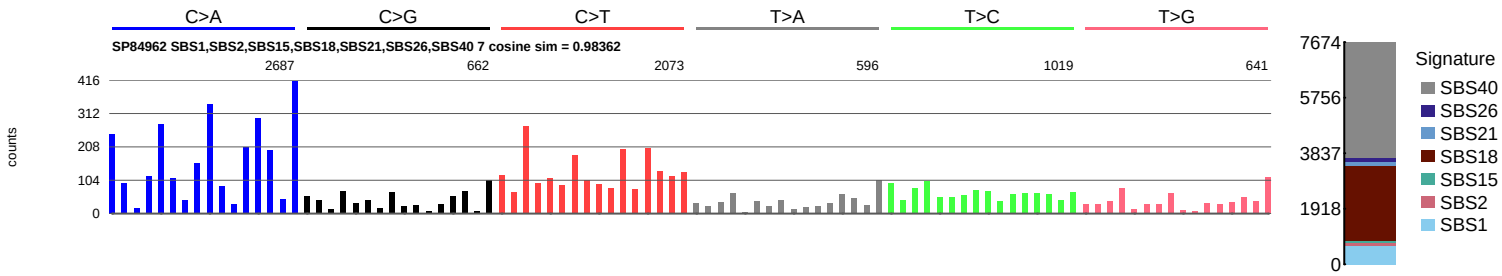
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS18,SBS21,SBS26,SBS40,SBS44 7 cosine sim = 0.98362
2690
716
2033
603
999
633
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS18,SBS21,SBS41 6 cosine sim = 0.98362
2590
513
2177
630
1255
508
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
SBS41
SBS21
SBS18
SBS5
SBS2
SBS1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C>G
C>T
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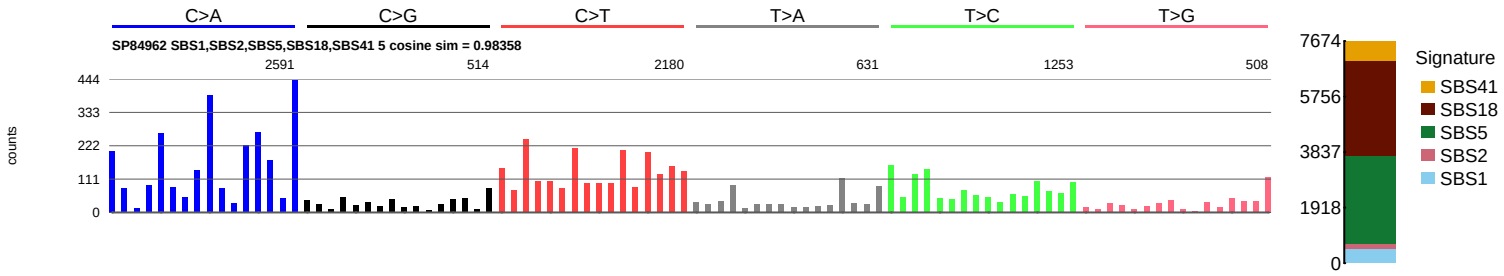
T>C
T>G
SP84962 SBS1,SBS2,SBS15,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98362
2687
662
2073
596
1019
641
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS15
SBS2
SBS1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C>G
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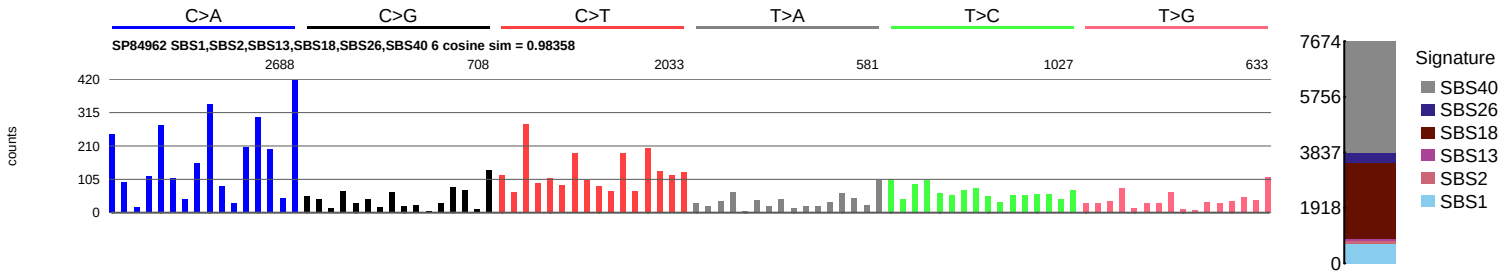
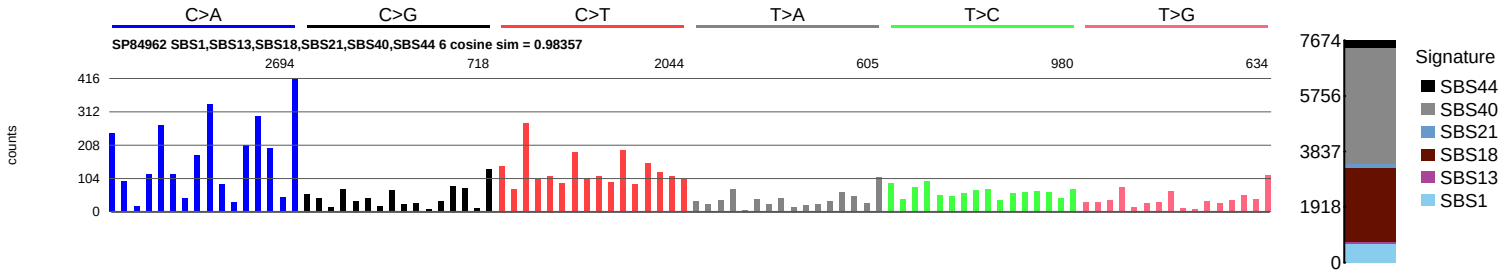
SP84962 SBS1,SBS2,SBS5,SBS18,SBS41 5 cosine sim = 0.98358
2591
514
2180
631
1253
508
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
SBS41
SBS18
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS13,SBS18,SBS26,SBS40 6 cosine sim = 0.98358
2688
708
2033
581
1027
633
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS40
SBS26
SBS18
SBS13
SBS2
SBS1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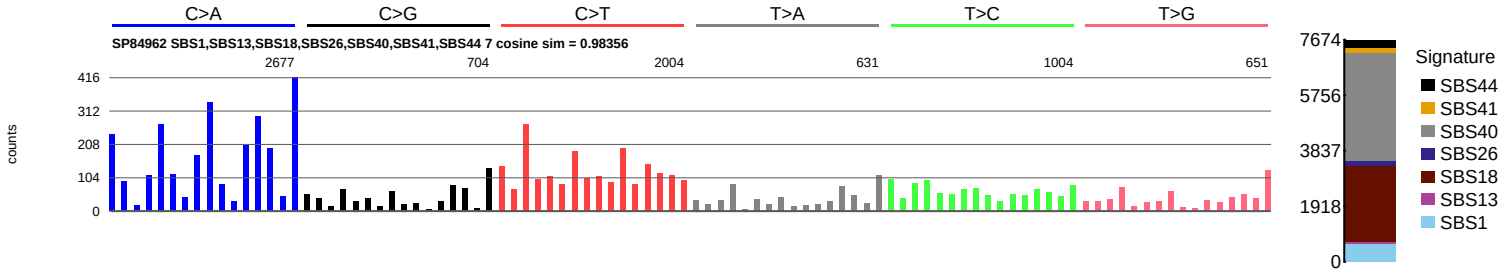
T>G
SP84962 SBS1,SBS13,SBS18,SBS21,SBS40,SBS44 6 cosine sim = 0.98357
2694
718
2044
605
980
634
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS13
SBS1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C>G
C>T
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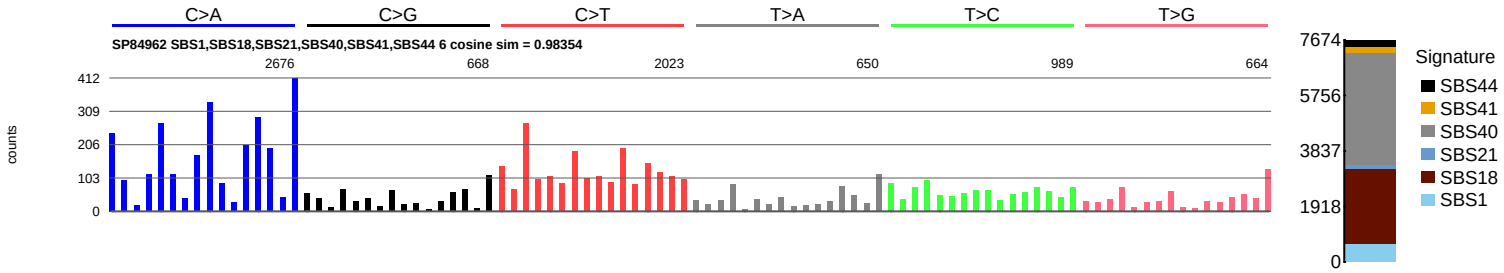
SP84962 SBS1,SBS13,SBS18,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98356
2677
704
2004
631
1004
651
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS13
SBS1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C>G
C>T
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T>G
SP84962 SBS1,SBS18,SBS21,SBS40,SBS41,SBS44 6 cosine sim = 0.98354
2676
668
2023
650
989
664
counts
412
309
206
103
0
7674
5756
3837
1918
0
Signature
SBS44
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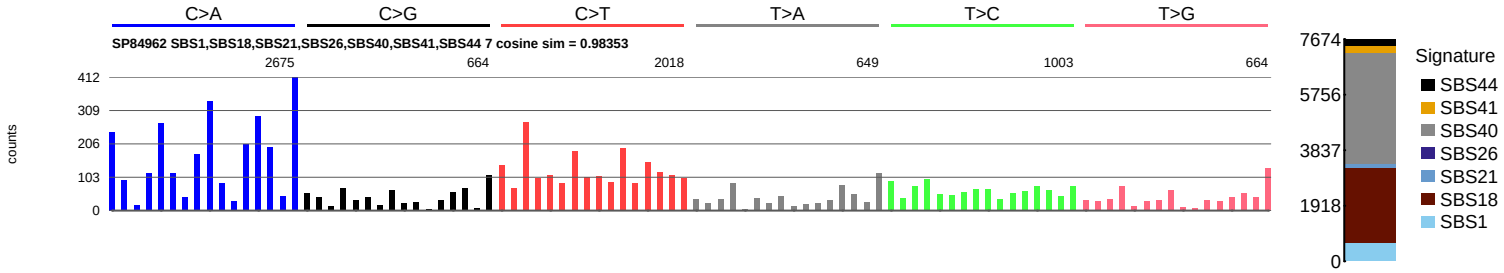
SBS21
SBS18
SBS1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C>G
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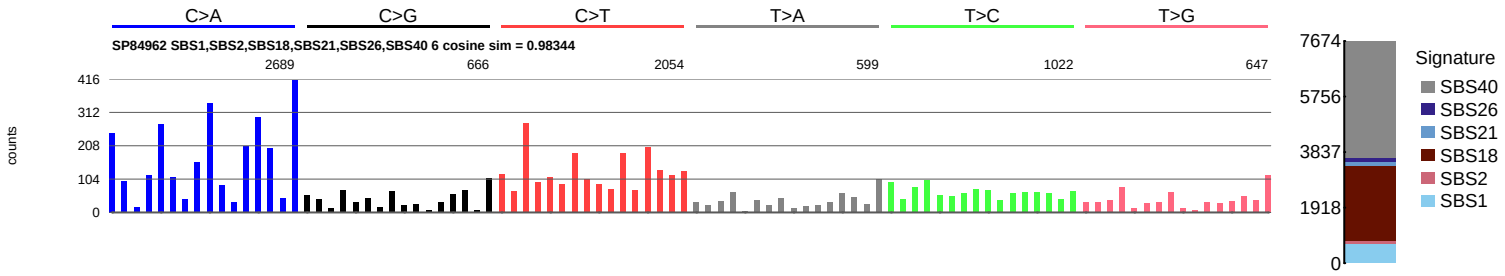
T>A
T>C
T>G
SP84962 SBS1,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98353
2675
664
2018
649
1003
664
counts
412
309
206
103
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS1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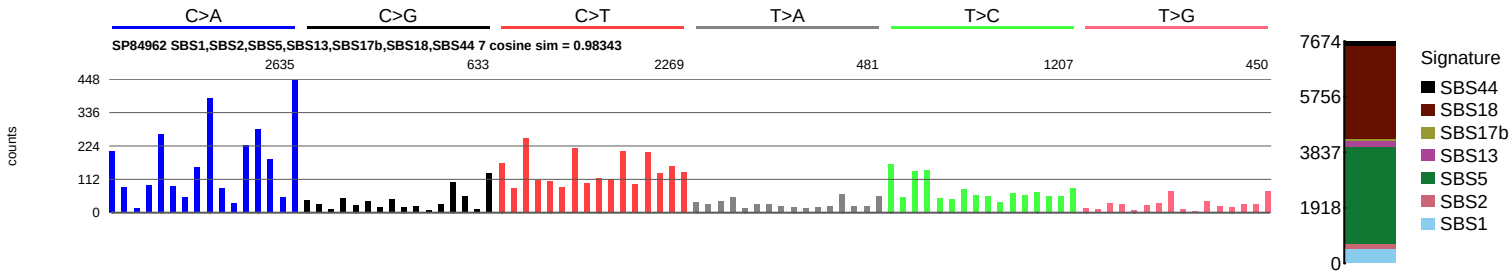
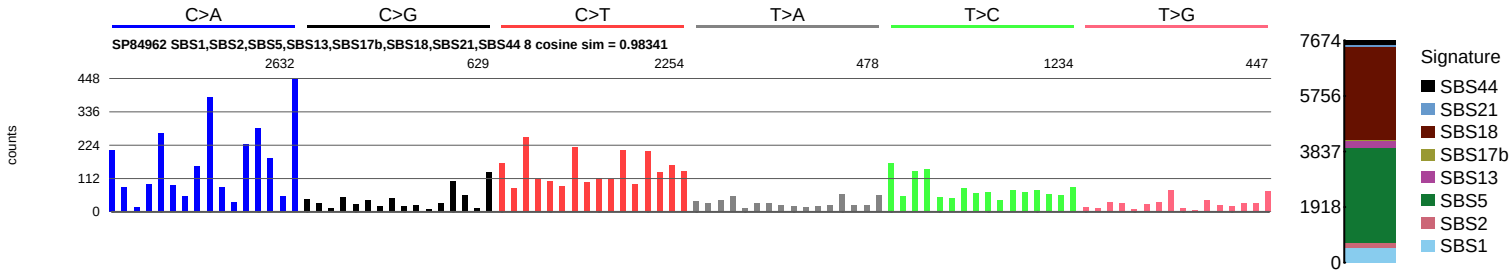
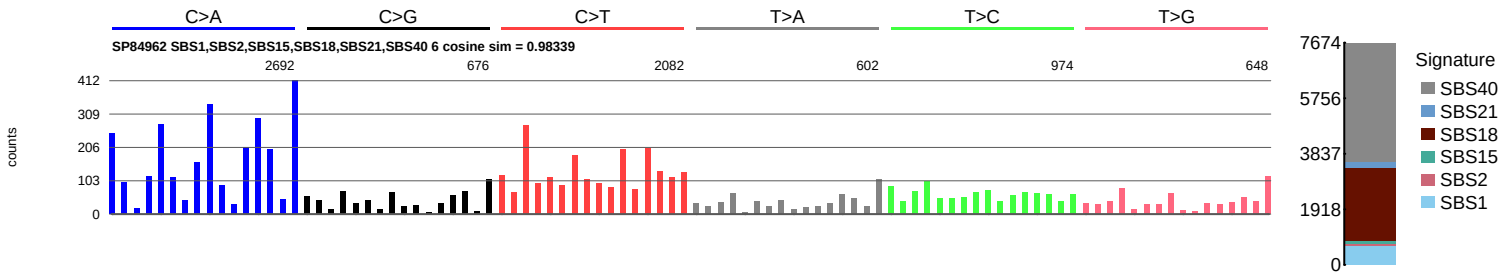
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS18,SBS21,SBS26,SBS40 6 cosine sim = 0.98344
2689
666
2054
599
1022
647
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS44 7 cosine sim = 0.98343
2635
633
2269
481
1207
450
counts
448
336
224
112
0
7674
5756
3837
1918
0
Signature
SBS44
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98341
2632
629
2254
478
1234
447
counts
448
336
224
112
0
7674
5756
3837
1918
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS15,SBS18,SBS21,SBS40 6 cosine sim = 0.98339
2692
676
2082
602
974
648
counts
412
309
206
103
0
7674
5756
3837
1918
0
Signature
SBS40
SBS21
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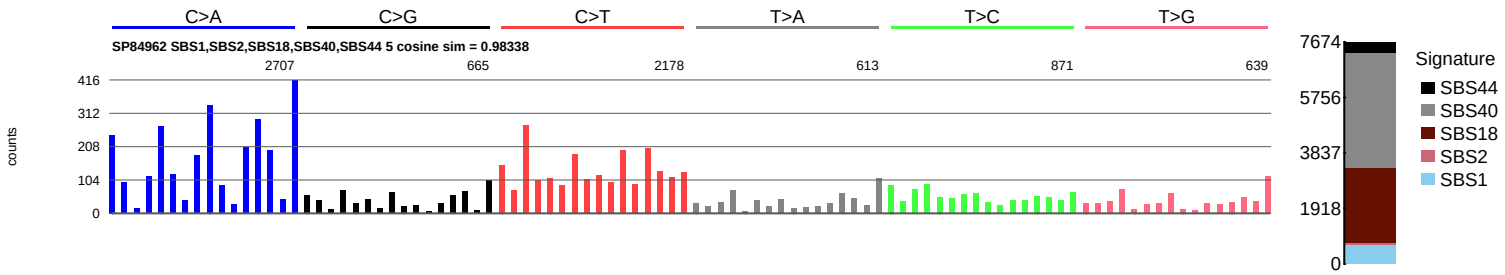
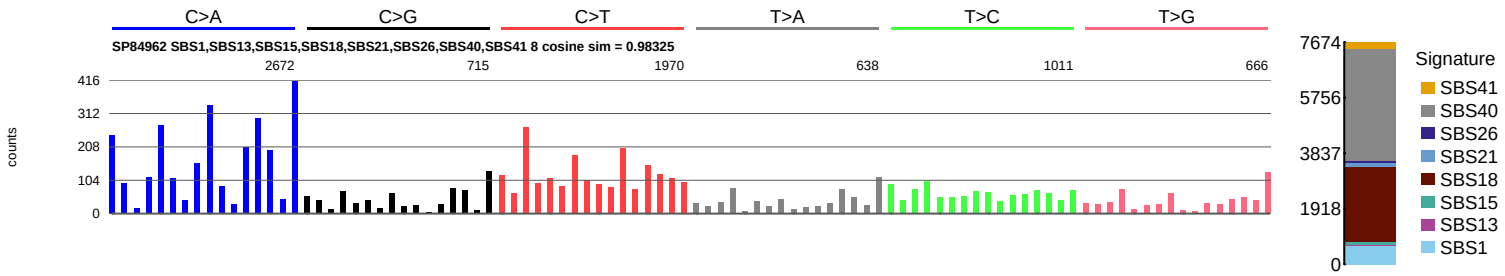
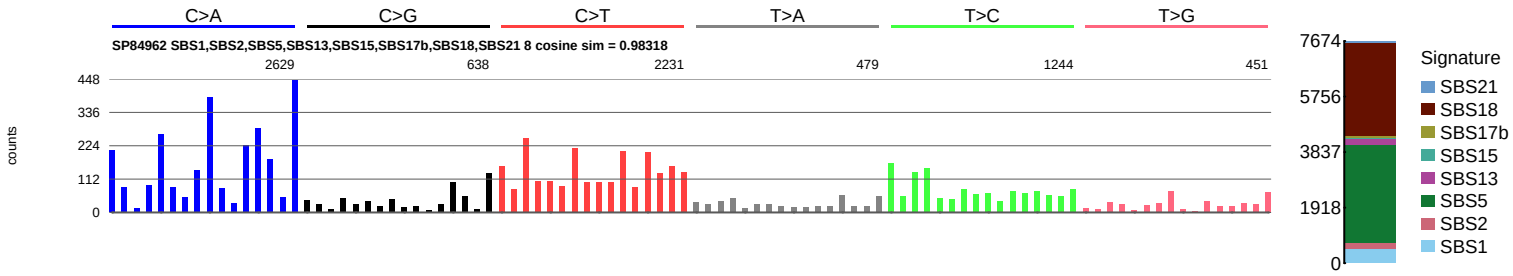
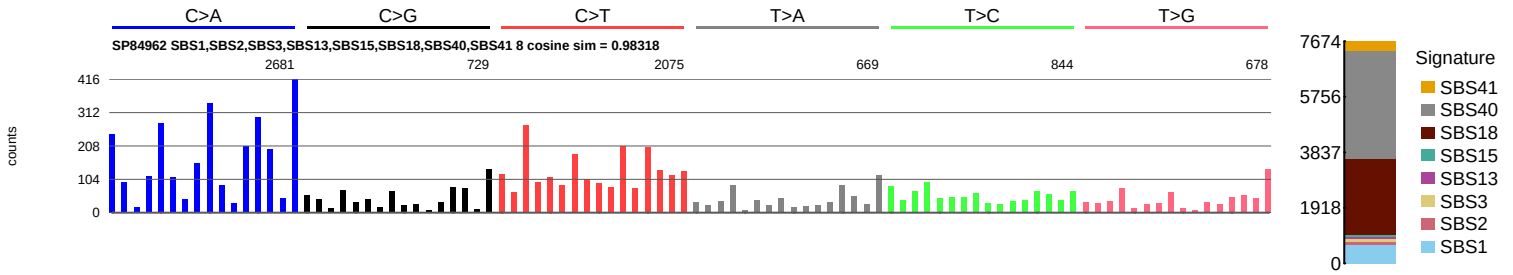
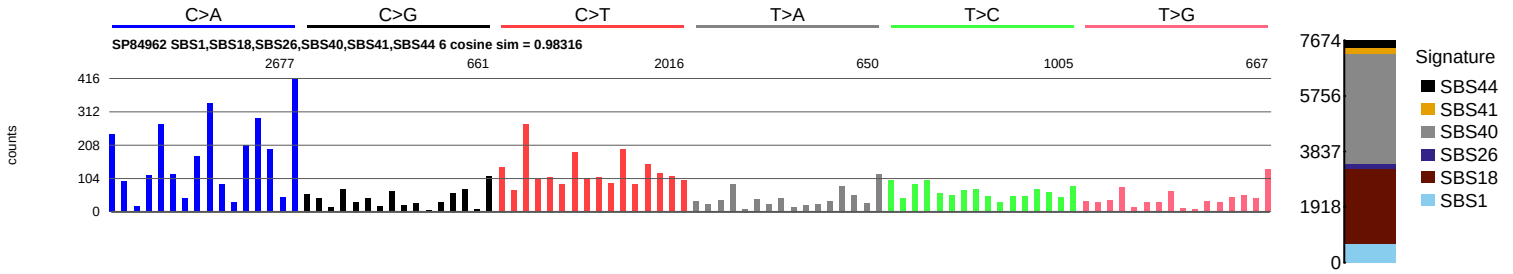
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS18,SBS40,SBS44 5 cosine sim = 0.98338
2707
665
2178
613
871
639
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98325
2672
715
1970
638
1011
666
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21 8 cosine sim = 0.98318
2629
638
2231
479
1244
451
counts
448
336
224
112
0
7674
5756
3837
1918
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS40,SBS41 8 cosine sim = 0.98318
2681
729
2075
669
844
678
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1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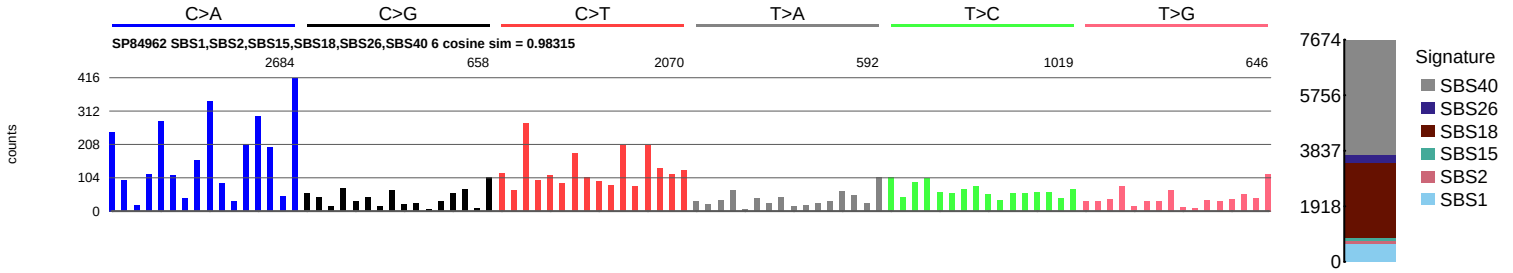
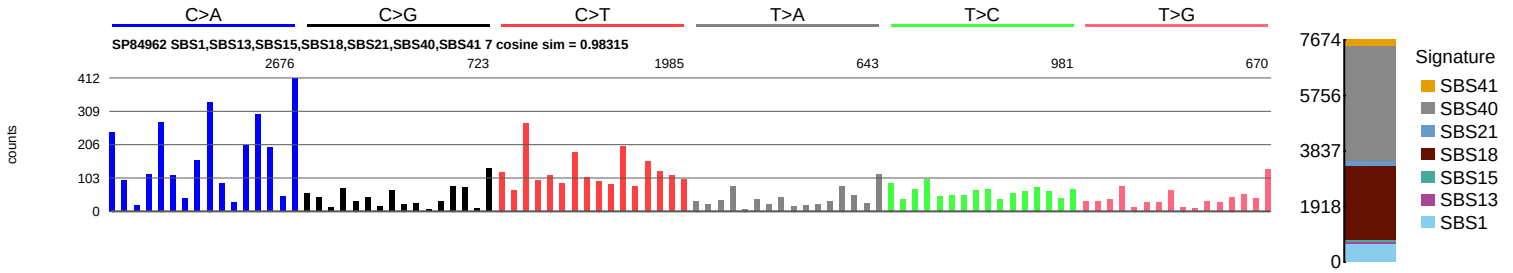
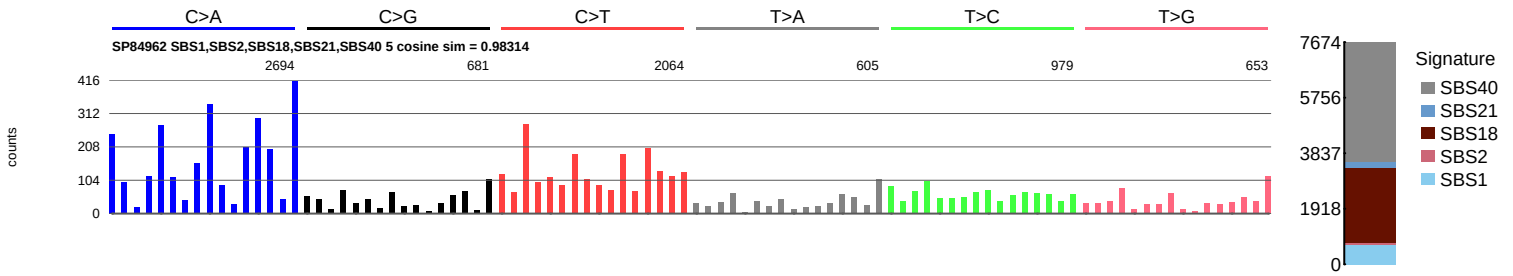
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS18,SBS26,SBS40,SBS41,SBS44 6 cosine sim = 0.98316
2677
661
2016
650
1005
667
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS15,SBS18,SBS26,SBS40 6 cosine sim = 0.98315
2684
658
2070
592
1019
646
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98315
2676
723
1985
643
981
670
counts
412
309
206
103
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS13
SBS1


C>A
C>G
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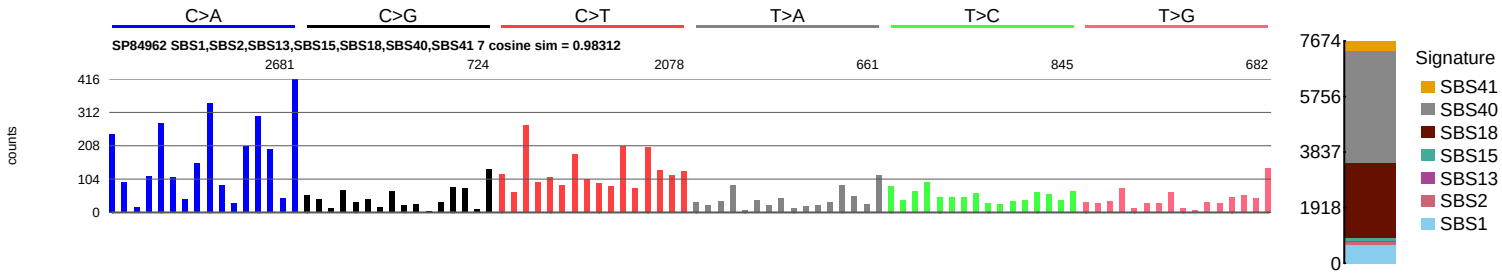
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS18,SBS21,SBS40 5 cosine sim = 0.98314
2694
681
2064
605
979
653
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS40
SBS21
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS13,SBS15,SBS18,SBS40,SBS41 7 cosine sim = 0.98312
2681
724
2078
661
845
682
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
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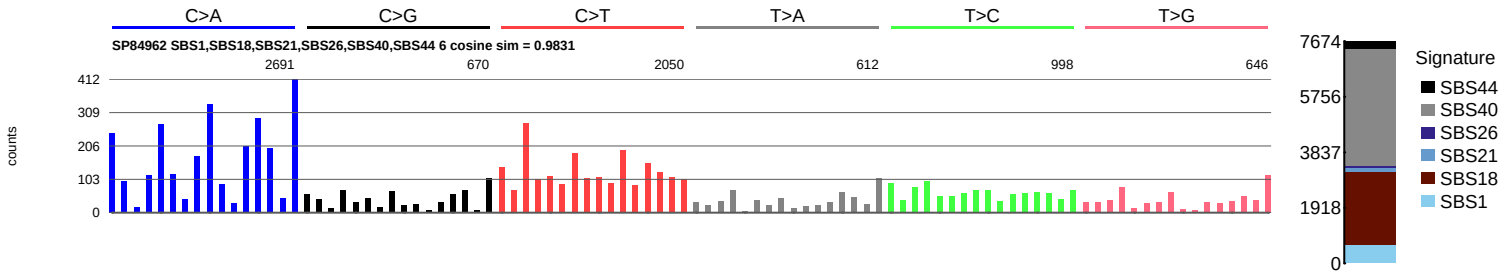
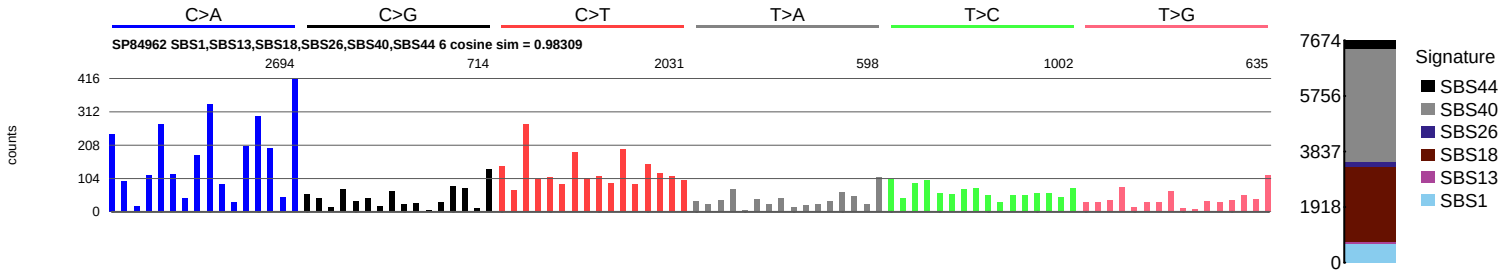
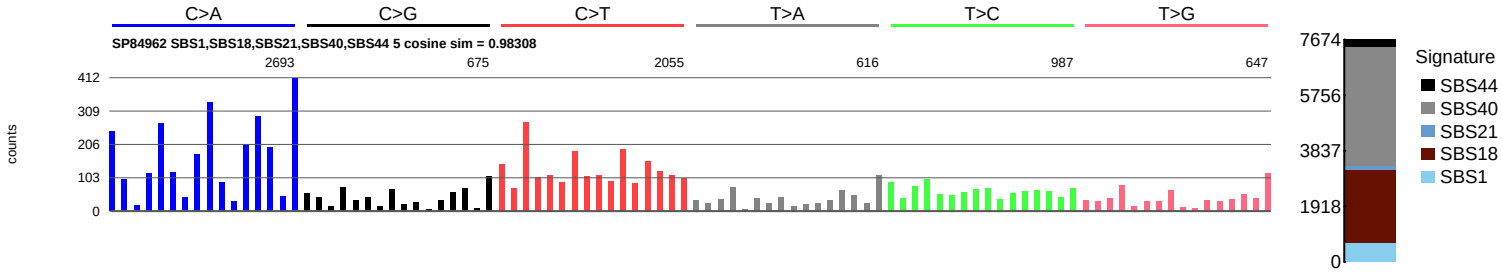
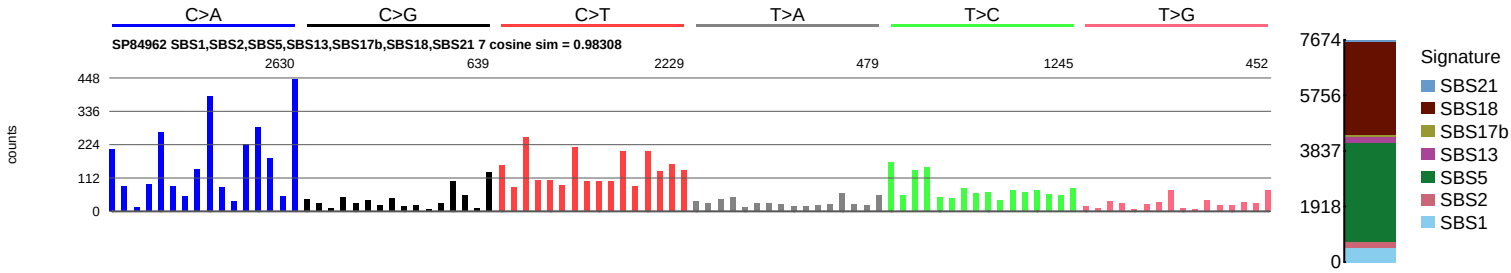
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS18,SBS21,SBS26,SBS40,SBS44 6 cosine sim = 0.9831
2691
670
2050
612
998
646
counts
412
309
206
103
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS18,SBS26,SBS40,SBS44 6 cosine sim = 0.98309
2694
714
2031
598
1002
635
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS18,SBS21,SBS40,SBS44 5 cosine sim = 0.98308
2693
675
2055
616
987
647
counts
412
309
206
103
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21 7 cosine sim = 0.98308
2630
639
2229
479
1245
452
counts
448
336
224
112
0
7674
5756
3837
1918
0
Signature
SBS21
SBS18
SBS17b
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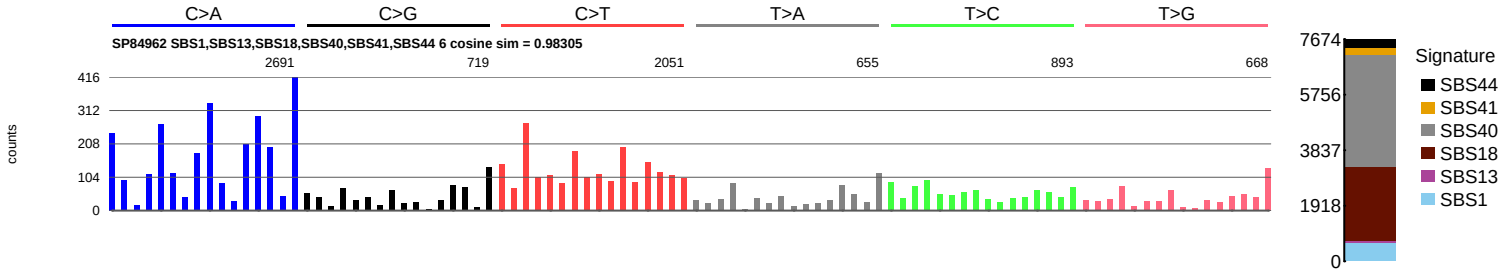
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS18,SBS40,SBS41,SBS44 6 cosine sim = 0.98305
2691
719
2051
655
893
668
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
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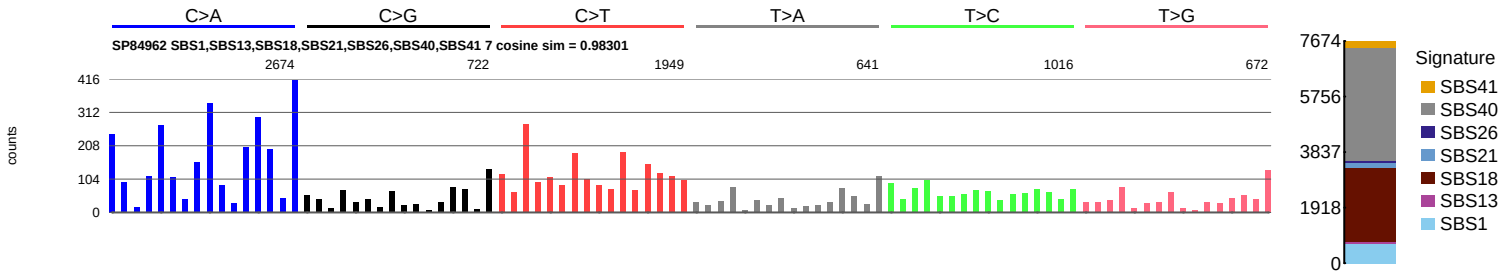
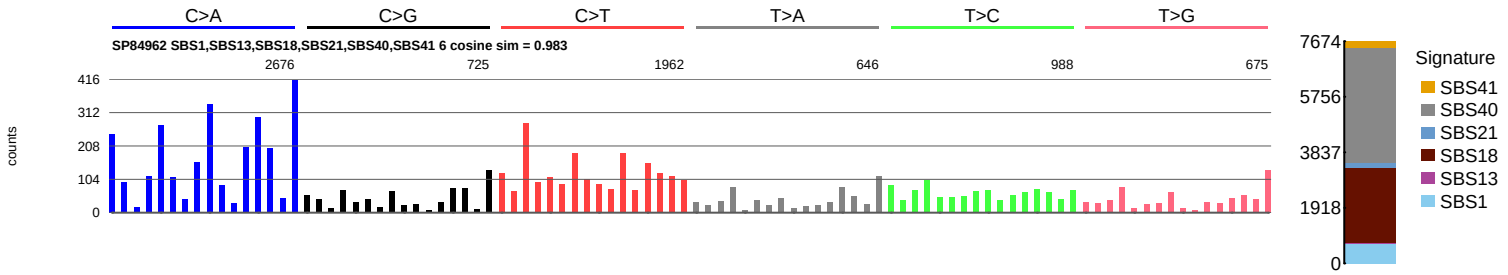
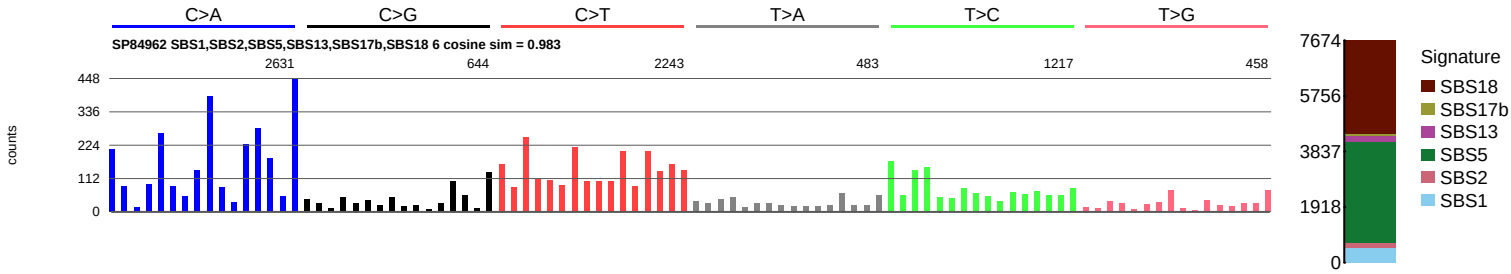
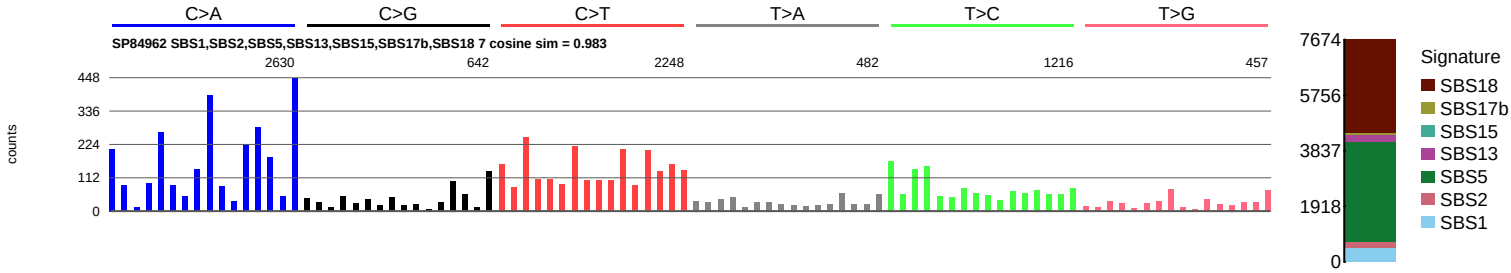
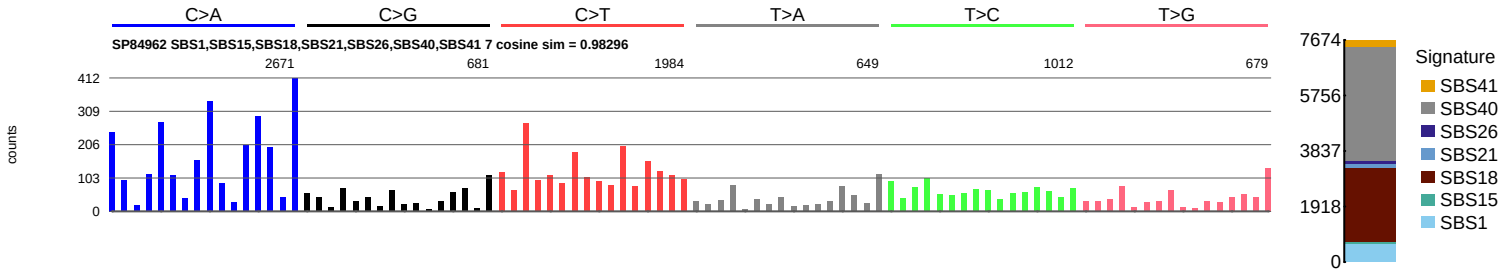
SBS18
SBS13
SBS1

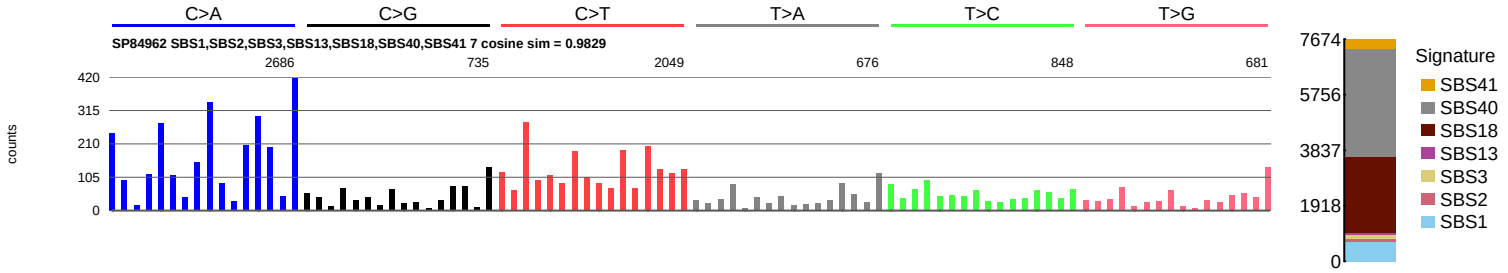

C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS18,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.98301
2674
722
1949
641
1016
672
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS18,SBS21,SBS40,SBS41 6 cosine sim = 0.983
2676
725
1962
646
988
675
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18 6 cosine sim = 0.983
2631
644
2243
483
1217
458
counts
448
336
224
112
0
7674
5756
3837
1918
0
Signature
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18 7 cosine sim = 0.983
2630
642
2248
482
1216
457
counts
448
336
224
112
0
7674
5756
3837
1918
0
Signature
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.98296
2671
681
1984
649
1012
679
counts
412
309
206
103
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS13,SBS18,SBS40,SBS41 7 cosine sim = 0.9829
2686
735
2049
676
848
681
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS18
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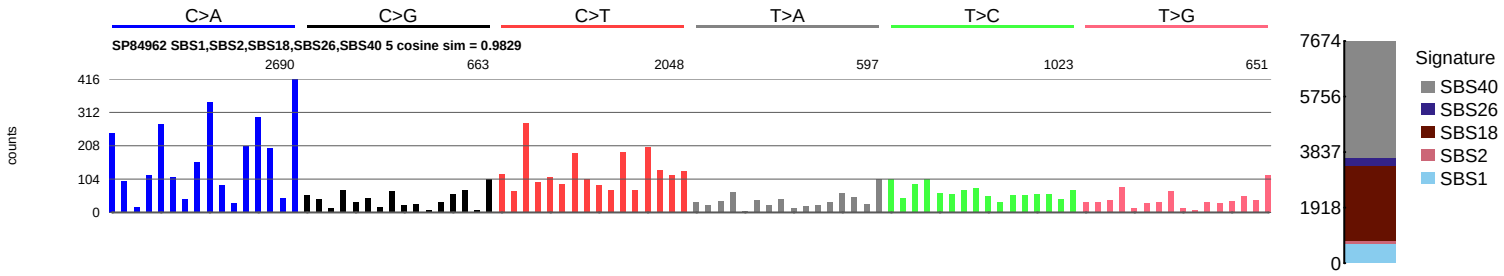
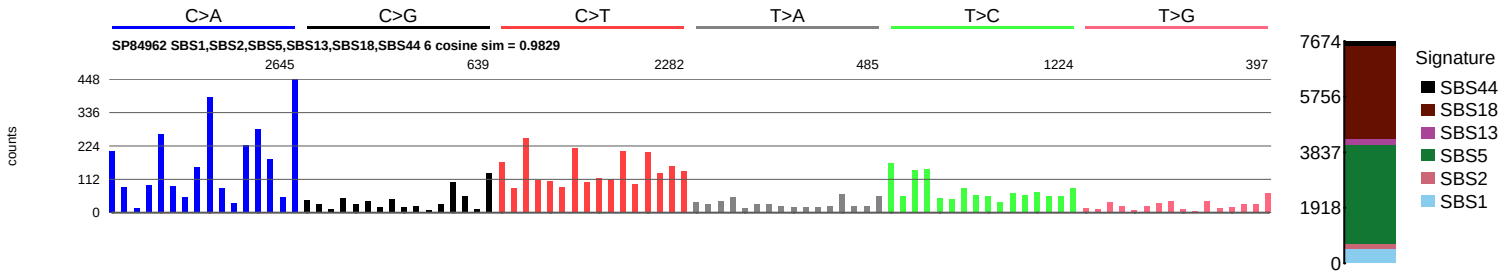
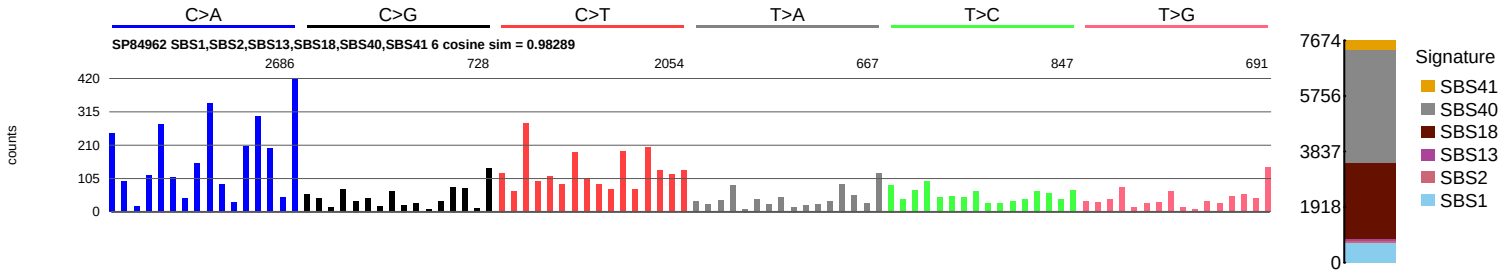
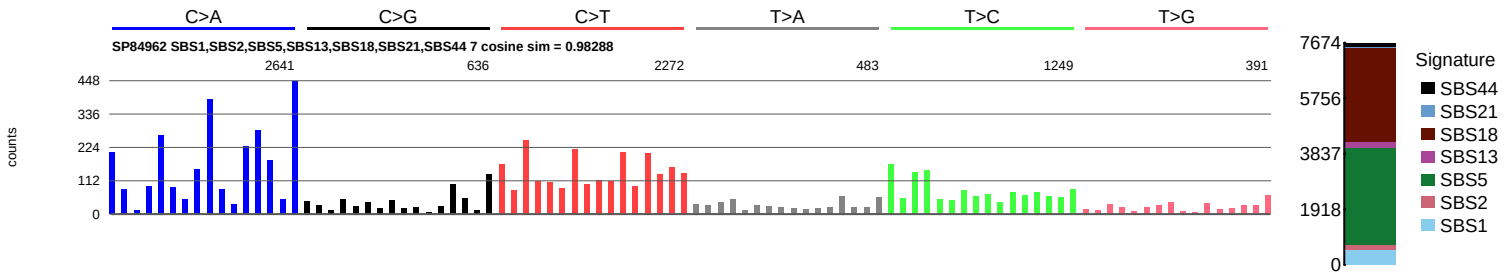
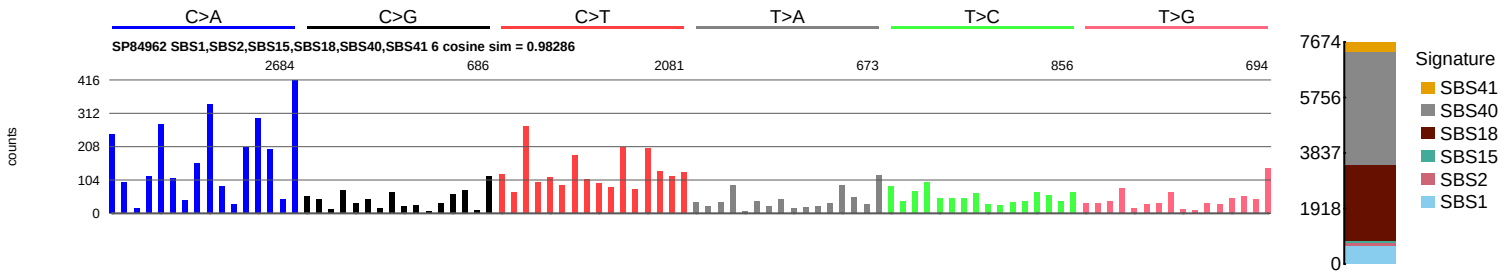
SBS3
SBS2
SBS1

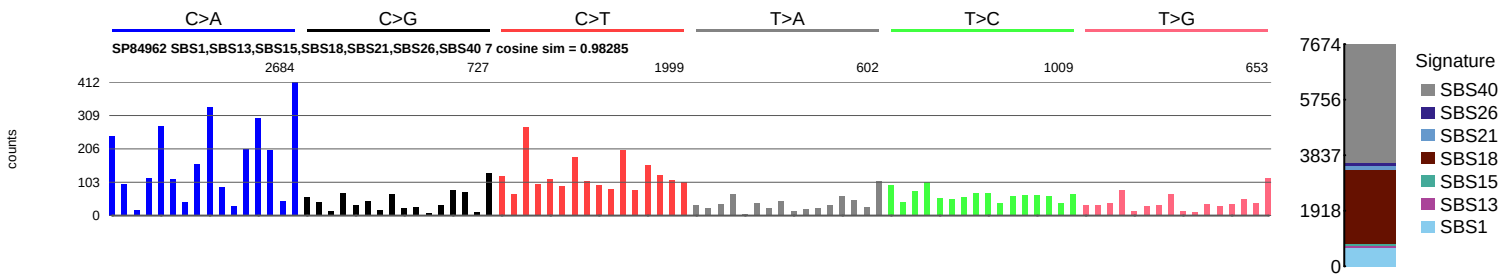

C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS18,SBS26,SBS40 5 cosine sim = 0.9829
2690
663
2048
597
1023
651
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS40
SBS26
SBS18
SBS2
SBS1

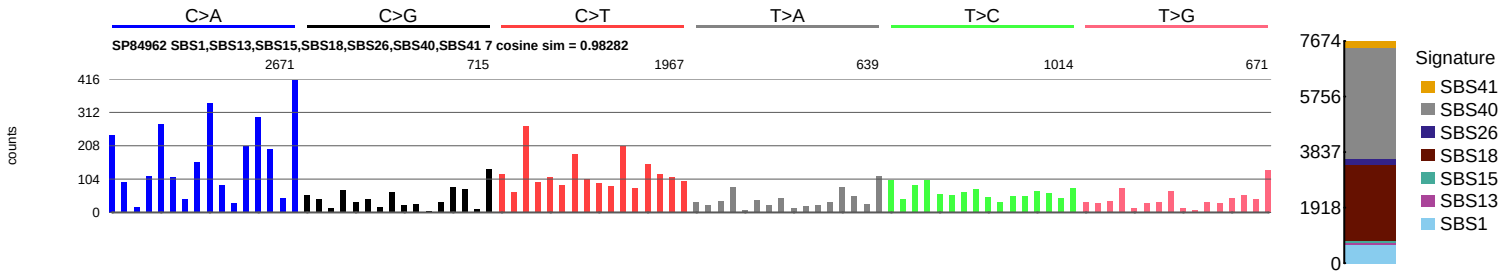

C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS18,SBS44 6 cosine sim = 0.9829
2645
639
2282
485
1224
397
counts
448
336
224
112
0
7674
5756
3837
1918
0
Signature
SBS44
SBS18
SBS13
SBS5
SBS2
SBS1

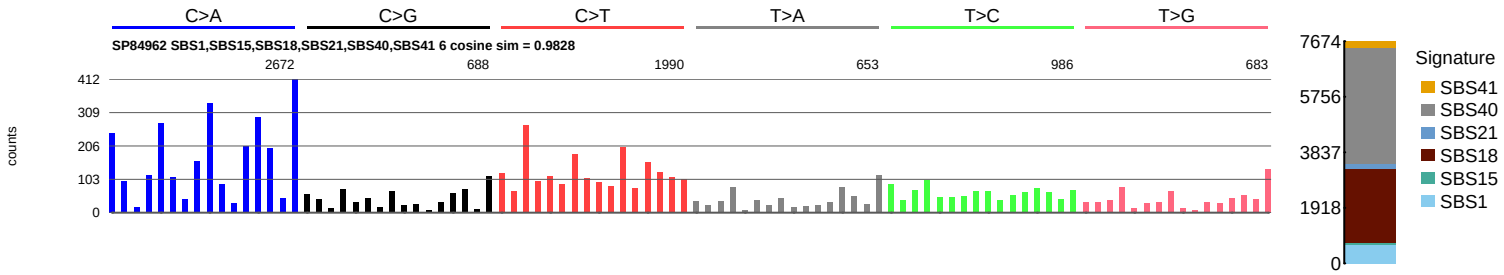

C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS13,SBS18,SBS40,SBS41 6 cosine sim = 0.98289
2686
728
2054
667
847
691
counts
420
315
210
105
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS18
SBS13
SBS2
SBS1

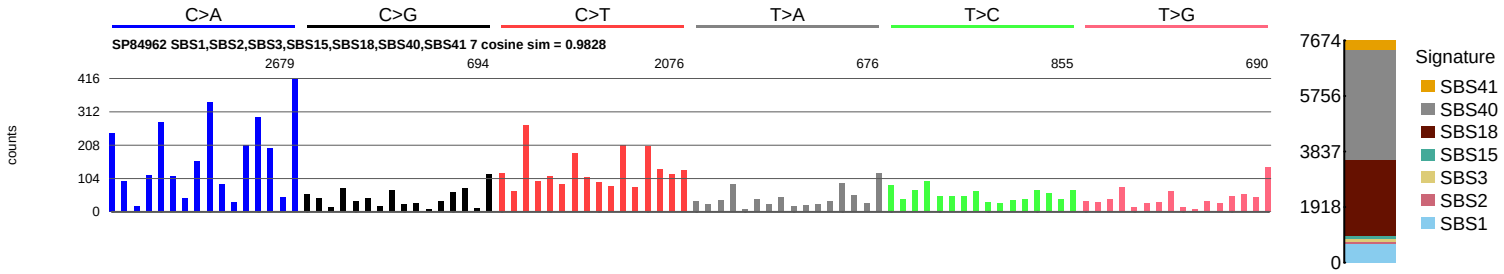

C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS44 7 cosine sim = 0.98288
2641
636
2272
483
1249
391
counts
448
336
224
112
0
7674
5756
3837
1918
0
Signature
SBS44
SBS21
SBS18
SBS13
SBS5
SBS2
SBS1

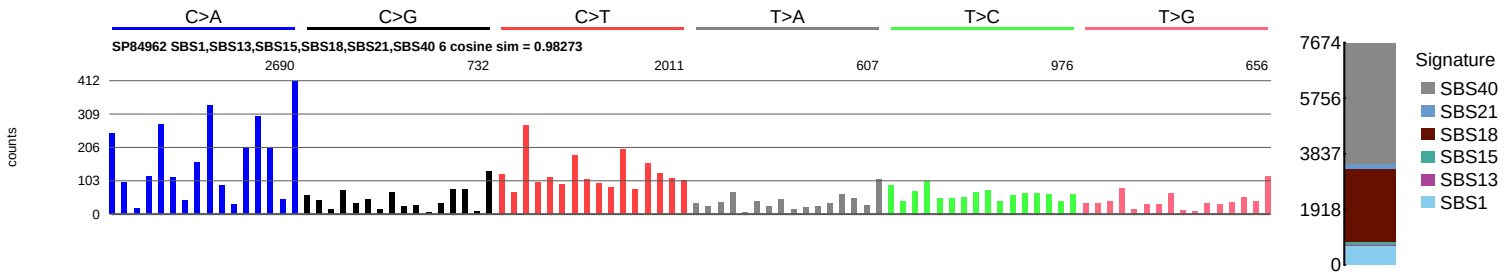

C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS15,SBS18,SBS40,SBS41 6 cosine sim = 0.98286
2684
686
2081
673
856
694
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS2
SBS1

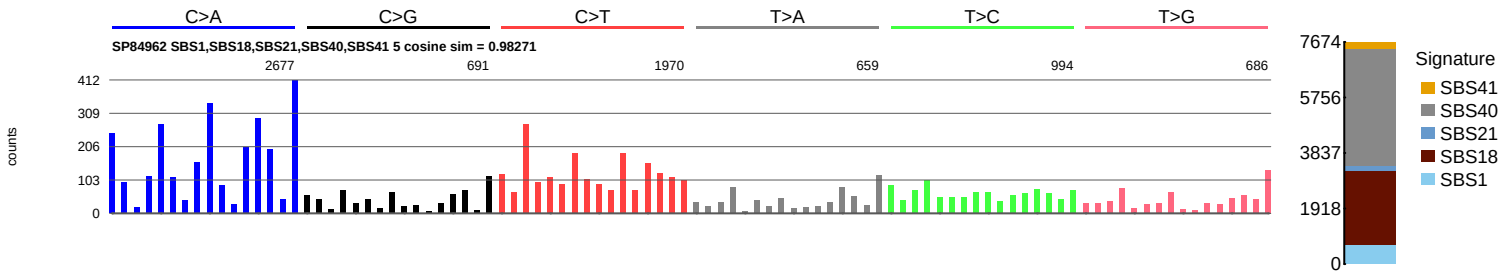

C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98285
2684
727
1999
602
1009
653
counts
412
309
206
103
0
7674
5756
3837
1918
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS1

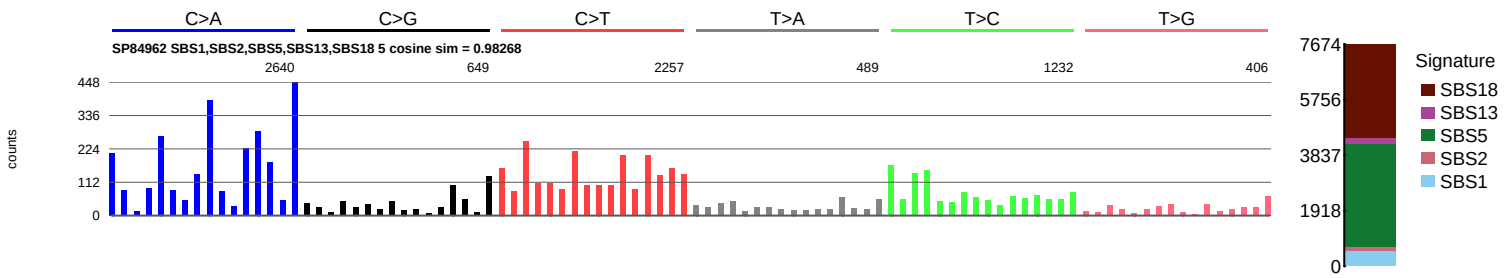

C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98282
2671
715
1967
639
1014
671
416
312
208
104
0
counts
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS13
SBS1

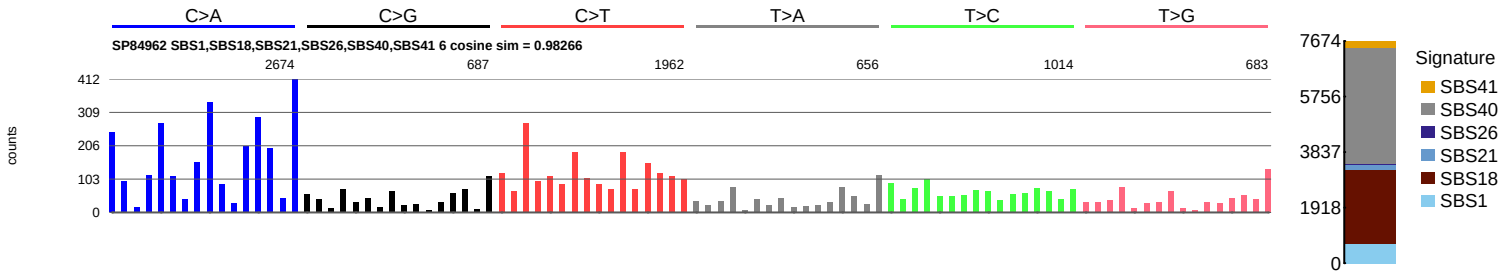

C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS15,SBS18,SBS21,SBS40,SBS41 6 cosine sim = 0.9828
2672
688
1990
653
986
683
412
309
206
103
0
counts
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS1

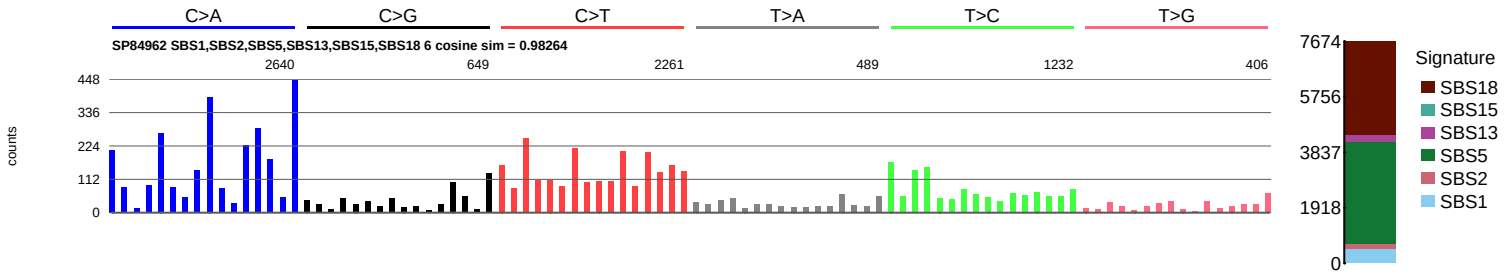

C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS15,SBS18,SBS40,SBS41 7 cosine sim = 0.9828
2679
694
2076
676
855
690
416
312
208
104
0
counts
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS3
SBS2
SBS1

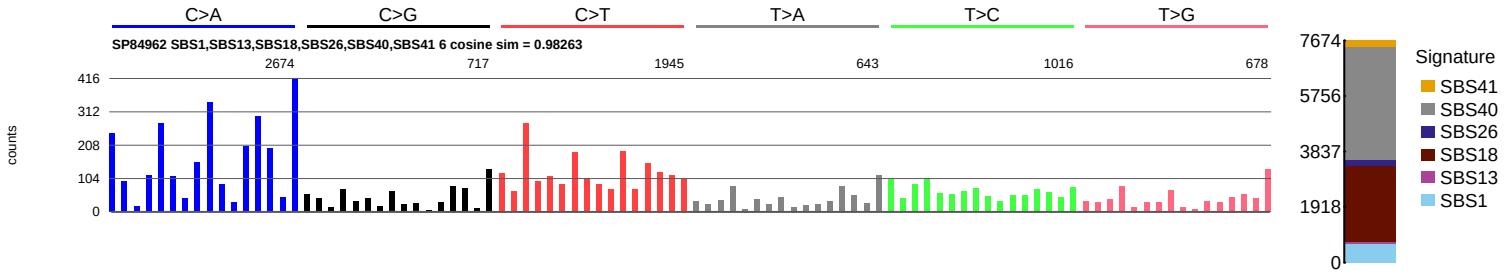

C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS15,SBS18,SBS21,SBS40 6 cosine sim = 0.98273
2690
732
2011
607
976
656
412
309
206
103
0
counts
7674
5756
3837
1918
0
Signature
SBS40
SBS21
SBS18
SBS15
SBS13
SBS1

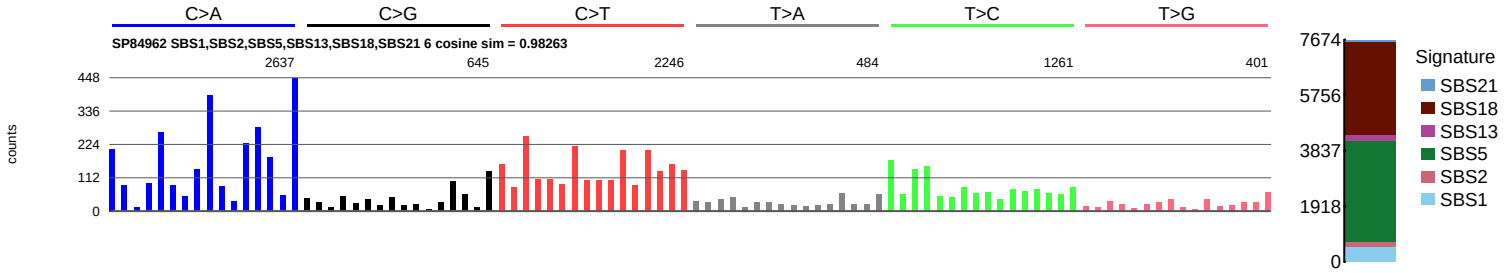

C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS18,SBS21,SBS40,SBS41 5 cosine sim = 0.98271
2677
691
1970
659
994
686
412
309
206
103
0
counts
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS18 5 cosine sim = 0.98268
2640
649
2257
489
1232
406
448
336
224
112
0
counts
7674
5756
3837
1918
0
Signature
SBS18
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS18,SBS21,SBS26,SBS40,SBS41 6 cosine sim = 0.98266
2674
687
1962
656
1014
683
counts
412
309
206
103
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18 6 cosine sim = 0.98264
2640
649
2261
489
1232
406
counts
448
336
224
112
0
7674
5756
3837
1918
0
Signature
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS18,SBS26,SBS40,SBS41 6 cosine sim = 0.98263
2674
717
1945
643
1016
678
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21 6 cosine sim = 0.98263
2637
645
2246
484
1261
401
counts
448
336
224
112
0
7674
5756
3837
1918
0
Signature
SBS21
SBS18
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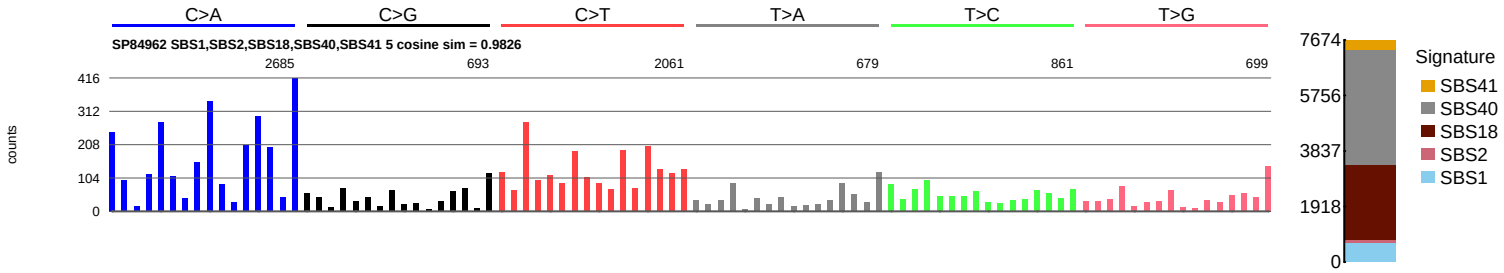
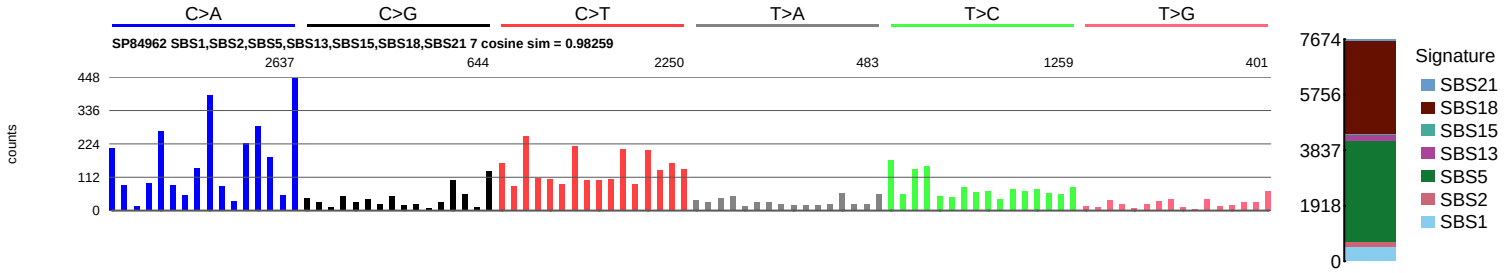
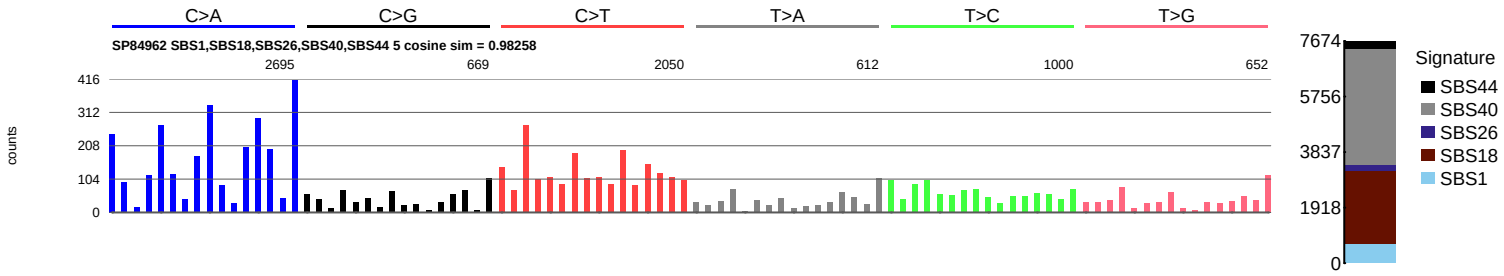
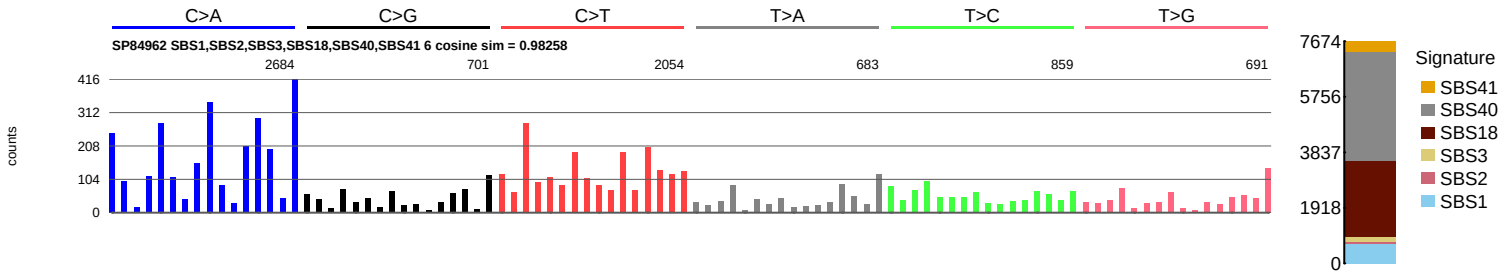
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS18,SBS40,SBS41 5 cosine sim = 0.9826
2685
693
2061
679
861
699
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21 7 cosine sim = 0.98259
2637
644
2250
483
1259
401
counts
448
336
224
112
0
7674
5756
3837
1918
0
Signature
SBS21
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS18,SBS26,SBS40,SBS44 5 cosine sim = 0.98258
2695
669
2050
612
1000
652
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS18,SBS40,SBS41 6 cosine sim = 0.98258
2684
701
2054
683
859
691
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
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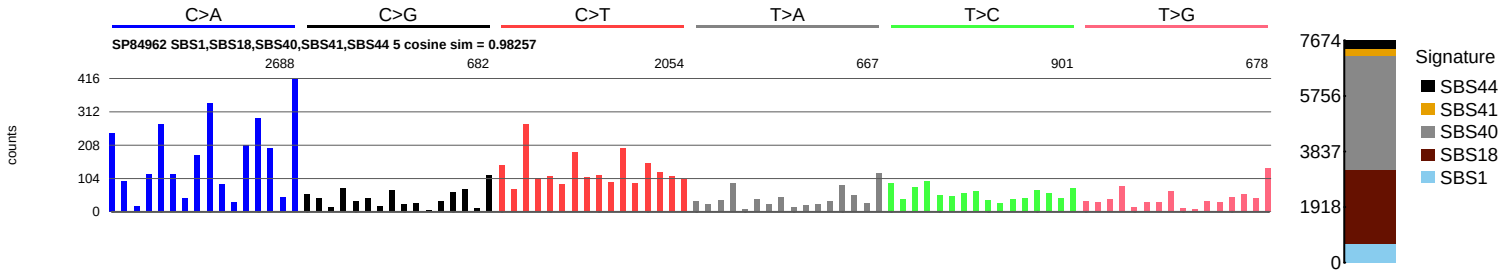
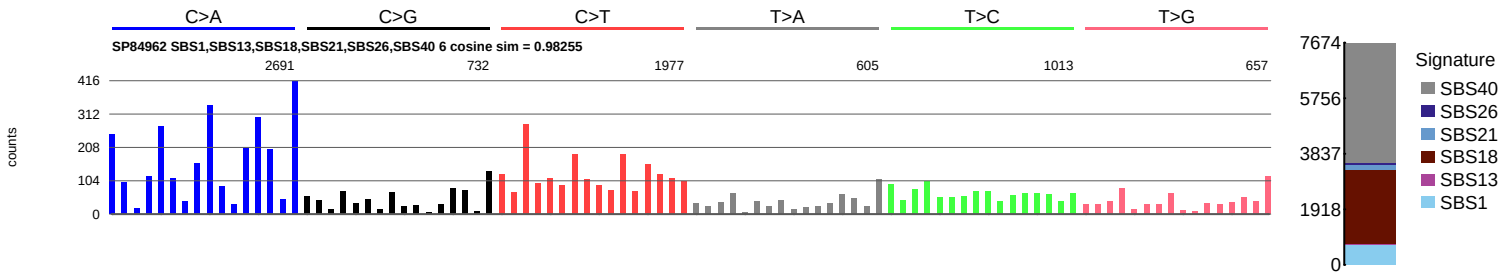
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS18,SBS40,SBS41,SBS44 5 cosine sim = 0.98257
2688
682
2054
667
901
678
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS18,SBS21,SBS26,SBS40 6 cosine sim = 0.98255
2691
732
1977
605
1013
657
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS40
SBS26
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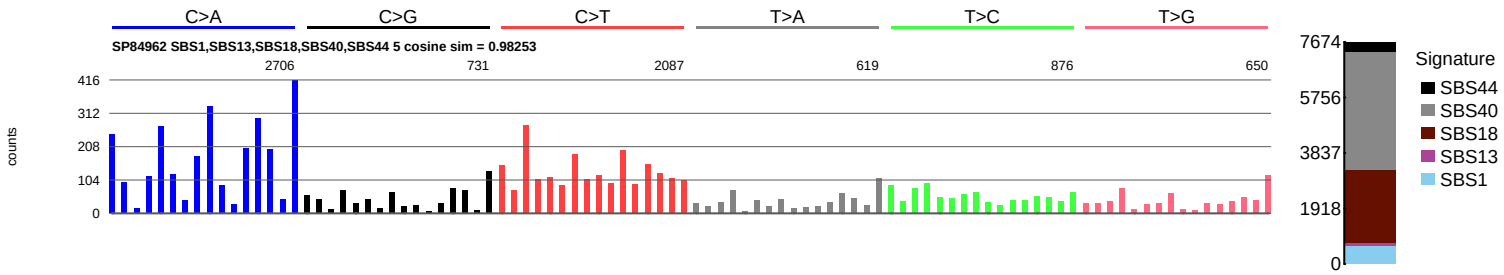
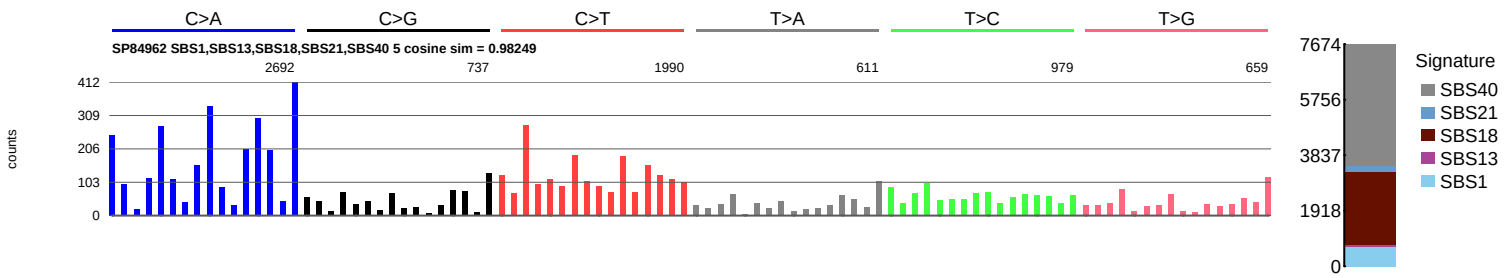
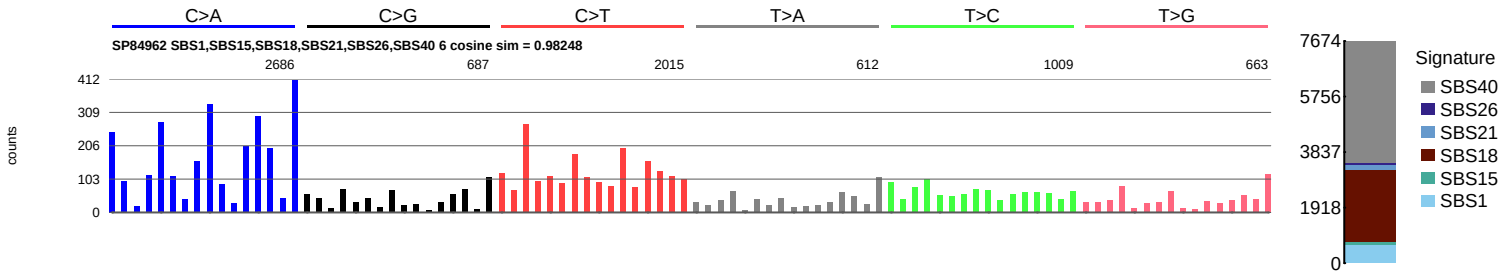
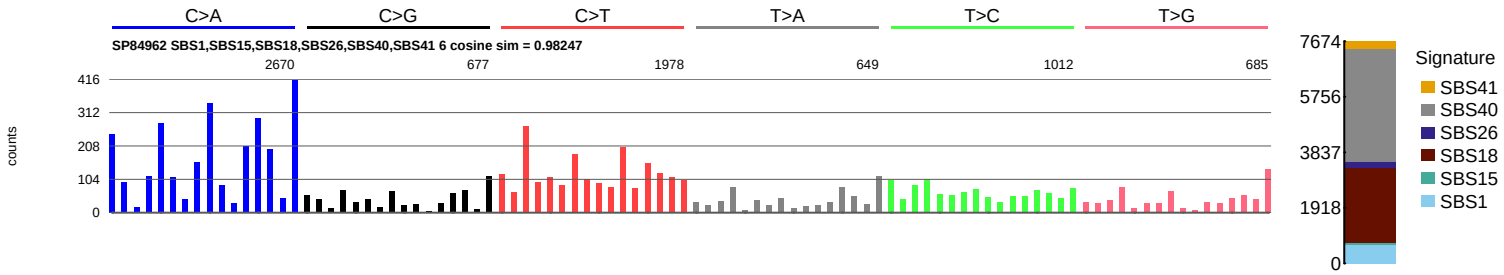
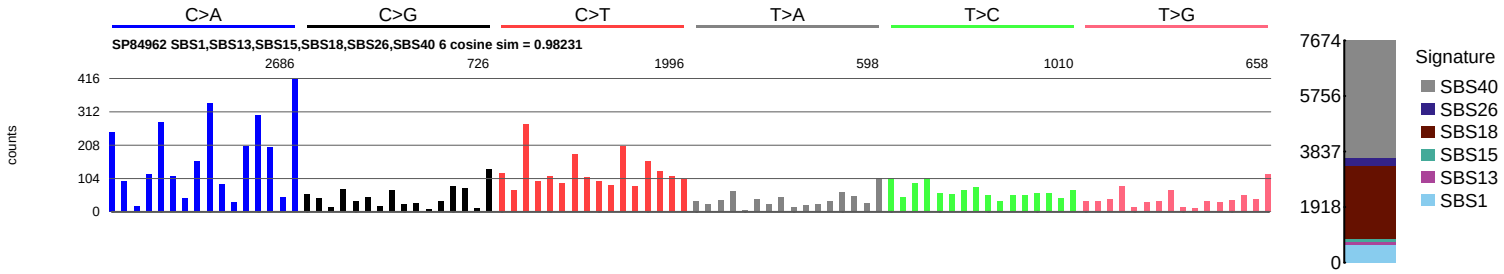
SBS18
SBS13
SBS1

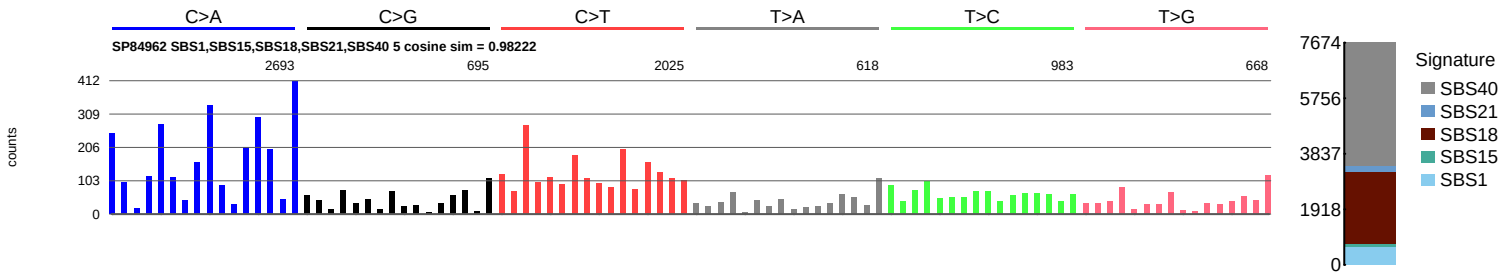

C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS18,SBS40,SBS44 5 cosine sim = 0.98253
2706
731
2087
619
876
650
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS18
SBS13
SBS1

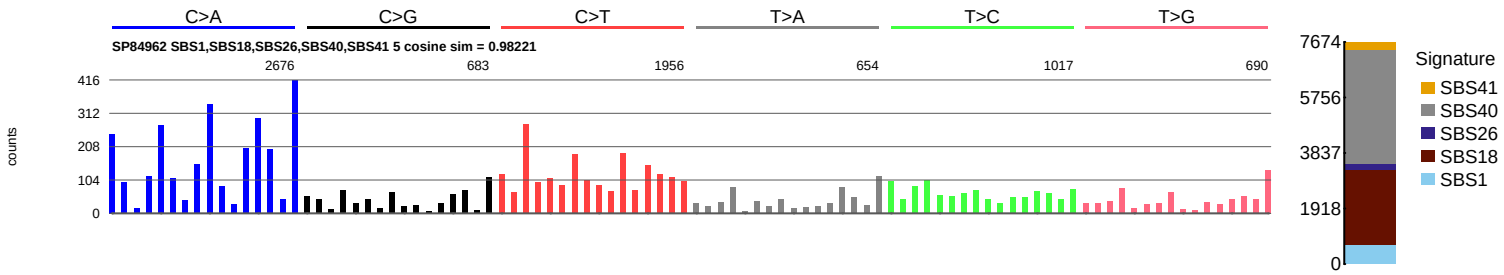

C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS18,SBS21,SBS40 5 cosine sim = 0.98249
2692
737
1990
611
979
659
counts
412
309
206
103
0
7674
5756
3837
1918
0
Signature
SBS40
SBS21
SBS18
SBS13
SBS1

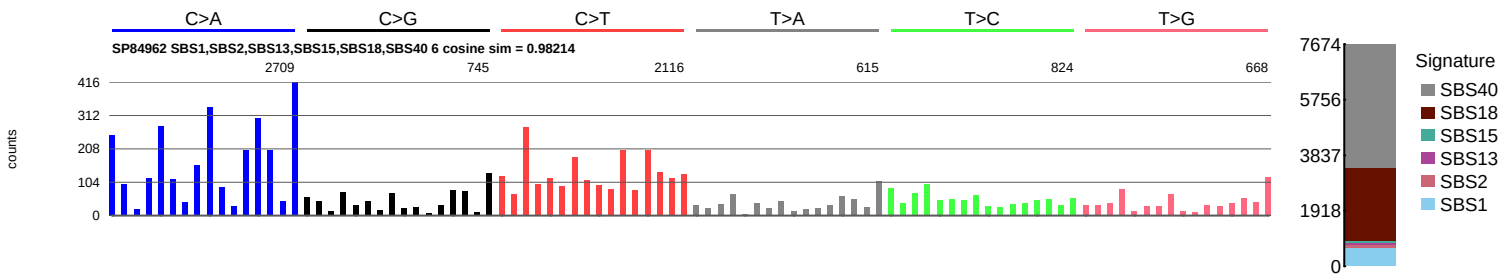

C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS15,SBS18,SBS21,SBS26,SBS40 6 cosine sim = 0.98248
2686
687
2015
612
1009
663
counts
412
309
206
103
0
7674
5756
3837
1918
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS15,SBS18,SBS26,SBS40,SBS41 6 cosine sim = 0.98247
2670
677
1978
649
1012
685
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS15,SBS18,SBS26,SBS40 6 cosine sim = 0.98231
2686
726
1996
598
1010
658
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS15,SBS18,SBS21,SBS40 5 cosine sim = 0.98222
2693
695
2025
618
983
668
counts
412
309
206
103
0
7674
5756
3837
1918
0
Signature
SBS40
SBS21
SBS18
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS18,SBS26,SBS40,SBS41 5 cosine sim = 0.98221
2676
683
1956
654
1017
690
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS13,SBS15,SBS18,SBS40 6 cosine sim = 0.98214
2709
745
2116
615
824
668
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS40
SBS18
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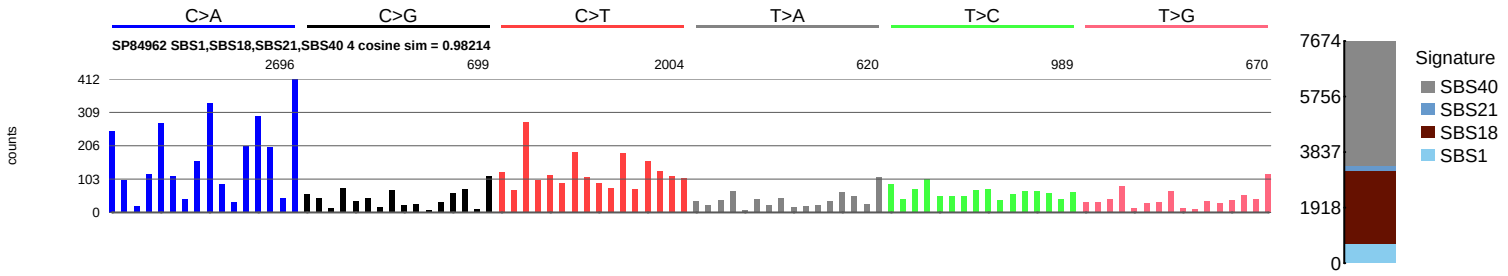
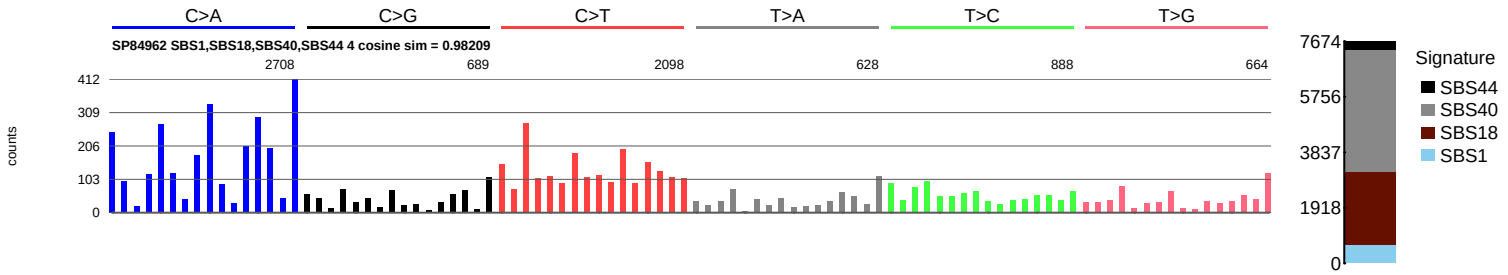
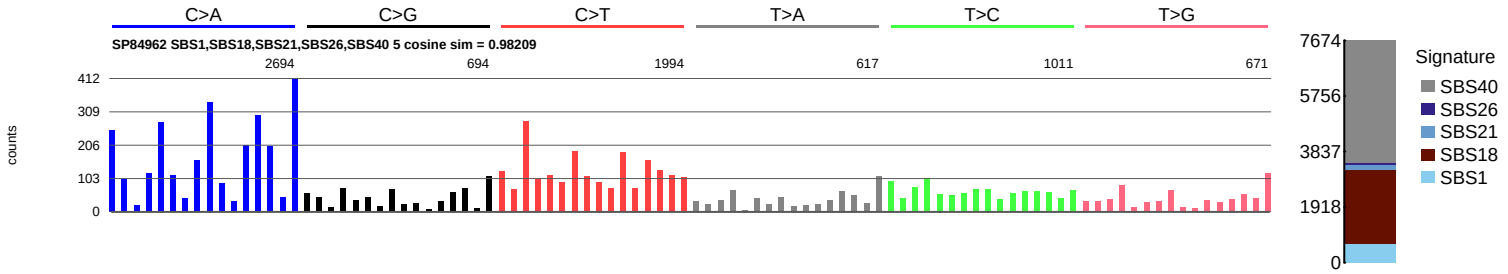
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS18,SBS21,SBS40 4 cosine sim = 0.98214
2696
699
2004
620
989
670
counts
412
309
206
103
0
7674
5756
3837
1918
0
Signature
SBS40
SBS21
SBS18
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS18,SBS40,SBS44 4 cosine sim = 0.98209
2708
689
2098
628
888
664
counts
412
309
206
103
0
7674
5756
3837
1918
0
Signature
SBS44
SBS40
SBS18
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS18,SBS21,SBS26,SBS40 5 cosine sim = 0.98209
2694
694
1994
617
1011
671
counts
412
309
206
103
0
7674
5756
3837
1918
0
Signature
SBS40
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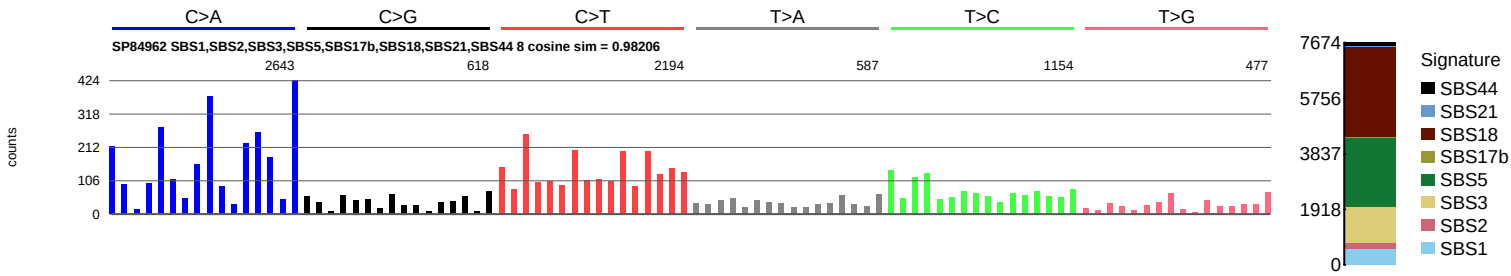
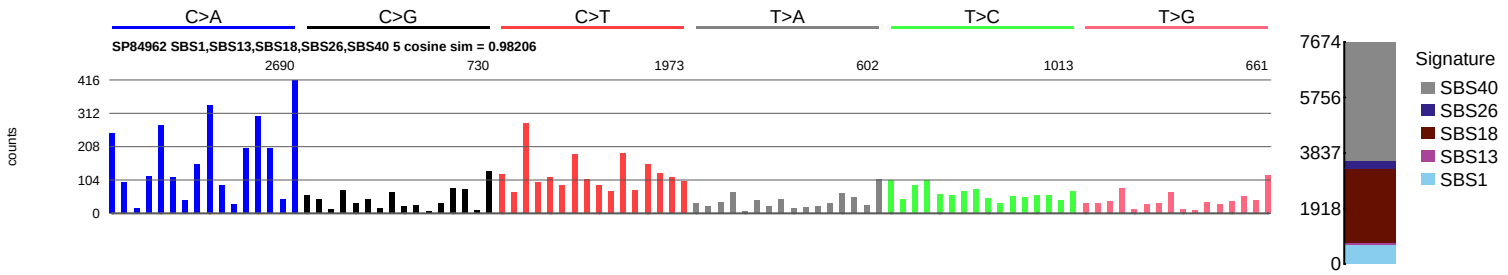
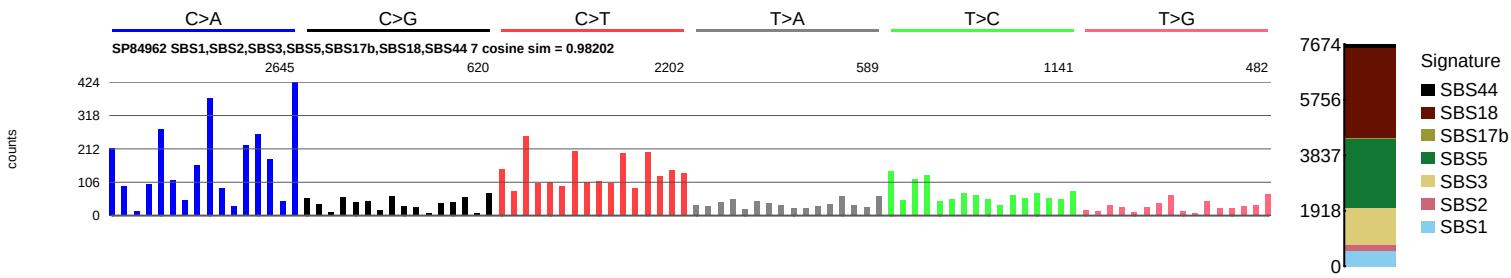
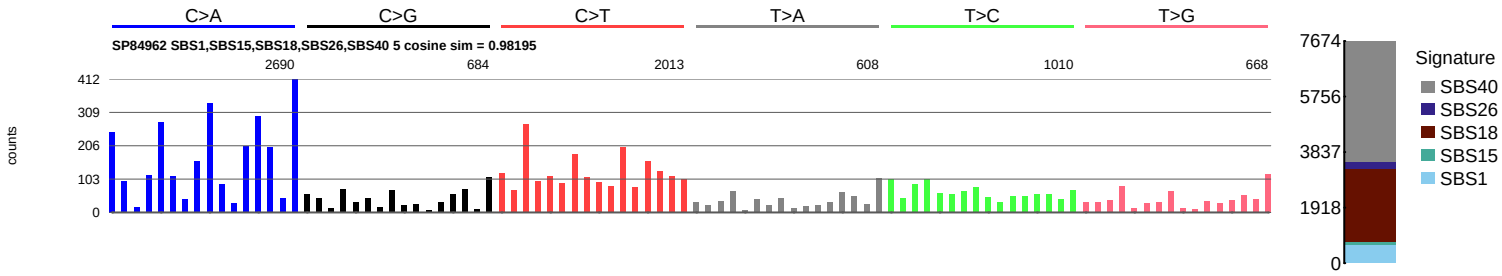
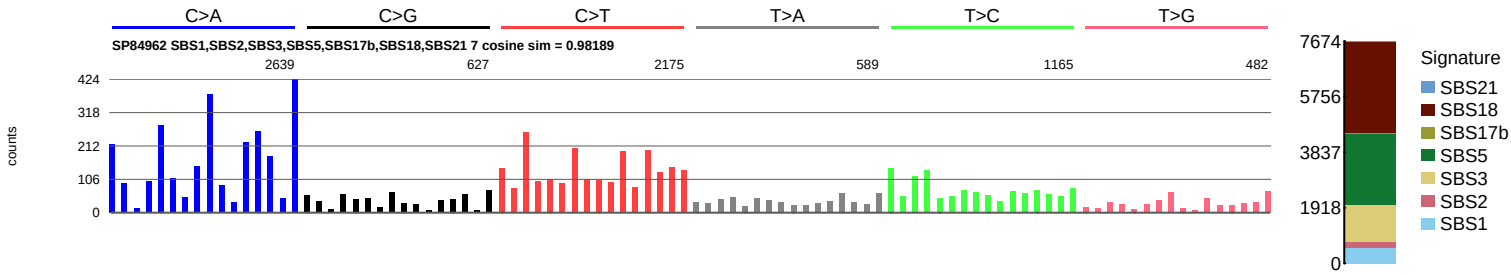
SBS21
SBS18
SBS1

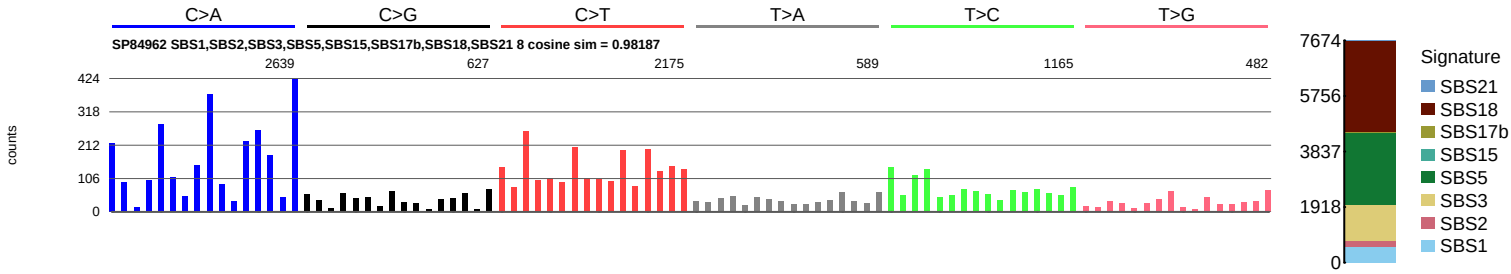

C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98206
2643
618
2194
587
1154
477
counts
424
318
212
106
0
7674
5756
3837
1918
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

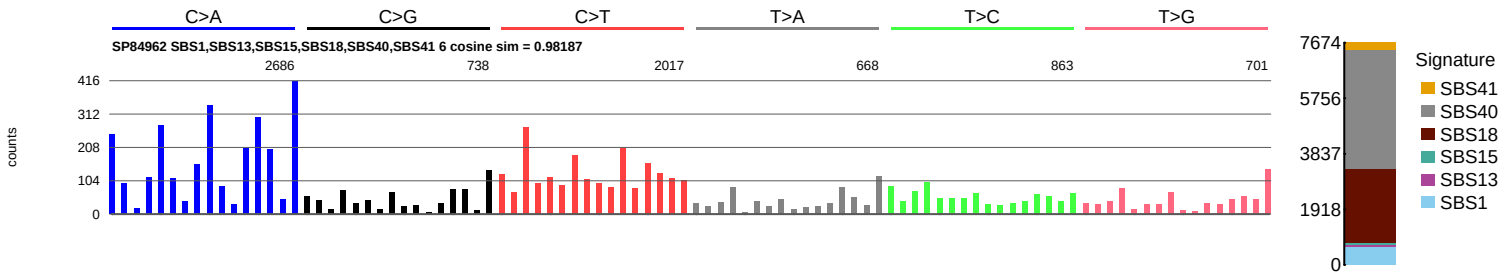

C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS18,SBS26,SBS40 5 cosine sim = 0.98206
2690
730
1973
602
1013
661
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS40
SBS26
SBS18
SBS13
SBS1

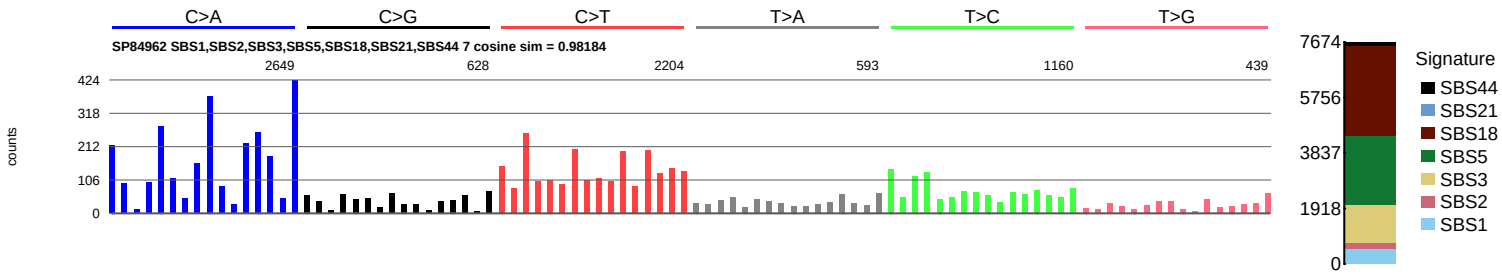

C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS44 7 cosine sim = 0.98202
2645
620
2202
589
1141
482
counts
424
318
212
106
0
7674
5756
3837
1918
0
Signature
SBS44
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS15,SBS18,SBS26,SBS40 5 cosine sim = 0.98195
2690
684
2013
608
1010
668
counts
412
309
206
103
0
7674
5756
3837
1918
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21 7 cosine sim = 0.98189
2639
627
2175
589
1165
482
counts
424
318
212
106
0
7674
5756
3837
1918
0
Signature
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21 8 cosine sim = 0.98187
2639
627
2175
589
1165
482
counts
424
318
212
106
0
7674
5756
3837
1918
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS15,SBS18,SBS40,SBS41 6 cosine sim = 0.98187
2686
738
2017
668
863
701
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS18,SBS21,SBS44 7 cosine sim = 0.98184
2649
628
2204
593
1160
439
counts
424
318
212
106
0
7674
5756
3837
1918
0
Signature
SBS44
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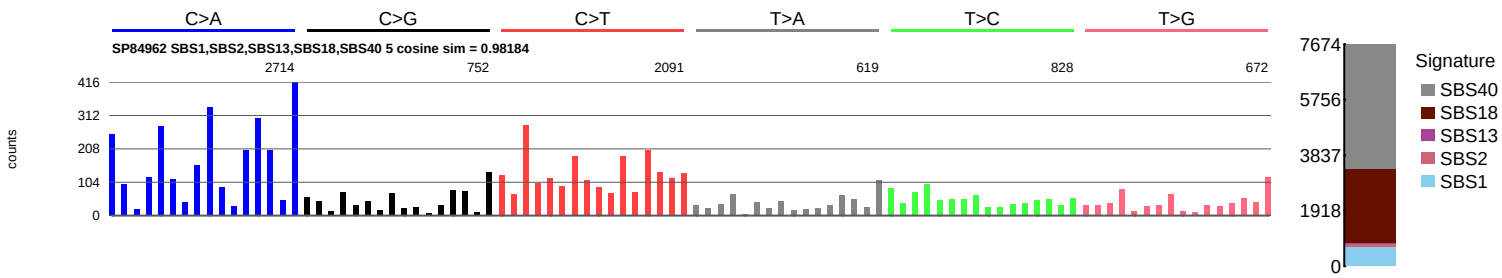
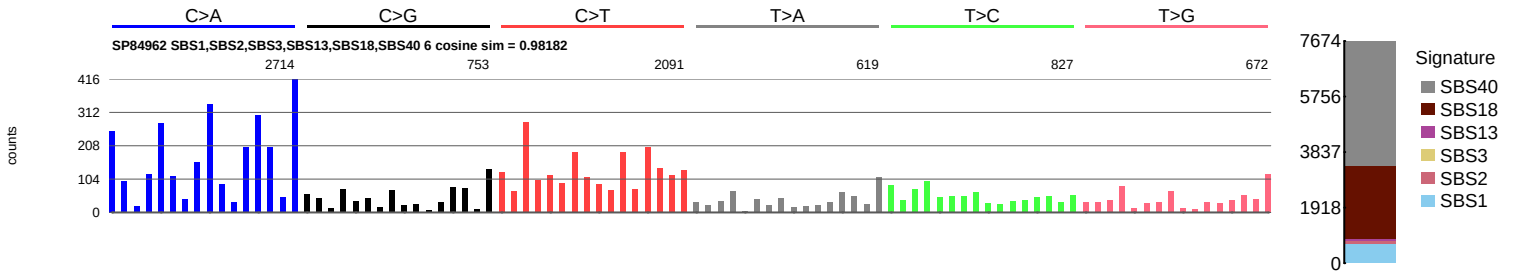
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS13,SBS18,SBS40 5 cosine sim = 0.98184
2714
752
2091
619
828
672
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS40
SBS18
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS13,SBS18,SBS40 6 cosine sim = 0.98182
2714
753
2091
619
827
672
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS40
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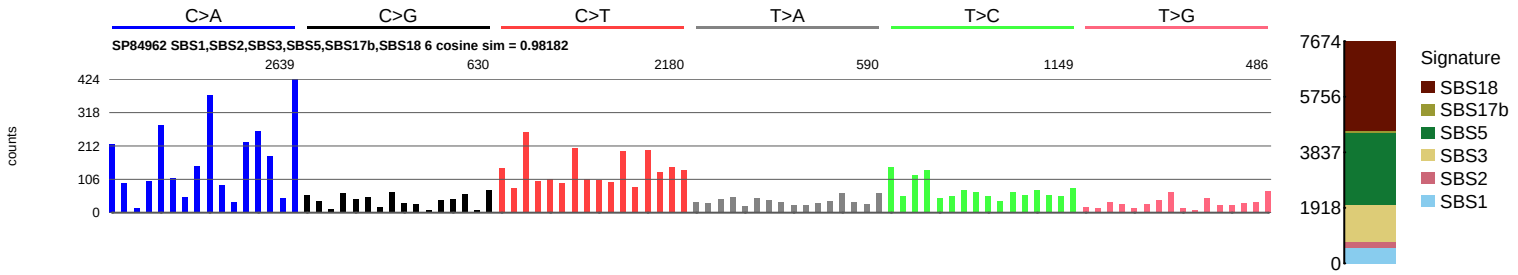
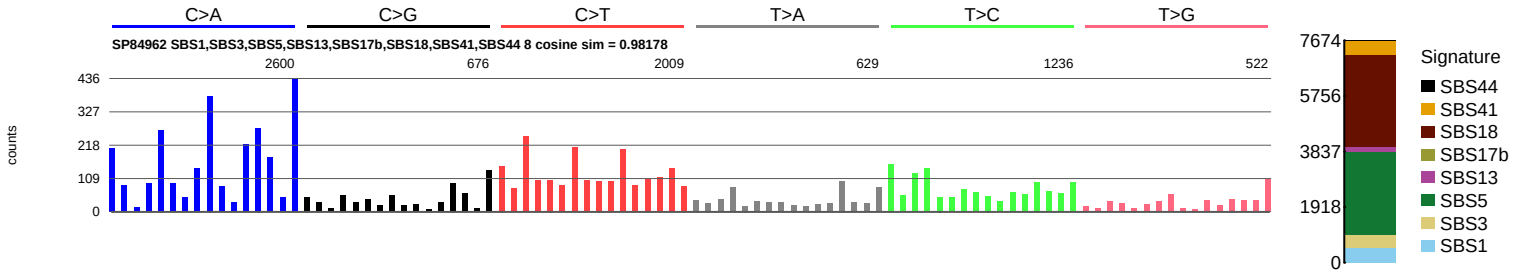
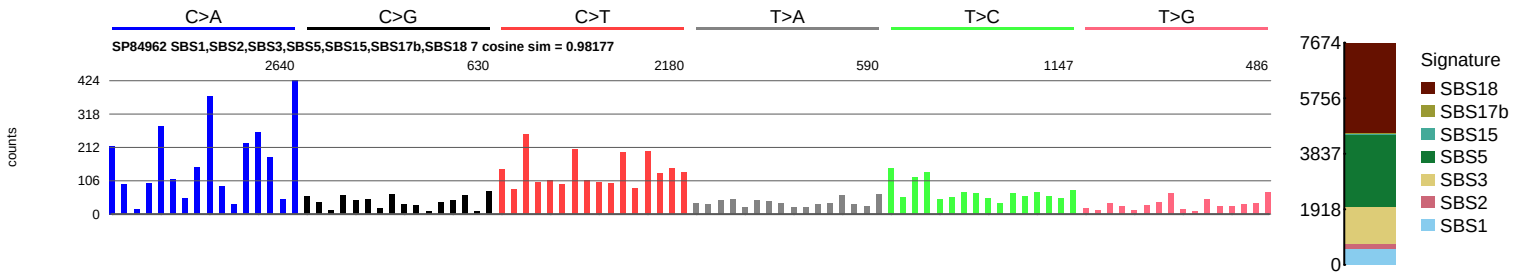
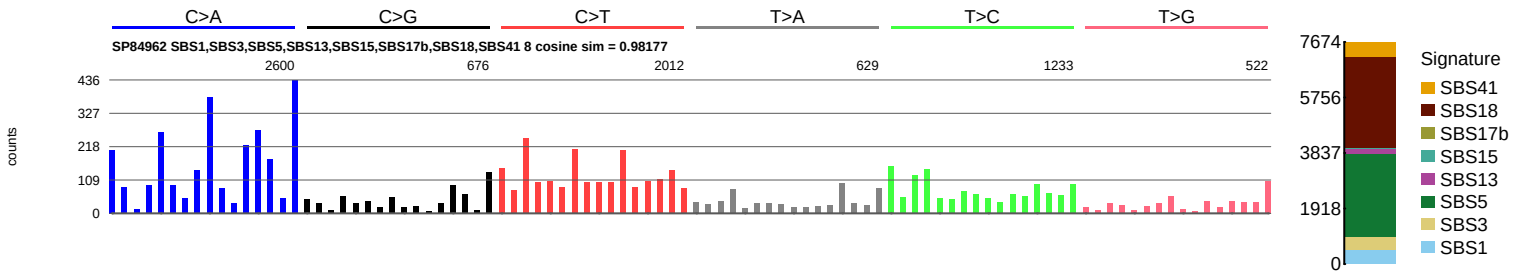
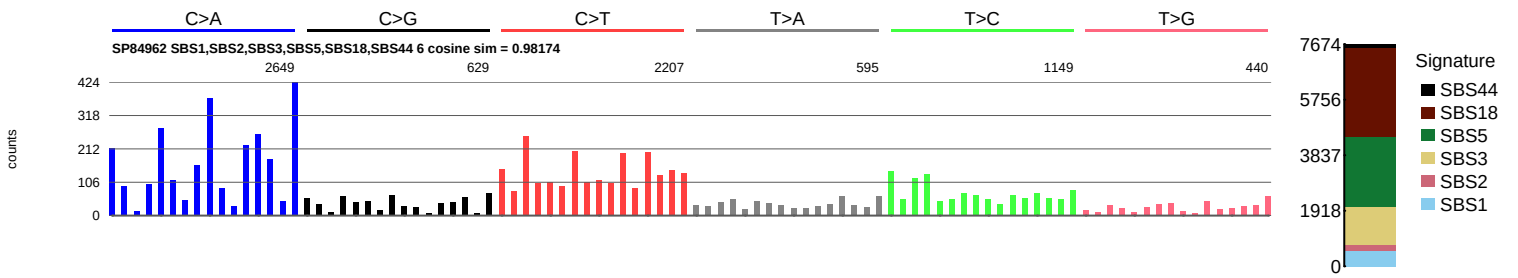
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18 6 cosine sim = 0.98182
2639
630
2180
590
1149
486
counts
424
318
212
106
0
7674
5756
3837
1918
0
Signature
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98178
2600
676
2009
629
1236
522
counts
436
327
218
109
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18 7 cosine sim = 0.98177
2640
630
2180
590
1147
486
counts
424
318
212
106
0
7674
5756
3837
1918
0
Signature
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.98177
2600
676
2012
629
1233
522
counts
436
327
218
109
0
7674
5756
3837
1918
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS18,SBS44 6 cosine sim = 0.98174
2649
629
2207
595
1149
440
counts
424
318
212
106
0
7674
5756
3837
1918
0
Signature
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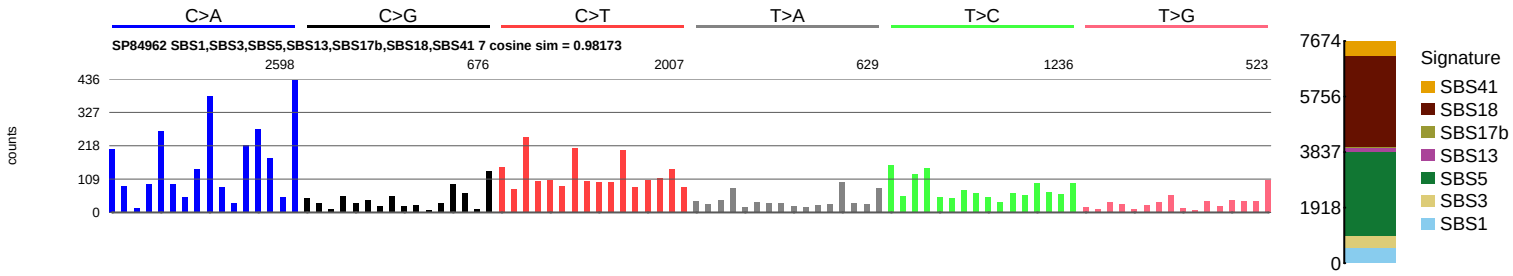
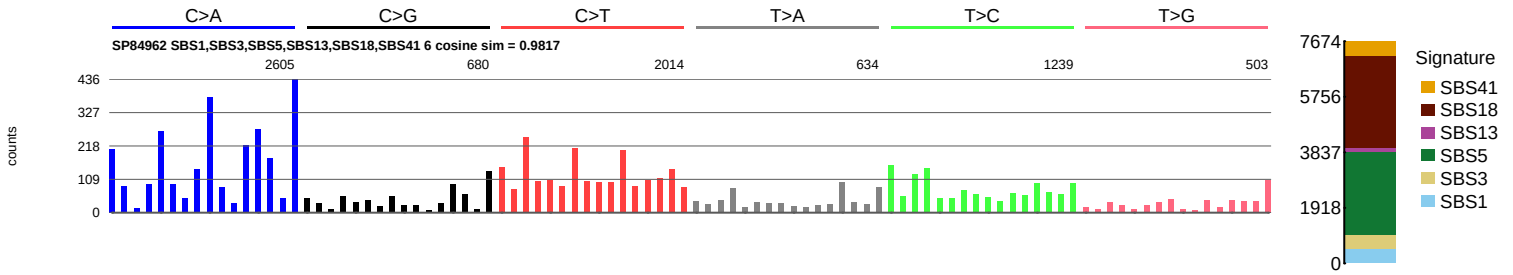
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41 7 cosine sim = 0.98173
2598
676
2007
629
1236
523
counts
436
327
218
109
0
7674
5756
3837
1918
0
Signature
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS1


C>A
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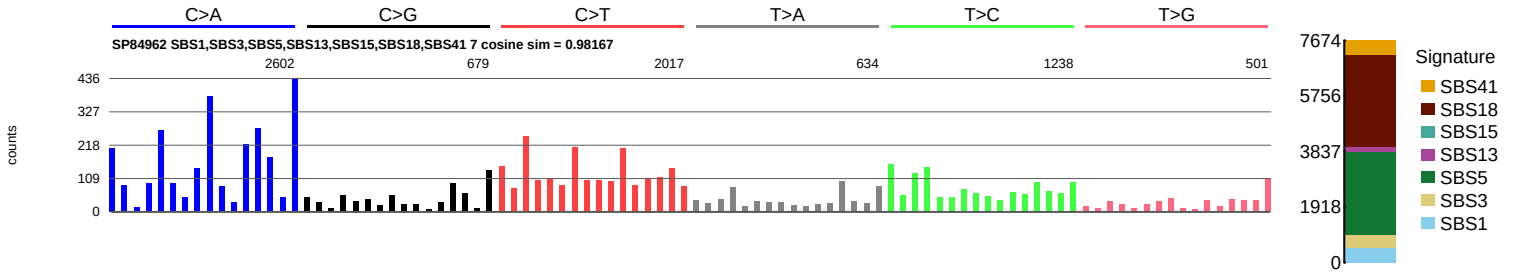
T>G
SP84962 SBS1,SBS3,SBS5,SBS13,SBS18,SBS41 6 cosine sim = 0.9817
2605
680
2014
634
1239
503
counts
436
327
218
109
0
7674
5756
3837
1918
0
Signature
SBS41
SBS18
SBS13
SBS5
SBS3
SBS1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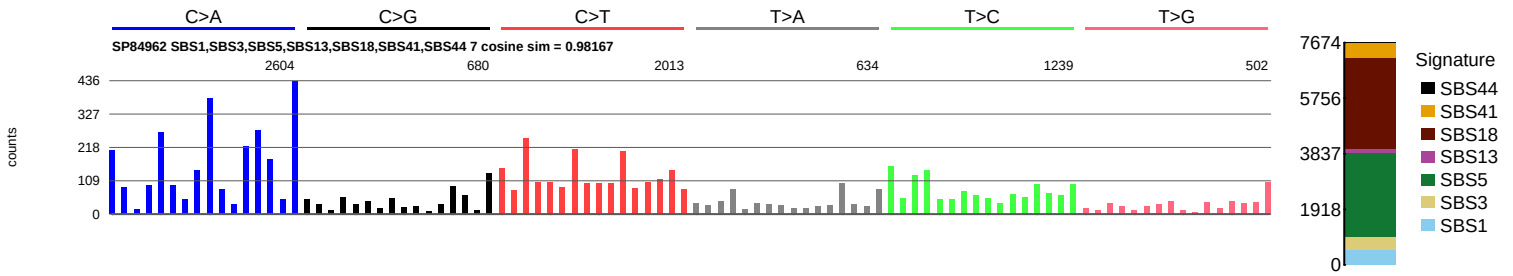
SP84962 SBS1,SBS3,SBS5,SBS13,SBS15,SBS18,SBS41 7 cosine sim = 0.98167
2602
679
2017
634
1238
501
counts
436
327
218
109
0
7674
5756
3837
1918
0
Signature
SBS41
SBS18
SBS15
SBS13
SBS5
SBS3
SBS1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C>G
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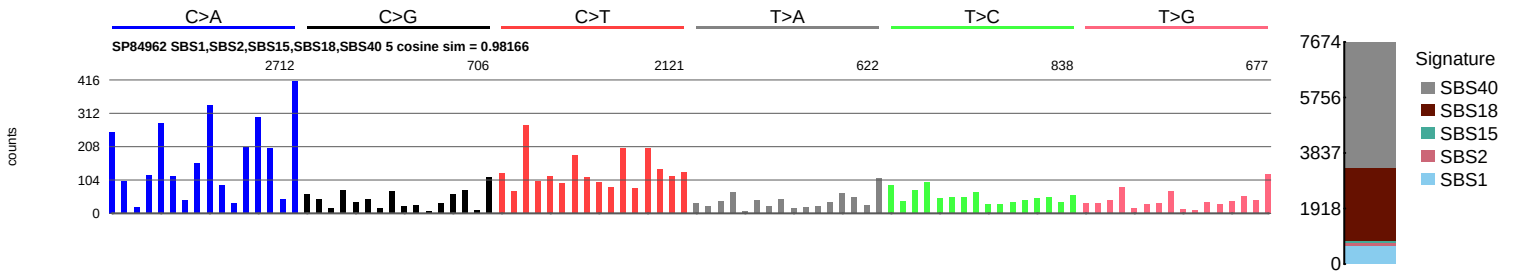
T>C
T>G
SP84962 SBS1,SBS3,SBS5,SBS13,SBS18,SBS41,SBS44 7 cosine sim = 0.98167
2604
680
2013
634
1239
502
counts
436
327
218
109
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS18
SBS13
SBS5
SBS3
SBS1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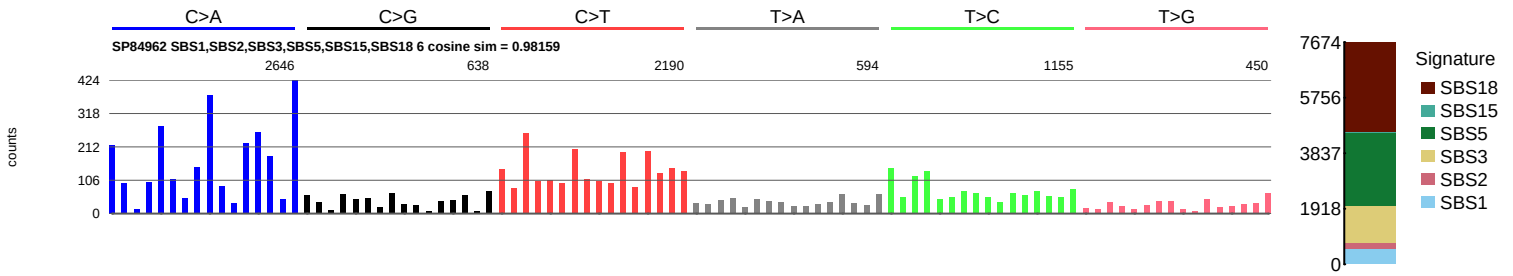
SP84962 SBS1,SBS2,SBS15,SBS18,SBS40 5 cosine sim = 0.98166
2712
706
2121
622
838
677
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS40
SBS18
SBS15
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18 6 cosine sim = 0.98159
2646
638
2190
594
1155
450
counts
424
318
212
106
0
7674
5756
3837
1918
0
Signature
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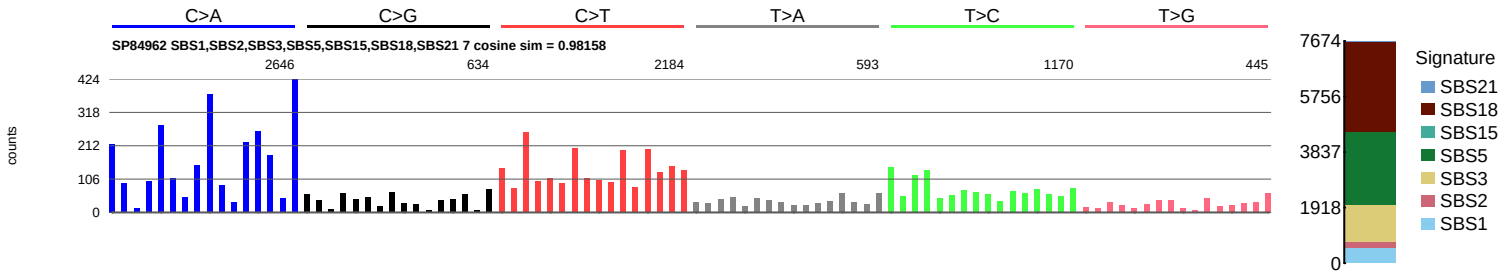
SBS2
SBS1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C>G
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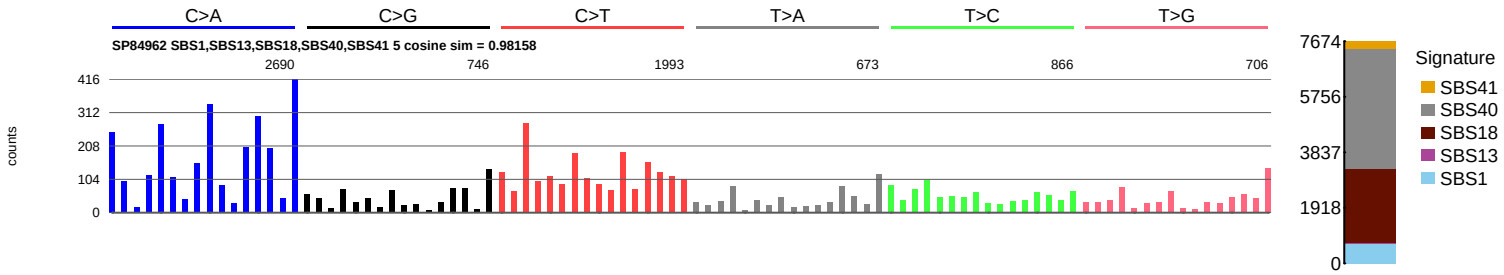
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS21 7 cosine sim = 0.98158
2646
634
2184
593
1170
445
counts
424
318
212
106
0
7674
5756
3837
1918
0
Signature
SBS21
SBS18
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
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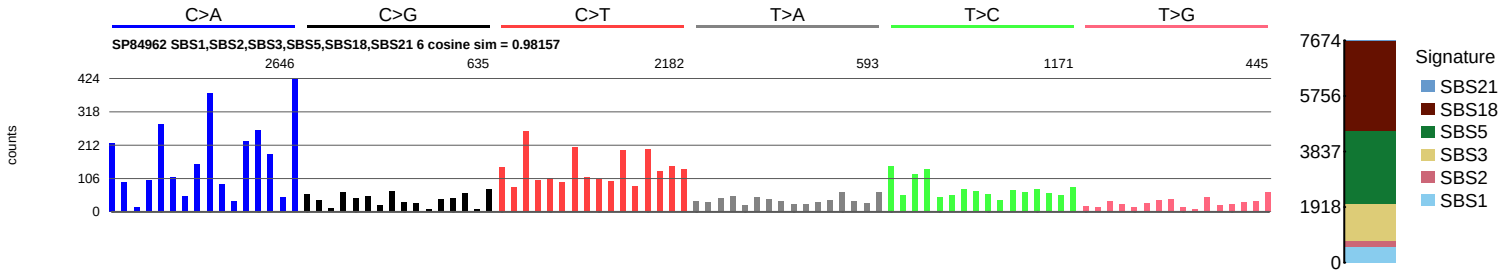
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS18,SBS40,SBS41 5 cosine sim = 0.98158
2690
746
1993
673
866
706
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS18
SBS13
SBS1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C>G
C>T
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T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS18,SBS21 6 cosine sim = 0.98157
2646
635
2182
593
1171
445
counts
424
318
212
106
0
7674
5756
3837
1918
0
Signature
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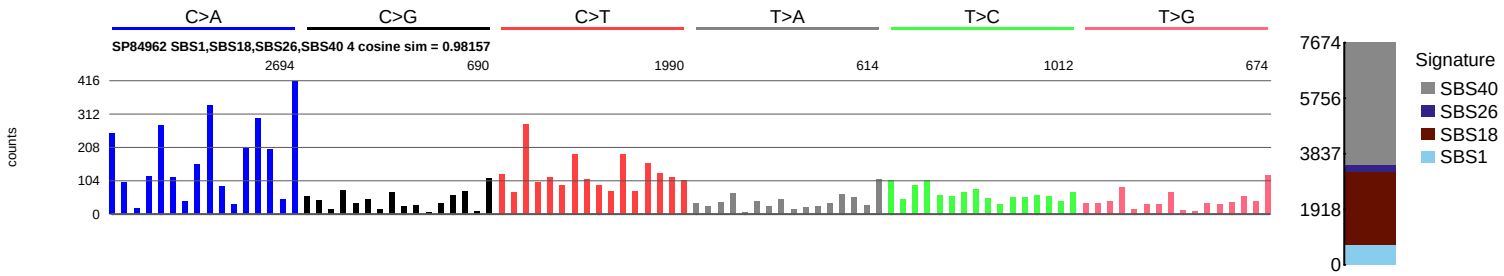
SBS2
SBS1


C>A
C>G
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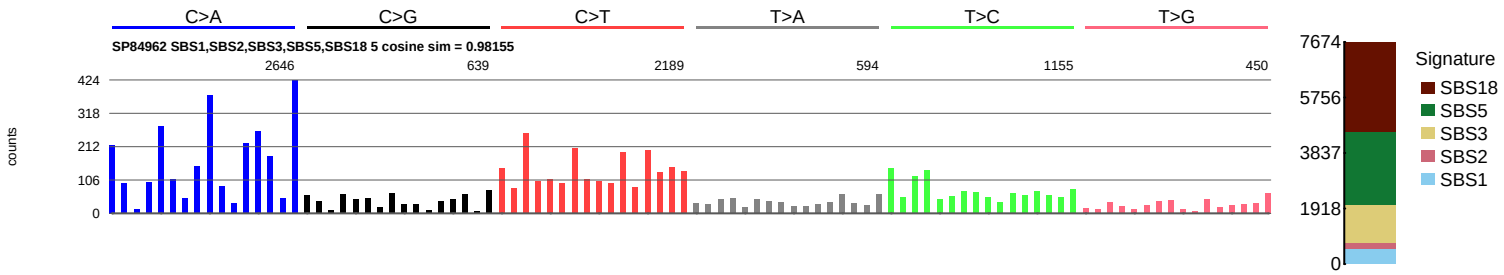
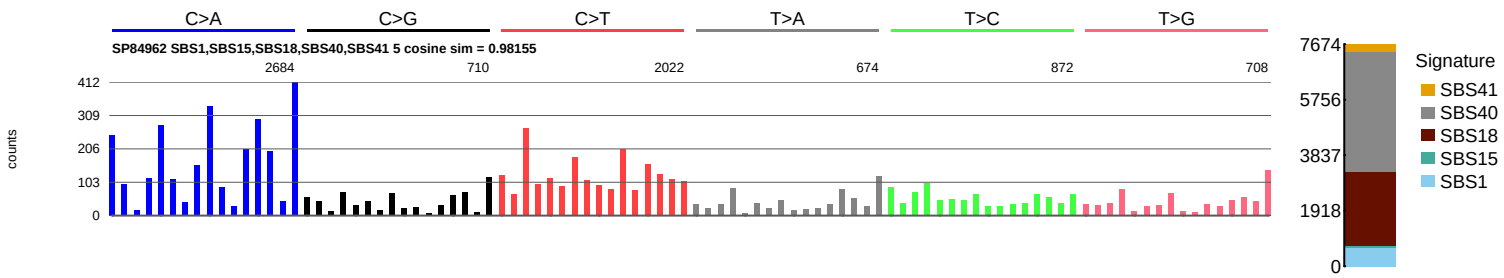
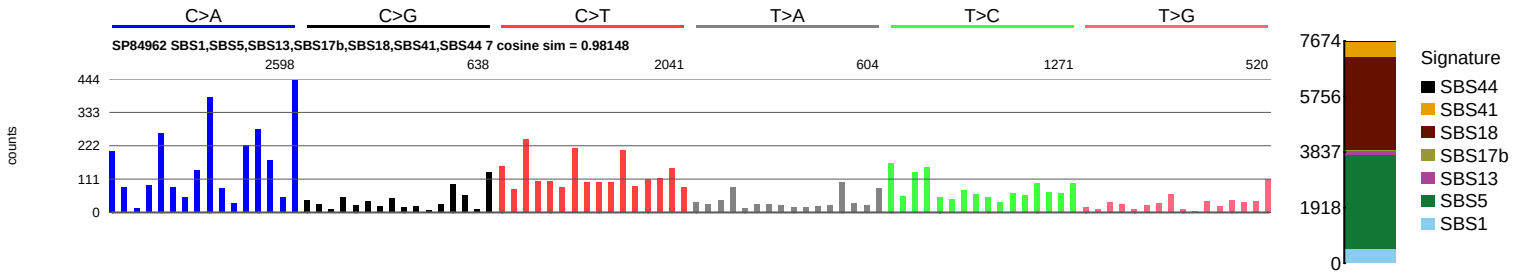
T>C
T>G
SP84962 SBS1,SBS18,SBS26,SBS40 4 cosine sim = 0.98157
2694
690
1990
614
1012
674
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS40
SBS26
SBS18
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS3,SBS5,SBS18 5 cosine sim = 0.98155
2646
639
2189
594
1155
450
counts
424
318
212
106
0
7674
5756
3837
1918
0
Signature
SBS18
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS15,SBS18,SBS40,SBS41 5 cosine sim = 0.98155
2684
710
2022
674
872
708
counts
412
309
206
103
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS1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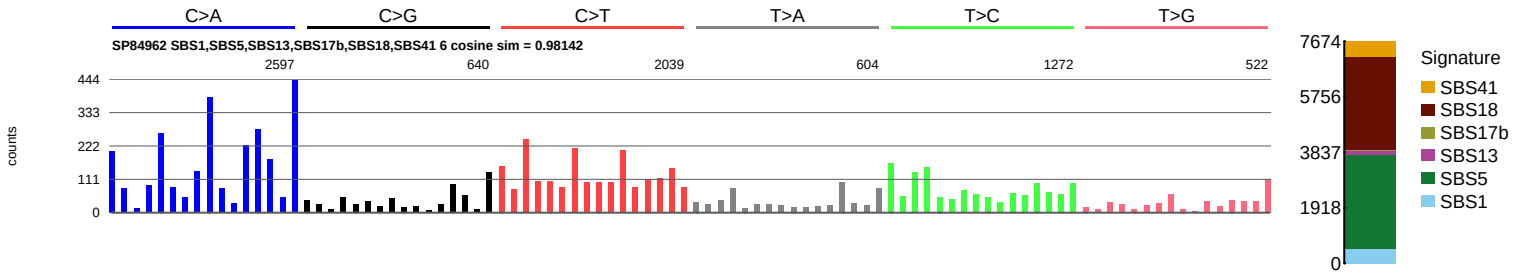
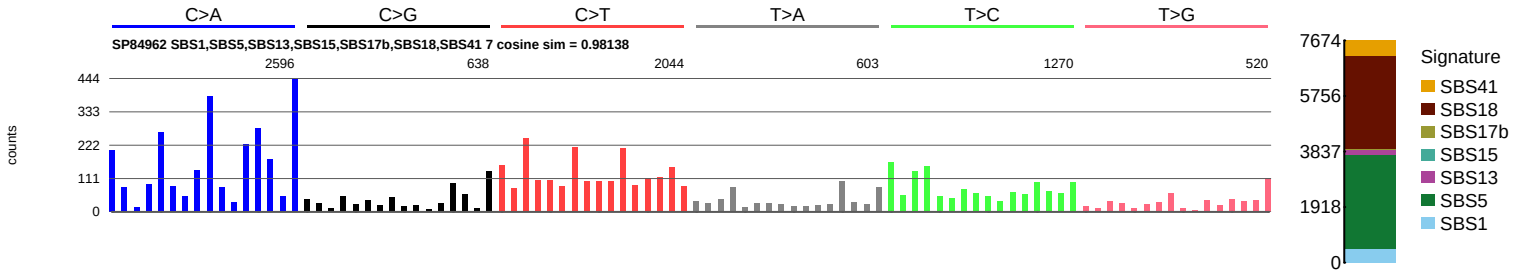
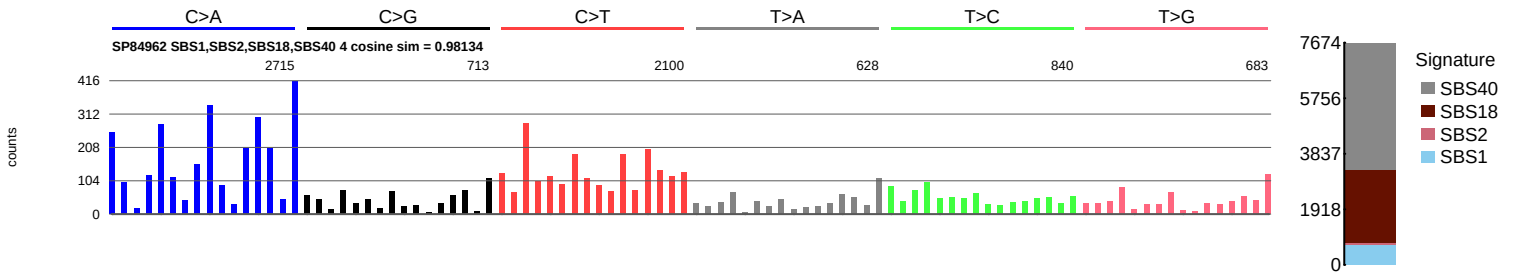
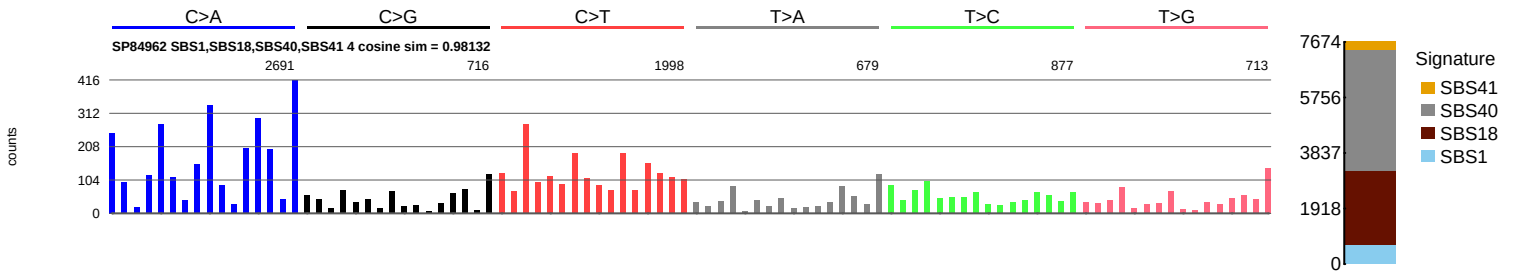
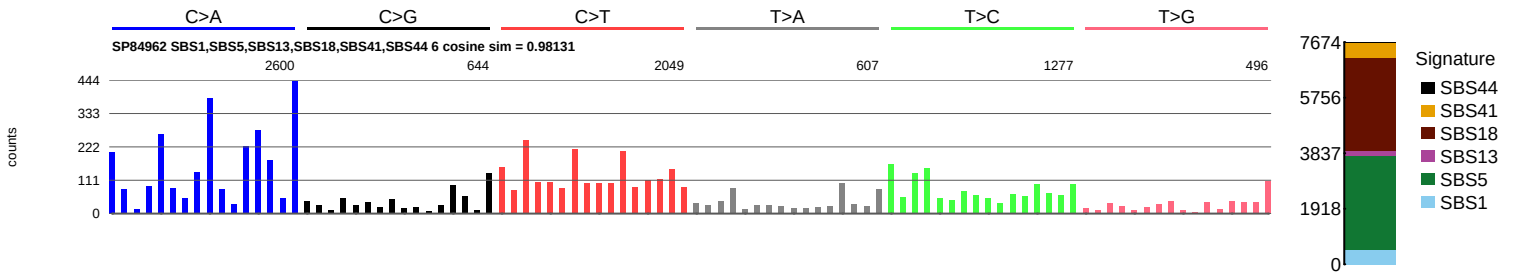
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98148
2598
638
2041
604
1271
520
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS1

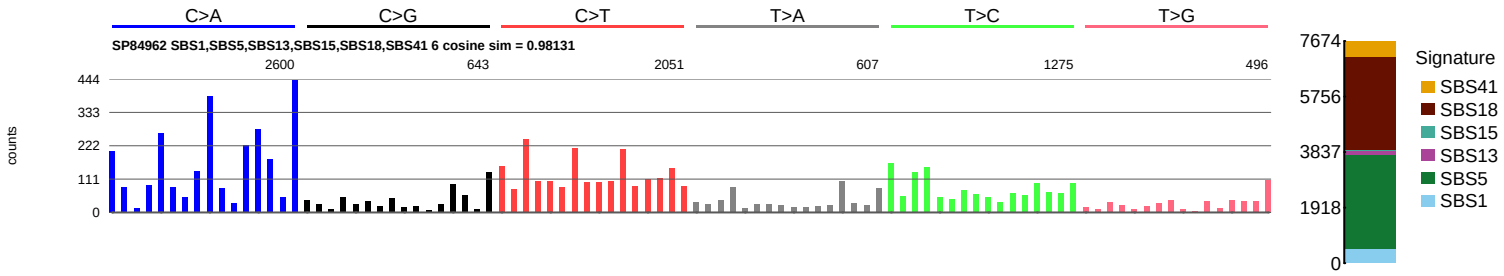

C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS41 6 cosine sim = 0.98142
2597
640
2039
604
1272
522
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 7 cosine sim = 0.98138
2596
638
2044
603
1270
520
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS2,SBS18,SBS40 4 cosine sim = 0.98134
2715
713
2100
628
840
683
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS40
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS18,SBS40,SBS41 4 cosine sim = 0.98132
2691
716
1998
679
877
713
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS41
SBS40
SBS18
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS13,SBS18,SBS41,SBS44 6 cosine sim = 0.98131
2600
644
2049
607
1277
496
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
SBS44
SBS41
SBS18
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS13,SBS15,SBS18,SBS41 6 cosine sim = 0.98131
2600
643
2051
607
1275
496
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
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SBS18
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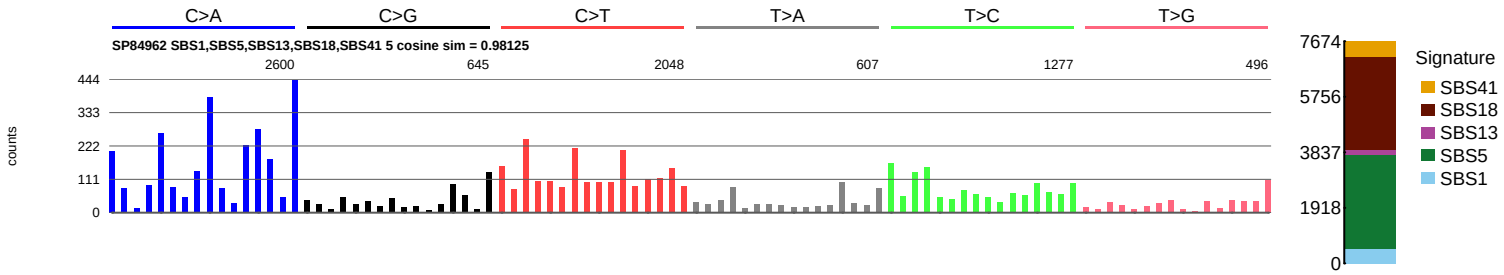
SBS13
SBS5
SBS1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T>A
T>C
T>G
SP84962 SBS1,SBS5,SBS13,SBS18,SBS41 5 cosine sim = 0.98125
2600
645
2048
607
1277
496
counts
444
333
222
111
0
7674
5756
3837
1918
0
Signature
SBS41
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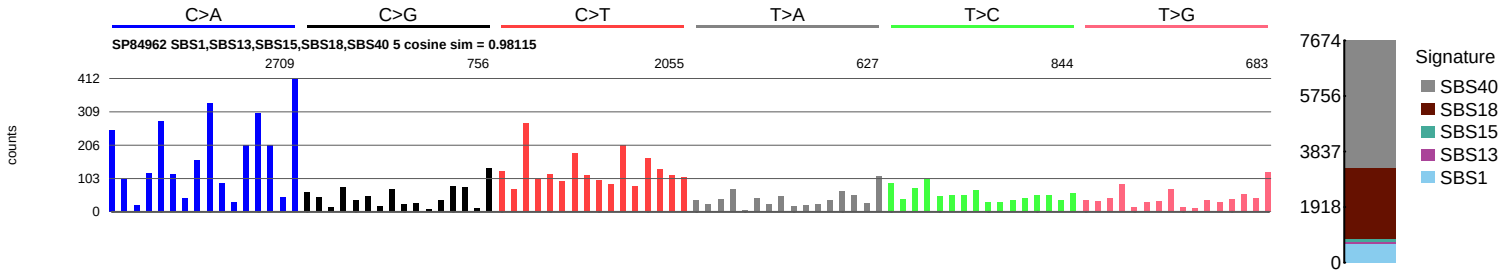
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS15,SBS18,SBS40 5 cosine sim = 0.98115
2709
756
2055
627
844
683
counts
412
309
206
103
0
7674
5756
3837
1918
0
Signature
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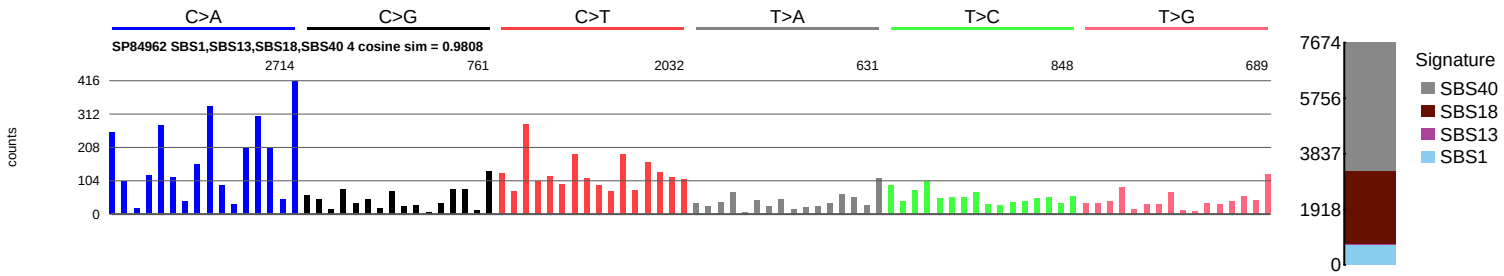
SBS13
SBS1


C>A
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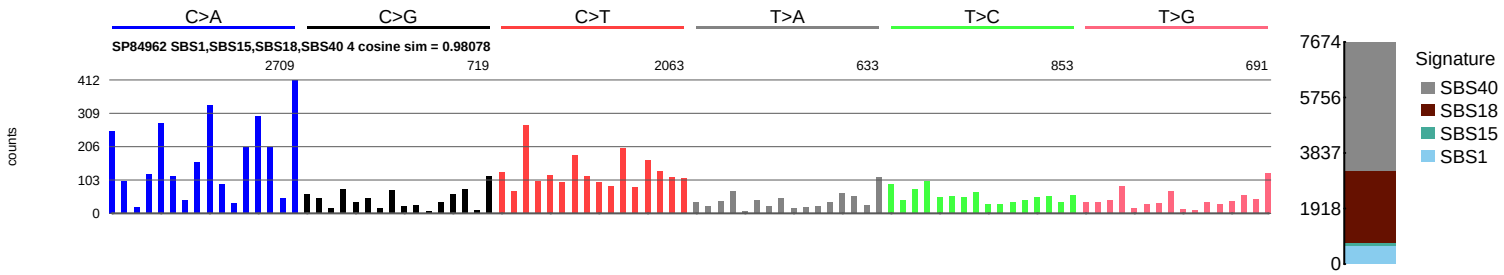
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS13,SBS18,SBS40 4 cosine sim = 0.9808
2714
761
2032
631
848
689
counts
416
312
208
104
0
7674
5756
3837
1918
0
Signature
SBS40
SBS18
SBS13
SBS1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C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS15,SBS18,SBS40 4 cosine sim = 0.98078
2709
719
2063
633
853
691
counts
412
309
206
103
0
7674
5756
3837
1918
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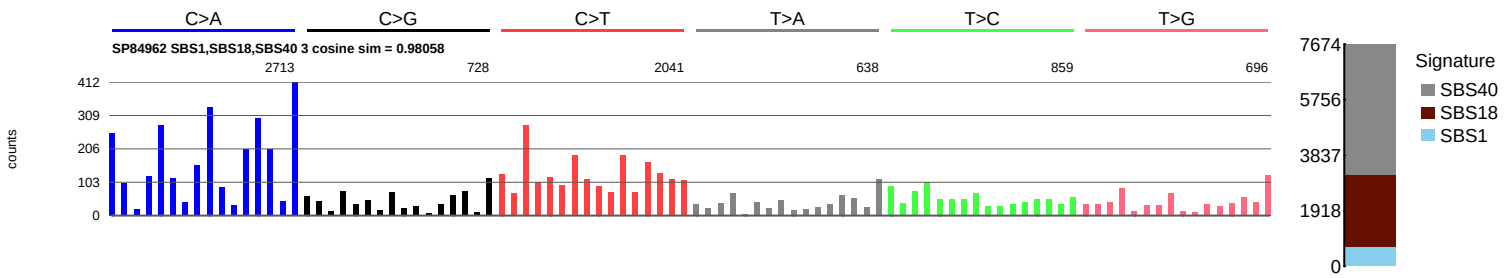
Signature
SBS40
SBS18
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84962 SBS1,SBS18,SBS40 3 cosine sim = 0.98058
2713
728
2041
638
859
696
counts
412
309
206
103
0
7674
5756
3837
1918
0
Signature
SBS40
SBS18
SBS1